



# Hospital Management System

## Multi-tenant SaaS Platform

Product, User Flow and Technical Specification

React · Node.js · MongoDB · AWS S3 + CloudFront

Delivery: 1 month · Price: INR 1,00,000 + GST

Version 1.3 · 09 October 2026

### What Phase 1 delivers

- 1 Security and Foundations**  
Roles and logins, maker-checker, Super Admin approvals, department registration
- 2 Patient Operations**  
Registration, OPD appointments, IPD admission and discharge, real-time beds
- 3 Clinical Workflows**  
Nursing stations and notes, shift rosters, doctor scheduling, staff directory
- 4 Diagnostics and Core Departments**  
Laboratory, radiology, pharmacy
- 5 Billing and Reporting**  
Cash, billing, printing, finance, inventory, HR, payroll
- 6 Support Services, Quality and Engagement**  
Medical records, diet and kitchen, housekeeping and facility, quality, patient portal, CRM, ABDM

**SaaS promise.** One cloud product replaces separate billing, pharmacy, lab, HR, payroll, inventory and accounting software. Hospitals sign up online, subscribe module by module, and switch modules on as they grow. Nothing to install.

# Contents

|                                                                |           |
|----------------------------------------------------------------|-----------|
| <b>1. Executive Summary and Scope</b>                          | <b>6</b>  |
| 1.1 What this document contains                                | 6         |
| 1.2 Phase 1 scope                                              | 6         |
| 1.3 Out of scope for Phase 1                                   | 7         |
| 1.4 Delivery and commercials                                   | 7         |
| 1.5 Key assumptions                                            | 7         |
| <b>2. Product Model and Module Subscription</b>                | <b>8</b>  |
| 2.1 Module catalogue                                           | 8         |
| 2.2 Plans and indicative pricing                               | 9         |
| 2.3 Subscription rules                                         | 9         |
| 2.4 Subscription lifecycle                                     | 10        |
| <b>3. SaaS Platform: Tenants, Subscriptions and Operations</b> | <b>11</b> |
| 3.1 Who uses what                                              | 11        |
| 3.2 Platform roles                                             | 11        |
| 3.3 Self-service signup and trial                              | 12        |
| 3.4 Tenant provisioning                                        | 12        |
| 3.5 Subscription billing engine                                | 12        |
| 3.6 Adding or removing a module (hospital side)                | 13        |
| 3.7 Limits and metering                                        | 14        |
| 3.8 Tenant isolation and fair use                              | 14        |
| 3.9 Custom domains and white-label                             | 14        |
| 3.10 Platform Console                                          | 15        |
| 3.11 SaaS business metrics                                     | 15        |
| 3.12 Releases, support and service levels                      | 15        |
| 3.13 Data ownership and offboarding                            | 16        |
| 3.14 Platform data model                                       | 16        |
| 3.15 Platform APIs                                             | 16        |
| 3.16 Platform reference code                                   | 17        |
| <b>4. Roles, Logins and Approvals</b>                          | <b>19</b> |
| 4.1 Permission model                                           | 19        |
| 4.2 System roles                                               | 19        |
| 4.3 Role to module access matrix                               | 20        |
| 4.4 Login and session security                                 | 21        |
| 4.5 Maker-checker approvals                                    | 22        |
| 4.6 Super Admin approvals                                      | 23        |
| 4.7 Audit trail                                                | 24        |

|                                                            |           |
|------------------------------------------------------------|-----------|
| <b>5. Core Platform: Setup, Departments and Patients</b>   | <b>25</b> |
| 5.1 Hospital and branch setup                              | 25        |
| 5.2 Department registration                                | 25        |
| 5.3 Master data                                            | 26        |
| 5.4 Patient registration                                   | 26        |
| 5.5 Front office: enquiries and visitor passes             | 27        |
| 5.6 ABDM integration (Ayushman Bharat Digital Mission)     | 27        |
| 5.7 Patient portal                                         | 28        |
| 5.8 Staff directory                                        | 28        |
| 5.9 Notifications and documents                            | 28        |
| 5.10 Dashboards                                            | 28        |
| <b>6. Patient Operations: OPD, IPD and Beds</b>            | <b>30</b> |
| 6.1 Doctor schedules and slots                             | 30        |
| 6.2 Booking channels                                       | 30        |
| 6.3 Consultation screen (doctor)                           | 31        |
| 6.4 Specialty templates and treatment plans                | 31        |
| 6.5 Health check-up packages                               | 32        |
| 6.6 Tele-consultation                                      | 32        |
| 6.7 Admission                                              | 33        |
| 6.8 During the stay                                        | 33        |
| 6.9 Discharge                                              | 34        |
| 6.10 Insurance, TPA and government schemes                 | 34        |
| 6.11 Packages and day care                                 | 35        |
| 6.12 OPD and IPD analytics                                 | 35        |
| 6.13 Real-time bed management                              | 35        |
| <b>7. Clinical Workflows: Nursing, Doctors and Rosters</b> | <b>37</b> |
| 7.1 Nursing station screen                                 | 37        |
| 7.2 Features                                               | 37        |
| 7.3 Doctor profiles and scheduling                         | 38        |
| 7.4 Shift rosters                                          | 38        |
| <b>8. Diagnostics and Pharmacy</b>                         | <b>40</b> |
| 8.1 Laboratory features                                    | 40        |
| 8.2 Radiology workflows                                    | 41        |
| 8.3 Pharmacy features                                      | 43        |
| <b>9. Billing, Finance, Inventory, HR and Payroll</b>      | <b>45</b> |
| 9.1 Cash, billing and printing (CORE)                      | 45        |
| 9.2 Financial reporting and accounts                       | 46        |
| 9.3 Inventory and stock tracking                           | 47        |

|                                                                      |           |
|----------------------------------------------------------------------|-----------|
| 9.4 HR management                                                    | 48        |
| 9.5 Payroll management                                               | 49        |
| <b>10. Support Services, Records, Quality and Patient Engagement</b> | <b>50</b> |
| 10.1 Medical records features                                        | 50        |
| 10.2 Diet and kitchen features                                       | 51        |
| 10.3 Housekeeping and linen                                          | 52        |
| 10.4 Maintenance and biomedical equipment                            | 52        |
| 10.5 Biomedical waste                                                | 53        |
| 10.6 Quality features                                                | 53        |
| 10.7 Patient CRM features                                            | 54        |
| <b>11. End-to-End Journeys and Screen Map</b>                        | <b>55</b> |
| 11.1 In-patient journey across modules                               | 55        |
| 11.2 Money journey: charge to ledger                                 | 55        |
| 11.3 People journey: hire to pay                                     | 55        |
| 11.4 Stock journey: purchase to patient                              | 56        |
| 11.5 Screen map                                                      | 56        |
| <b>12. Technical Architecture</b>                                    | <b>57</b> |
| 12.1 Technology stack                                                | 57        |
| 12.2 Deployment architecture on AWS                                  | 57        |
| 12.3 Multi-tenancy                                                   | 59        |
| 12.4 Backend structure: modular monolith                             | 59        |
| 12.5 Request lifecycle                                               | 59        |
| 12.6 Domain events                                                   | 60        |
| <b>13. Database Design (MongoDB)</b>                                 | <b>61</b> |
| 13.1 Design rules                                                    | 61        |
| 13.2 Collections by module                                           | 61        |
| 13.3 Critical indexes                                                | 63        |
| 13.4 Sample documents                                                | 63        |
| <b>14. Reference Source Code</b>                                     | <b>64</b> |
| 14.1 Backend foundation                                              | 64        |
| 14.2 Numbering, transactions and approvals                           | 67        |
| 14.3 Patient, bed and pharmacy services                              | 68        |
| 14.4 Billing and payroll services                                    | 70        |
| 14.5 Frontend foundation                                             | 71        |
| <b>15. REST API Documentation</b>                                    | <b>73</b> |
| 15.1 Conventions                                                     | 73        |
| 15.2 Authentication                                                  | 74        |
| 15.3 Platform, settings and approvals                                | 74        |

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| 15.4 Patients . . . . .                                               | 75        |
| 15.5 OPD and appointments . . . . .                                   | 75        |
| 15.6 IPD and beds . . . . .                                           | 76        |
| 15.7 Nursing . . . . .                                                | 77        |
| 15.8 Laboratory and radiology . . . . .                               | 77        |
| 15.9 Pharmacy and inventory . . . . .                                 | 78        |
| 15.10 Billing . . . . .                                               | 78        |
| 15.11 HR, attendance and payroll . . . . .                            | 79        |
| 15.12 Finance and reports . . . . .                                   | 80        |
| 15.13 Medical records, diet, facility and quality . . . . .           | 80        |
| 15.14 CRM, ABDM and clinic features . . . . .                         | 81        |
| 15.15 Front office, portal and other additions . . . . .              | 81        |
| 15.16 WebSocket events . . . . .                                      | 82        |
| <b>16. Deployment on AWS (S3, CloudFront, ECS) . . . . .</b>          | <b>83</b> |
| 16.1 Environments . . . . .                                           | 83        |
| 16.2 One-time AWS setup . . . . .                                     | 83        |
| 16.3 React app on S3 and CloudFront . . . . .                         | 83        |
| 16.4 API and worker containers . . . . .                              | 84        |
| 16.5 CI/CD with GitHub Actions . . . . .                              | 84        |
| 16.6 Environment variables . . . . .                                  | 85        |
| 16.7 Release and rollback . . . . .                                   | 86        |
| <b>17. Security, Compliance and Operations . . . . .</b>              | <b>87</b> |
| 17.1 Security controls . . . . .                                      | 87        |
| 17.2 Compliance . . . . .                                             | 87        |
| 17.3 Backup and disaster recovery . . . . .                           | 87        |
| 17.4 Monitoring and support . . . . .                                 | 88        |
| <b>18. Non-Functional Requirements and Testing . . . . .</b>          | <b>89</b> |
| 18.1 Non-functional requirements . . . . .                            | 89        |
| 18.2 Testing strategy . . . . .                                       | 89        |
| <b>19. Phase 1 Delivery Plan, Commercial and Acceptance . . . . .</b> | <b>90</b> |
| 19.1 Delivery summary . . . . .                                       | 90        |
| 19.2 Delivery squads . . . . .                                        | 90        |
| 19.3 Four-week plan . . . . .                                         | 90        |
| 19.4 Delivery assumptions . . . . .                                   | 91        |
| 19.5 Commercial . . . . .                                             | 91        |
| 19.6 Go-live checklist for each hospital . . . . .                    | 91        |
| 19.7 Phase 1 acceptance criteria . . . . .                            | 92        |
| 19.8 Glossary . . . . .                                               | 92        |

## SECTION 01

# Executive Summary and Scope

This document is the complete Phase 1 specification for the **Hospital Management System**, a multi-tenant SaaS product that runs every operational, clinical, financial and people process of a hospital from one product. The goal is simple: once a hospital goes live, it should not need a separate billing tool, pharmacy package, HR tool, payroll tool, inventory tool or accounting package.

The product is sold as a **module-wise subscription**. A hospital always buys the Core platform, then switches on only the modules it needs, such as Pharmacy, Laboratory or Payroll. Modules can be added or removed later without migration or reinstallation. The same codebase serves a 10-bed clinic and a 500-bed hospital.

The stack is **React** for the web application, **Node.js** for the API and background workers, and **MongoDB** for data. The React build is hosted on **Amazon S3 behind CloudFront**. Patient documents, reports and payslips are stored in a separate private S3 bucket. The API runs in containers on AWS.

## 1.1 What this document contains

| Section | Contents                                                                                                            | Primary reader                       |
|---------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 1-2     | Scope, product model, module catalogue and subscription rules                                                       | Founders, sales, product             |
| 3       | SaaS platform: signup, provisioning, subscription billing, metering, custom domains, platform console, SaaS metrics | Founders, sales, finance, developers |
| 4       | Roles, permissions, maker-checker and Super Admin approvals                                                         | Product, security, QA                |
| 5-10    | Module-by-module features, user flows, business rules, data and screens                                             | Product, developers, QA, trainers    |
| 11      | End-to-end patient, money and people journeys across modules                                                        | Everyone                             |
| 12-14   | Architecture, database design and reference source code                                                             | Developers, architects               |
| 15      | REST API documentation with request and response examples                                                           | Frontend and backend developers      |
| 16-17   | AWS deployment on S3 and CloudFront, CI/CD, security and operations                                                 | DevOps, security                     |
| 18-19   | Non-functional requirements, testing, delivery plan and acceptance                                                  | Project managers, QA, client         |

## 1.2 Phase 1 scope

Phase 1 delivers the five pillars from the roadmap, plus the people, money and support-service modules a hospital needs so that it can retire every other tool.

| Roadmap pillar                          | Phase 1 modules in this document                                                                                                                                                                                                                                       |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Security and Foundations</b>         | System roles and logins, maker-checker approvals, Super Admin approvals, department registration, hospital setup and masters, audit trail, notifications                                                                                                               |
| <b>Patient Operations</b>               | Patient registration (UHID), OPD appointment scheduling and queue, IPD admission, transfer and discharge, real-time bed management, insurance and TPA pre-authorisation and claims, government schemes (PM-JAY, CGHS, ECHS), packages, day care, OPD and IPD analytics |
| <b>Clinical Workflows</b>               | Nursing stations and nursing notes, shift rosters, doctor profiles and scheduling, staff directory, OPD consultation and e-prescription                                                                                                                                |
| <b>Diagnostics and Core Departments</b> | Laboratory and diagnostics tracking, radiology workflows, pharmacy back-end and dispensing                                                                                                                                                                             |
| <b>Billing and Reporting</b>            | Cash, billing and printing, financial reporting and accounts, inventory and stock tracking, purchase, HR, recruitment, training, attendance, leave and payroll management                                                                                              |

| Roadmap pillar                                                                | Phase 1 modules in this document                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Support services and quality</b><br>(added so no other software is needed) | Medical records with birth, death and medico-legal registers; diet and kitchen; housekeeping, linen, maintenance, biomedical equipment and waste; quality, feedback, complaints, incident reporting and infection control; patient portal; visitor passes; patient CRM and outreach; ABDM integration |

## 1.3 Out of scope for Phase 1

These items belong to Phase 2. The Phase 1 design leaves hooks for them so they plug in later as new subscription modules without rework.

| Phase 2 item                                                                     | Phase 1 hook already in place                                                                                                                                 |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Emergency Room, Operation Theatre (with CSSD)                                    | Admission source field, bed categories, service master and billing engine accept new departments                                                              |
| Direct insurer exchange (National Health Claims Exchange) and scheme portal APIs | Phase 1 insurance desk keeps pre-authorisations, replies, documents and claims with insurer reference numbers; staff key them into insurer and scheme portals |
| Blood bank, ambulance, global referrals                                          | Generic order and service model, partner master, notification engine                                                                                          |
| Advanced analytics (staff, doctor and nurse performance scorecards, forecasting) | Every event is timestamped (check-in, consult start, consult end, billing, exit) and kept in an event log                                                     |
| AI features (follow-up bots, smart registration, dictation, OCR, remote care)    | Structured prescriptions, notification queue, document store on S3, open API                                                                                  |

## 1.4 Delivery and commercials

|                 |                                                                                  |
|-----------------|----------------------------------------------------------------------------------|
| <b>Delivery</b> | All Phase 1 modules delivered and live within 1 month of kickoff                 |
| <b>Price</b>    | INR 1,00,000 one-time for Phase 1 delivery, plus GST                             |
| <b>Details</b>  | Week-by-week plan, delivery assumptions and payment milestones are in Section 19 |

## 1.5 Key assumptions

- The first market is India. Examples use INR, GST, PF, ESI, Professional Tax and TDS. All taxes and statutory deductions are configuration, not code, so other countries can be supported by changing masters.
- The product is multi-tenant SaaS. Each hospital is a tenant. A hospital group can run several branches under one tenant.
- Users work on desktop browsers at counters and on tablets in wards. A responsive layout covers both. Native mobile apps are not part of Phase 1.
- Lab analysers and biometric devices connect through file import or a small connector service. Full HL7 / ASTM bidirectional interfacing is optional in Phase 1.
- Prices shown in this document are indicative and must be finalised by the business.
- Web addresses use example.com as a stand-in for the product's own domain, chosen when the brand is decided.

## SECTION 02

## Product Model and Module Subscription

The product is one application with many modules. What a hospital sees is decided by its subscription. A module that is not subscribed is hidden from the menu and its APIs refuse requests, so there is no way to use it by accident or by calling the API directly.

### 2.1 Module catalogue

| Code | Module                     | Type      | Requires | Key capabilities                                                                                                                                                                                                                       |
|------|----------------------------|-----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CORE | Core Platform              | Mandatory | -        | Logins, roles, maker-checker, Super Admin approvals, hospital and department setup, patient registration, basic billing and receipts, staff directory, audit, notifications, dashboards                                                |
| OPD  | OPD and Appointments       | Add-on    | CORE     | Doctor schedules, slots, booking, walk-in tokens, queue screen, vitals, consultation notes, e-prescription, follow-ups, OPD analytics (wait time per stage)                                                                            |
| IPD  | IPD, ADT and Beds          | Add-on    | CORE     | Admission, transfer, discharge, real-time bed board, running bill, deposits, discharge summary, insurance and TPA desk (pre-authorisation, enhancement, final approval, claims), government schemes, packages, day care, IPD analytics |
| NUR  | Nursing Station            | Add-on    | IPD      | Ward census, vitals chart, medication administration, nursing notes, intake and output, handover, indents                                                                                                                              |
| LAB  | Laboratory (LIS)           | Add-on    | CORE     | Test masters, sample collection, barcodes, result entry, validation, reports, TAT tracking                                                                                                                                             |
| RAD  | Radiology (RIS)            | Add-on    | CORE     | Modality scheduling, study tracking, report templates, radiologist sign-off, PACS viewer link                                                                                                                                          |
| PHR  | Pharmacy                   | Add-on    | CORE     | Drug master, purchase and GRN, batch and expiry, FEFO dispensing, OPD sales, IPD issues, returns, controlled drugs register                                                                                                            |
| INV  | Inventory and Purchase     | Add-on    | CORE     | Item master, vendors, requisitions, purchase orders, GRN, stores, indents, transfers, audits, assets                                                                                                                                   |
| HRM  | HR, Rosters and Attendance | Add-on    | CORE     | Employee lifecycle, documents, shifts, rosters, attendance, leave, holidays, exit                                                                                                                                                      |
| PAY  | Payroll                    | Add-on    | HRM      | Salary structures, statutory deductions, loans, payroll run with approval, payslips, bank file, payroll journal                                                                                                                        |
| FIN  | Finance and Accounts       | Add-on    | CORE     | Chart of accounts, auto journals, vendor payables, ledgers, trial balance, P&L, balance sheet, GST reports, petty cash, expense claims, budgets                                                                                        |
| MRD  | Medical Records            | Add-on    | CORE     | Record completion and coding, file tracking, record release, birth, death, mortuary and medico-legal registers, hospital statistics                                                                                                    |
| DIET | Diet and Kitchen           | Add-on    | IPD      | Diet orders, NBM, dietitian notes, kitchen production sheet, tray labels, delivery, meal charges                                                                                                                                       |
| FAC  | Housekeeping and Facility  | Add-on    | CORE     | Bed cleaning tasks, area checklists, linen and laundry, maintenance tickets, biomedical equipment PM and AMC, biomedical waste                                                                                                         |
| QLT  | Quality and Feedback       | Add-on    | CORE     | Patient feedback, complaints, incident reporting with CAPA, NABH indicators, audits, document control, infection control                                                                                                               |
| CRM  | Patient CRM and Outreach   | Add-on    | CORE     | Call centre, leads, health camps, campaigns, memberships and wallet, referral partners, review requests                                                                                                                                |

**Bundling decisions.** Bed management is part of IPD because one cannot work without the other. Doctor profiles live in CORE because billing and lab reports need them even when OPD is not subscribed. Doctor schedules and slots are part of OPD.

**Clinic mode.** Clinics get a simplified screen layout: one screen to register, consult, prescribe, bill and print, and a doctor can also act as admin. The same modules run underneath, so a clinic can grow into a hospital without migrating.

## 2.2 Plans and indicative pricing

Hospitals can buy a bundled plan or pick modules one by one. Pricing scales by licensed beds and named users. All prices are monthly, exclude GST and are placeholders for the business to finalise.

|                         | Clinic                                  | Hospital               | Enterprise                              |
|-------------------------|-----------------------------------------|------------------------|-----------------------------------------|
| <b>Target</b>           | OPD clinics, polyclinics, up to 20 beds | 20-150 beds            | 150+ beds, groups, multi-branch         |
| <b>Included modules</b> | CORE, OPD, PHR, LAB                     | All 16 Phase 1 modules | All modules + dedicated database option |
| <b>Users included</b>   | 15                                      | 100                    | Unlimited                               |
| <b>Branches</b>         | 1                                       | Up to 3                | Unlimited                               |
| <b>Document storage</b> | 25 GB                                   | 250 GB                 | 1 TB, then metered                      |
| <b>Support</b>          | Email, business hours                   | Phone + email, 12x6    | 24x7, named manager                     |
| <b>Indicative price</b> | INR 6,000 / month                       | INR 35,000 / month     | Custom quote                            |

| A-la-carte module        | Indicative price / month | Metering                                  |
|--------------------------|--------------------------|-------------------------------------------|
| CORE (mandatory)         | INR 3,000                | Includes 10 users; INR 150 per extra user |
| OPD                      | INR 2,000                | Per branch                                |
| IPD (with beds)          | INR 40 per licensed bed  | Minimum 10 beds                           |
| NUR                      | INR 20 per licensed bed  | Follows IPD bed count                     |
| LAB, RAD, PHR, INV       | INR 2,500 each           | Per branch                                |
| HRM                      | INR 30 per employee      | Active employees in month                 |
| PAY                      | INR 25 per payslip       | Payslips generated in month               |
| FIN                      | INR 3,000                | Per legal entity                          |
| MRD, DIET, QLT           | INR 1,500 each           | Per branch                                |
| FAC, CRM                 | INR 2,000 each           | Per branch                                |
| Patient portal (in CORE) | Included                 | Online payments carry gateway fees        |

## 2.3 Subscription rules

- **Dependency check.** A module cannot be enabled unless its required modules are active. PAY needs HRM. NUR needs IPD. Disabling a module that others depend on is blocked with a clear message.
- **Trial.** A new tenant gets a 14-day trial of the Hospital plan with demo data that can be wiped in one click.
- **Billing cycle.** Monthly or annual. Annual gets two months free. Upgrades take effect immediately and are pro-rated. Downgrades apply at the next renewal.
- **Limits.** Users, beds, branches and storage are enforced. When a limit is reached, the action is blocked and the Hospital Super Admin is told how to upgrade.
- **Grace period.** On failed payment, the tenant gets 7 days of full access with a banner, then 15 days of read-only access, then suspension. Clinical data is never deleted during this window.

- **Data on cancellation.** Data is retained for 90 days and can be exported as JSON, CSV and PDF reports. After 90 days it is deleted, except where law requires longer retention, in which case it is archived encrypted.
- **Data on module removal.** Data of a removed module is kept and becomes visible again if the module is re-enabled. Historical bills and reports that reference it still print.
- **Payment collection.** Subscription invoices are generated by the platform and paid via Razorpay (India) or Stripe (international), with auto-debit mandates.

## 2.4 Subscription lifecycle

### Tenant subscription states



| State     | What users can do                                         | Moves to                                     |
|-----------|-----------------------------------------------------------|----------------------------------------------|
| TRIAL     | Everything in the trial plan                              | ACTIVE on payment, CLOSED after 30 days idle |
| ACTIVE    | Everything in subscribed modules                          | PAST_DUE when an invoice is unpaid           |
| PAST_DUE  | Everything, with a payment banner                         | ACTIVE on payment, READ_ONLY after 7 days    |
| READ_ONLY | View and print only. No new bills, admissions or orders   | ACTIVE on payment, SUSPENDED after 15 days   |
| SUSPENDED | Only the Hospital Super Admin can log in to pay or export | ACTIVE on payment, CLOSED after 90 days      |
| CLOSED    | Nothing. Data deleted or archived per retention policy    | -                                            |

**Clinical safety rule.** Even in READ\_ONLY and SUSPENDED states, an admitted patient's current medication chart and lab results remain viewable. Patient safety overrides commercial enforcement.

**SaaS operations.** Signup, tenant provisioning, subscription billing, metering, custom domains and the Platform Console are specified in Section 3.

## SECTION 03

## SaaS Platform: Tenants, Subscriptions and Operations

The Hospital Management System is a **multi-tenant SaaS** product. You, the platform owner, run one cloud deployment. Every hospital that signs up becomes a **tenant** on it. Hospitals never install anything. They open a browser, pay a subscription, and get updates automatically. This section covers everything on the platform-owner side: how hospitals sign up, how they are provisioned, how they are billed, and how the platform is operated.

### 3.1 Who uses what

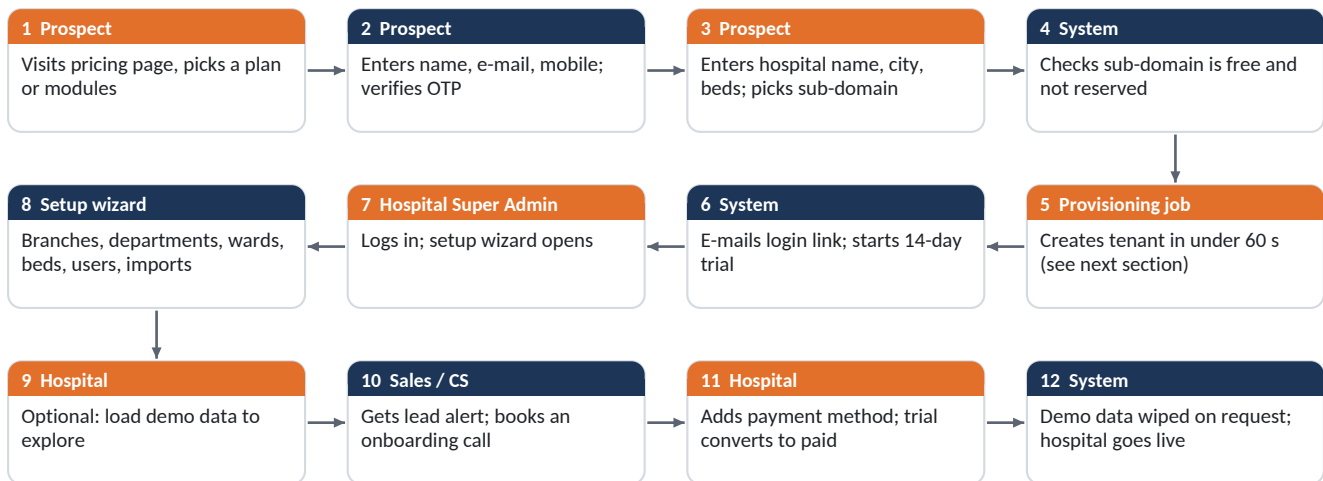
| Party                | Application                            | URL                                     | Hosting                        |
|----------------------|----------------------------------------|-----------------------------------------|--------------------------------|
| Prospects            | Marketing site, pricing, signup        | www.example.com                         | S3 + CloudFront                |
| Platform owner staff | Platform Console                       | console.example.com                     | S3 + CloudFront, IP-restricted |
| Hospital staff       | Hospital app (all modules)             | {hospital}.example.com or custom domain | S3 + CloudFront                |
| Patients             | Booking page, report and receipt links | {hospital}.example.com/book             | S3 + CloudFront                |
| Integrations         | Public REST API with API keys          | api.example.com/v1                      | ECS behind ALB                 |

**One codebase.** All hospitals run the same version of the same code. A hospital's subscription only changes which modules, limits and features are switched on for it. There are no per-hospital code branches.

### 3.2 Platform roles

| Role                      | Can do                                                                   | Cannot do                                             |
|---------------------------|--------------------------------------------------------------------------|-------------------------------------------------------|
| Platform Super Admin      | Everything on the console, manage platform staff, plans and prices       | Read patient data without a consented support session |
| Sales                     | Create leads and trial tenants, send quotes, apply approved discounts    | Change prices, see platform finance                   |
| Implementation Consultant | Run setup wizard and data imports for assigned tenants during onboarding | Access tenants after go-live without consent          |
| Support (L1 / L2)         | View tenant health, usage, errors; open consented support sessions       | Change subscriptions or billing                       |
| Platform Finance          | Invoices, payments, credit notes, refunds, revenue reports               | Change modules or tenant settings                     |
| Platform Ops / DevOps     | Feature flags, releases, maintenance windows, tenant migrations          | Change billing                                        |

### 3.3 Self-service signup and trial



- **Sales-assisted path.** Enterprise and multi-branch groups are created by Sales from the console with a custom quote, longer trial and an assigned consultant.
- **Abuse protection.** OTP on mobile and e-mail, CAPTCHA, one trial per mobile number and domain, and manual review for unusual signups.
- **Reserved sub-domains.** www, api, console, admin, app, status, mail, and names of existing tenants cannot be taken.

### 3.4 Tenant provisioning

Provisioning is a background job. Every step is idempotent, so a failed job can be retried safely without creating duplicates.

| Step               | What it creates                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| 1. Tenant record   | Tenant in platform database with status PROVISIONING, plan, modules, limits, sub-domain                     |
| 2. Data location   | Shared cluster by default; dedicated database for Enterprise                                                |
| 3. System roles    | All standard roles and permissions from Section 4                                                           |
| 4. Default masters | Departments template, bed categories, tax codes, payment modes, ICD-10, units, leave types, shift templates |
| 5. Finance seed    | Hospital chart of accounts and number series for the current financial year                                 |
| 6. Templates       | Print layouts, SMS, e-mail and WhatsApp templates                                                           |
| 7. First user      | Hospital Super Admin with forced password set and 2FA                                                       |
| 8. Storage         | S3 prefix tenants/{tenantId}/ and storage quota counter                                                     |
| 9. Subscription    | Trial subscription, trial end date, metering baseline                                                       |
| 10. Activate       | Status TRIAL, tenant cache warmed, welcome e-mail sent                                                      |

### 3.5 Subscription billing engine

The platform bills hospitals itself. No separate billing tool is needed. Plans, modules and prices are data in the platform database, so Sales can launch a new plan or price without a code release.

| Concept   | Description                                                                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Catalogue | Modules, plans (bundles of modules and limits) and add-ons (extra users, beds, storage, SMS packs). Prices are versioned; existing customers keep their price until renewal |

| Concept                      | Description                                                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>          | One per tenant: plan, add-on modules, quantities, billing cycle (monthly or annual), currency, renewal date, payment method                      |
| <b>Fixed charges</b>         | Plan and module prices, billed in advance each cycle                                                                                             |
| <b>Metered charges</b>       | Extra users, beds, employees, payslips, storage, SMS and WhatsApp messages, billed in arrears from daily usage snapshots                         |
| <b>Prepaid credits</b>       | SMS and WhatsApp credit packs bought in advance; low-balance alert at 10%; sending pauses at zero except OTPs                                    |
| <b>Proration</b>             | Upgrades and module additions charge the unused part of the cycle immediately. Removals take effect at renewal; no refunds for the current cycle |
| <b>Coupons and discounts</b> | Percentage or fixed, for N cycles; Sales discounts above a limit need Platform Super Admin approval                                              |
| <b>Invoices</b>              | GST invoice with 18% GST and the platform's GSTIN, hospital GSTIN for input credit, PDF on S3, e-mailed; e-invoice IRN where applicable          |
| <b>Payments</b>              | Razorpay Subscriptions with e-mandate (UPI AutoPay, card, NACH) in India; Stripe Billing abroad; bank transfer recorded manually                 |
| <b>Dunning</b>               | Retry failed charges on day 1, 3 and 6 with e-mail, SMS and in-app banner; then status moves per the lifecycle in Section 2                      |
| <b>Credit notes</b>          | For billing errors and goodwill credits, adjusted on the next invoice                                                                            |

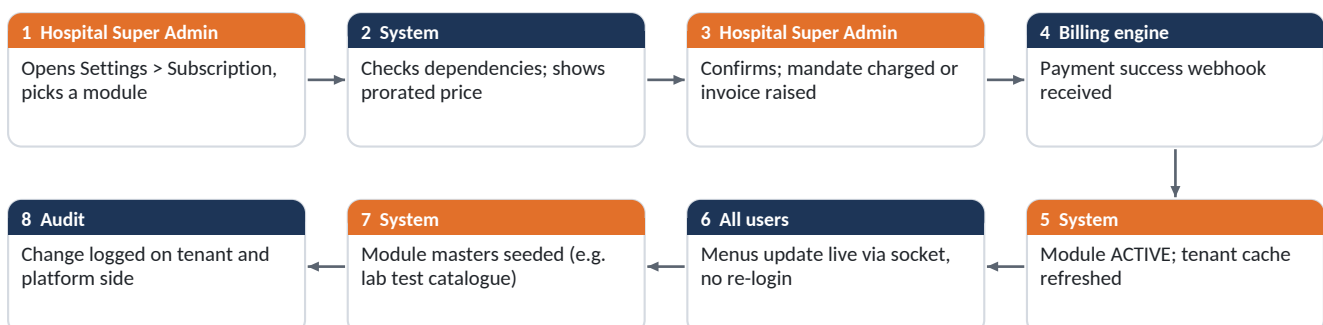
### Proration example

Hospital on monthly billing, cycle 1 Oct - 30 Oct (30 days). On 11 Oct it adds Laboratory (INR 2,500 / month). Days remaining including today = 20.

|                                   |                |
|-----------------------------------|----------------|
| Prorated charge = 2,500 × 20 / 30 | = INR 1,666.67 |
| GST 18%                           | = INR 300.00   |
| Invoice now                       | = INR 1,966.67 |

From the 31 Oct renewal invoice, Laboratory is billed in full at INR 2,500 + GST.

## 3.6 Adding or removing a module (hospital side)



- Removing a module is scheduled for the renewal date. Until then it stays usable. Afterwards it is hidden and its data is kept for re-activation.
- Downgrading below current usage is blocked with a list of what to reduce, e.g. 'deactivate 12 users to move to 100 users'.

### 3.7 Limits and metering

| Metric          | How it is counted                    | When the limit is hit                                    |
|-----------------|--------------------------------------|----------------------------------------------------------|
| Named users     | Active user accounts, daily snapshot | Cannot create new users; add-on offered                  |
| Licensed beds   | Active beds in bed master            | Cannot activate more beds                                |
| Branches        | Active branches                      | Cannot add a branch                                      |
| Employees (HRM) | Active employees in the month        | Billed as metered overage                                |
| Payslips (PAY)  | Payslips generated in the month      | Billed as metered overage                                |
| Storage         | Bytes in tenant S3 prefix, nightly   | Uploads blocked at 110%; generated documents still saved |
| SMS / WhatsApp  | Messages sent, real time             | Non-critical messages pause at zero credit               |
| API calls       | Per API key per minute               | HTTP 429 with retry time                                 |

**Clinical safety rule.** A limit never blocks patient care. Admissions, orders, results and medication charting always work even when a user or storage limit is exceeded; only administrative growth is blocked.

### 3.8 Tenant isolation and fair use

- **Data.** Every query is scoped by tenant through the Mongoose plugin (Section 14). Automated tests try to read other tenants' data on every endpoint.
- **Noisy neighbours.** Per-tenant API rate limits, per-tenant job queues so one hospital's payroll run cannot delay another's SMS, and 30-second query timeouts.
- **Large tenants.** A tenant can be moved to a dedicated database or cluster with a scripted, near-zero-downtime migration. Code does not change.
- **Per-tenant backup and restore.** A single tenant can be restored to a point in time into a side database, without affecting others.

### 3.9 Custom domains and white-label

| Feature                                    | Clinic | Hospital | Enterprise |
|--------------------------------------------|--------|----------|------------|
| Sub-domain (name.example.com)              | ✓      | ✓        | ✓          |
| Logo, colours, print letterhead            | ✓      | ✓        | ✓          |
| Custom domain (his.cityhospital.com)       | –      | ✓        | ✓          |
| Own e-mail sender domain and SMS sender ID | –      | ✓        | ✓          |
| Remove platform 'Powered by' branding      | –      | –        | ✓          |
| Own mobile-friendly booking page domain    | –      | ✓        | ✓          |

Custom domains: the hospital adds a CNAME record pointing to the platform. The platform requests an ACM certificate, validates it by DNS, and attaches the domain to CloudFront. With many custom domains, CloudFront SaaS Manager (multi-tenant distributions) is used so each tenant domain gets its own certificate without hitting per-distribution limits.

### 3.10 Platform Console

| Area             | Screens and functions                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------|
| Dashboard        | MRR, ARR, active tenants, trials, churn, failed payments, system health                                                   |
| Tenants          | List and search; tenant profile with plan, modules, usage, invoices, health score, activity, notes; suspend or reactivate |
| Leads and trials | Signups, trial progress (setup steps done), conversion follow-ups                                                         |
| Catalogue        | Modules, plans, add-ons, price versions, coupons                                                                          |
| Subscriptions    | Change plan, add or remove modules, extend trial, apply credit                                                            |
| Billing          | Invoices, payments, refunds, credit notes, dunning queue, GST reports for the platform company                            |
| Usage            | Per-tenant usage trends and limit warnings                                                                                |
| Support          | Consented support sessions, tenant error logs, job failures, announcements to tenants                                     |
| Releases         | Feature flags per tenant or plan, rollout stages, maintenance windows, in-app changelog                                   |
| Platform users   | Platform staff, roles, 2FA, access log                                                                                    |

### 3.11 SaaS business metrics

| Metric                | Definition                                                              | Why it matters                |
|-----------------------|-------------------------------------------------------------------------|-------------------------------|
| MRR / ARR             | Recurring revenue per month / year, excluding one-time fees             | Core growth measure           |
| Trial conversion      | Trials that become paid within 30 days                                  | Onboarding quality            |
| Logo churn            | Tenants cancelled in the month / tenants at start                       | Retention                     |
| Net revenue retention | Revenue this year from last year's tenants / their revenue last year    | Upsell of modules             |
| Module attach rate    | Share of tenants using each add-on module                               | Product and pricing decisions |
| ARPA                  | Average revenue per tenant account                                      | Pricing health                |
| Tenant health score   | Logins, daily bills, admissions, open tickets, failed payments combined | Early churn warning           |

### 3.12 Releases, support and service levels

- **Release stages.** Internal tenant, then pilot hospitals, then 10% of tenants, then everyone, controlled by feature flags. Hospitals see an in-app changelog.
- **Support access.** Platform staff can enter a tenant only through a support session that the Hospital Super Admin approves, limited to 2 hours and fully audited on both sides.
- **Customer communication.** Status page, scheduled-maintenance banner 72 hours ahead, incident e-mails to Super Admins.

| Service level       | Clinic                | Hospital          | Enterprise                         |
|---------------------|-----------------------|-------------------|------------------------------------|
| Uptime commitment   | 99.5%                 | 99.9%             | 99.9% with service credits         |
| Support hours       | Business hours        | 12 x 6            | 24 x 7                             |
| Severity 1 response | 4 hours               | 1 hour            | 30 minutes                         |
| Onboarding          | Self-service + videos | Remote consultant | On-site consultant                 |
| Data export         | Self-service          | Self-service      | Self-service + scheduled to own S3 |

### 3.13 Data ownership and offboarding

- The hospital owns its data. The platform is a data processor under a Data Processing Agreement signed at subscription.
- Self-service export at any time: full JSON per collection, CSV for masters and transactions, and all documents as a ZIP from S3.
- On cancellation, data is kept for 90 days, then deleted from the live database, search indexes and S3. Backups age out within 35 days. A deletion certificate is issued.
- Legal documents on signup: Terms of Service, Privacy Policy, Data Processing Agreement and SLA, accepted with timestamp and IP.

### 3.14 Platform data model

Platform data lives in a separate platform database. Hospital data lives in tenant-scoped collections. Hospital users can never query the platform database.

| Collection                         | Key fields                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| tenants                            | name, subdomain, customDomains[], status, dataLocation, branding, createdBy, trialEndsAt                                     |
| catalogModules                     | code, name, dependsOn[], seedJob, active                                                                                     |
| plans, planPrices                  | code, modules[], limits{}; price per currency, cycle, validFrom, version                                                     |
| subscriptions                      | tenantId, plan, items[{module or addon, qty, priceVersion}], cycle, status, currentPeriod{start, end}, gatewaySubscriptionId |
| usageSnapshots                     | tenantId, date, users, beds, employees, payslips, storageBytes, sms, whatsapp                                                |
| platformInvoices, platformPayments | number, tenantId, lines[], gst, total, status, gateway refs                                                                  |
| coupons, creditNotes               | discount rules; adjustments                                                                                                  |
| featureFlags                       | key, rules (tenantIds, plans, percentage), default                                                                           |
| supportSessions                    | tenantId, agentId, approvedBy, startsAt, endsAt, actions[]                                                                   |
| platformUsers                      | name, e-mail, roles, mfa, lastLoginAt                                                                                        |

### 3.15 Platform APIs

| Method | Path                                  | Who                  | Description                       |
|--------|---------------------------------------|----------------------|-----------------------------------|
| GET    | /public/plans                         | anyone               | Plans and prices for pricing page |
| GET    | /public/subdomains/{name}             | anyone               | Availability check                |
| POST   | /public/signup                        | anyone (OTP)         | Create trial tenant               |
| GET    | /api/v1/subscription                  | Hospital Super Admin | Plan, modules, usage              |
| POST   | /api/v1/subscription/preview          | Hospital Super Admin | Price a change                    |
| POST   | /api/v1/subscription/changes          | Hospital Super Admin | Apply a change                    |
| GET    | /api/v1/subscription/invoices         | Hospital Super Admin | Platform invoices                 |
| POST   | /api/v1/support-sessions/{id}/approve | Hospital Super Admin | Allow support access              |
| CRUD   | /platform/tenants                     | Platform staff       | Manage tenants                    |
| CRUD   | /platform/plans                       | Platform Super Admin | Catalogue and prices              |
| POST   | /platform/tenants/{id}/subscription   | Sales / Finance      | Change on behalf of tenant        |
| GET    | /platform/metrics?from=&to=           | Platform Super Admin | MRR, churn, conversion            |

| Method | Path               | Who               | Description                |
|--------|--------------------|-------------------|----------------------------|
| POST   | /webhooks/razorpay | Razorpay (signed) | Payment and mandate events |
| POST   | /webhooks/stripe   | Stripe (signed)   | Payment events             |

### Signup and module change preview (amounts in paise)

```
POST https://www.example.com/public/signup
{ "contact": { "name": "Dr. Arjun Rao", "email": "arjun@cityhospital.in",
  "mobile": "9876543210", "otpToken": "otp_8f2..." },
  "hospital": { "name": "City Hospital", "city": "Nagpur", "beds": 60 },
  "subdomain": "cityhospital", "plan": "HOSPITAL", "acceptTermsVersion": "2026-08" }
```

```
202 Accepted
{ "tenantId": "670lab...", "status": "PROVISIONING",
  "loginUrl": "https://cityhospital.example.com/welcome?token=...",
  "trialEndsAt": "2026-10-23T23:59:59+05:30" }
```

```
POST /api/v1/subscription/preview
{ "add": [ { "module": "LAB" } ], "remove": [ { "module": "RAD" } ] }
```

```
200 OK
{ "effective": { "add": "IMMEDIATE", "remove": "2026-10-31" },
  "chargeNow": { "subtotal": 166667, "gst": 30000, "total": 196667, "currency": "INR" },
  "nextInvoiceEstimate": { "total": 4366000 },
  "blockedBy": [] }
```

## 3.16 Platform reference code

### apps/api/src/platform/billing/subscription.service.ts

```
/** Applies a subscription change. Additions start now (prorated); removals at renewal. */
export async function changeSubscription(tenantId: string, change: SubscriptionChange) {
  const sub = await Subscription.findOne({ tenantId, status: { $ne: 'CANCELLED' } });
  if (!sub) throw new AppError(404, 'NO_SUBSCRIPTION', 'Subscription not found');

  const target = applyChange(sub.items, change);
  const missing = unmetDependencies(target); // e.g. PAY needs HRM
  if (missing.length) {
    throw new AppError(422, 'DEPENDENCY_MISSING', 'Required modules missing', { missing });
  }
  await assertUsageFits(tenantId, target); // blocks unsafe downgrades

  const now = new Date();
  const ratio = remainingRatio(now, sub.currentPeriod); // e.g. 20 / 30
  const chargeLines = change.add.map((item) => ({
    item, amount: Math.round(priceOf(item, sub.priceBook) * ratio),
  }));

  return withPlatformTransaction(async (session) => {
    sub.items = mergeAdds(sub.items, change.add);
    sub.pendingRemovals.push(...change.remove.map((r) => ({ ...r, at: sub.currentPeriod.end })));
    await sub.save({ session });
    if (chargeLines.length) {
      await createProrationInvoice(tenantId, chargeLines, session); // GST added inside
    }
    await syncTenantModules(tenantId, sub, session); // tenants.modules[] updated
    outbox.add('tenant.modulesChanged', { tenantId }); // cache bust + socket refresh
    return sub;
  });
}
```

### apps/api/src/platform/billing/razorpay.webhook.ts

```
import crypto from 'node:crypto';

/** Razorpay webhook: verify signature, store event once, then process asynchronously. */
export const razorpayWebhook: RequestHandler = async (req, res) => {
  const signature = req.header('x-razorpay-signature') ?? '';
  const expected = crypto.createHmac('sha256', env.RAZORPAY_WEBHOOK_SECRET)
    .update(req.body as Buffer).digest('hex'); // raw body middleware
  const ok = signature.length === expected.length
    && crypto.timingSafeEqual(Buffer.from(signature), Buffer.from(expected));
  if (!ok) return res.status(400).end();
}
```

```

const event = JSON.parse(String(req.body));
const inserted = await WebhookEvent.updateOne(
  { provider: 'razorpay', eventId: req.header('x-razorpay-event-id') },
  { $setOnInsert: { type: event.event, payload: event, receivedAt: new Date() } },
  { upsert: true });
if (inserted.upsertedCount) {
  await queues.billing.add(event.event, { eventId: req.header('x-razorpay-event-id') });
}
res.status(200).end(); // ack fast, process in worker
};

// Worker handlers (excerpt)
billingWorker.on('subscription.charged', markInvoicePaidAndActivate);
billingWorker.on('payment.failed', startDunning); // day 1, 3, 6 retries
billingWorker.on('subscription.halted', moveTenantToPastDue);

```

#### **apps/api/src/platform/metering/snapshot.job.ts**

```

/** Nightly 01:00 IST: one usage snapshot per tenant, used for metering and limits. */
export async function snapshotUsage(tenantId: string, day: Date) {
  return runAs({ tenantId }, async () => {
    const [users, beds, employees, payslips, storageBytes, msgs] = await Promise.all([
      User.countDocuments({ status: 'ACTIVE' }),
      Bed.countDocuments({ status: { $ne: 'INACTIVE' } }),
      Employee.countDocuments({ status: 'ACTIVE' }),
      Payslip.countDocuments({ month: monthOf(day) }),
      storage.bytesForTenant(tenantId), // S3 inventory report
      MessageLog.aggregate(countByChannel(day)),
    ]);
    await UsageSnapshot.updateOne({ tenantId, date: day },
      { $set: { users, beds, employees, payslips, storageBytes, ...msgs } }, { upsert: true });
  });
}

```

## SECTION 04

## Roles, Logins and Approvals

Access is controlled at three levels. The **subscription** decides which modules exist for the hospital. The **role** decides what a user may do inside those modules. The **data scope** decides which records the user sees, for example only their own department. On top of this, sensitive actions go through **maker-checker** approval.

### 4.1 Permission model

- A permission is written as `module:resource:action`, for example `billing:invoice:create` or `pharmacy:stock:adjust`.
- A role is a named set of permissions plus a data scope: **OWN** (records the user created or is assigned to), **DEPARTMENT**, **BRANCH** or **ALL**.
- A user can hold several roles, for example a doctor who is also Head of Department. The effective permission set is the union of all roles.
- System roles ship with the product and cannot be deleted. Hospitals can clone them into custom roles and edit those.
- Permissions that belong to a module that is not subscribed are ignored, even if a role contains them.
- Every permission check happens on the server. The React app only hides buttons for convenience.

### 4.2 System roles

| Role                        | Scope         | Main responsibilities                                                                                    |
|-----------------------------|---------------|----------------------------------------------------------------------------------------------------------|
| Platform Super Admin        | All tenants   | SaaS owner. Tenants, plans, modules, platform invoices. Cannot see clinical data.                        |
| Hospital Super Admin        | Tenant        | Final approver for critical changes, user and role administration, subscription view, hospital settings. |
| Hospital Admin              | Branch        | Day-to-day setup: departments, masters, tariffs, users. Acts as maker for critical changes.              |
| Medical Superintendent      | Branch        | Clinical oversight, discharge approvals, bed escalations, clinical reports.                              |
| Department Head (HOD)       | Department    | Approves department rosters, leave and indents. Views department reports.                                |
| Consultant Doctor           | Own patients  | Consultation, orders, prescriptions, IPD rounds, discharge summary sign-off.                             |
| Resident / Duty Doctor      | Department    | Rounds notes, orders under a consultant, draft discharge summaries.                                      |
| Nursing Superintendent      | Branch        | All wards, nurse rosters, nursing audits.                                                                |
| Ward In-charge (Sister)     | Ward          | Ward census, bed status, nurse assignment, indents, handover sign-off.                                   |
| Staff Nurse                 | Assigned ward | Vitals, medication administration, nursing notes, intake and output.                                     |
| Front Office Executive      | Branch        | Registration, appointments, check-in, admission desk, enquiries.                                         |
| Cashier                     | Own counter   | Bills, receipts, deposits, refunds requests, day-end cash closing.                                       |
| Billing Manager             | Branch        | Approves discounts, refunds and cancellations. IPD final bills.                                          |
| Accountant                  | Entity        | Journals, payables, bank reconciliation, financial reports, GST.                                         |
| Finance Controller          | Entity        | Approves payments and journals above limits, closes periods.                                             |
| Lab Technician              | Lab           | Sample collection, accession, result entry.                                                              |
| Pathologist / Lab In-charge | Lab           | Result validation and report release.                                                                    |
| Radiology Technician        | Radiology     | Study scheduling, acquisition status.                                                                    |
| Radiologist                 | Radiology     | Reporting and sign-off.                                                                                  |

| Role                             | Scope          | Main responsibilities                                                                                      |
|----------------------------------|----------------|------------------------------------------------------------------------------------------------------------|
| Pharmacist                       | Pharmacy store | Dispensing, OPD sales, IPD issues, returns.                                                                |
| Pharmacy In-charge               | Pharmacy       | Purchase, GRN approval, stock adjustment requests, controlled drug register.                               |
| Store Keeper                     | Store          | GRN, issues, transfers, physical stock.                                                                    |
| Purchase Manager                 | Branch         | Vendors, quotations, purchase orders.                                                                      |
| HR Manager                       | Tenant         | Employee master, policies, approvals, exits.                                                               |
| HR Executive                     | Branch         | Onboarding, documents, attendance corrections.                                                             |
| Payroll Officer                  | Entity         | Payroll inputs, payroll run (maker).                                                                       |
| Employee (Self-service)          | Own            | Payslips, leave, attendance, roster, profile. Every staff login has this role.                             |
| Insurance Desk (TPA Coordinator) | Branch         | Pre-authorisations, enhancements, final approvals, claims and settlements with insurers, TPAs and schemes. |
| Auditor                          | Tenant         | Read-only access to finance, stock and audit logs.                                                         |
| MRD Officer / Coder              | Branch         | Record completion, coding, file tracking, record release, birth, death and MLC registers.                  |
| Dietitian                        | Branch         | Diet orders and plans, nutrition assessment, counselling.                                                  |
| Kitchen Supervisor               | Kitchen        | Production sheets, tray labels, delivery, kitchen indents.                                                 |
| Housekeeping Supervisor          | Branch         | Cleaning tasks and checklists, linen and laundry.                                                          |
| Maintenance Technician           | Branch         | Maintenance tickets assigned to them.                                                                      |
| Biomedical Engineer              | Branch         | Equipment register, PM, calibration, breakdowns, AMC.                                                      |
| Quality Manager                  | Tenant         | Feedback, complaints, incidents, indicators, audits, controlled documents.                                 |
| Security Desk                    | Gate           | Visitor and attendant pass check-in and check-out.                                                         |
| Call Centre / CRM Executive      | Branch         | Calls, appointments, leads, camps, campaigns, memberships.                                                 |
| Home Collection Phlebotomist     | Assigned area  | Home sample visits, barcodes, payment collection.                                                          |
| Infection Control Nurse          | Branch         | Infection surveillance, device days, hand hygiene audits.                                                  |

## 4.3 Role to module access matrix

C = create, R = read, U = update, A = approve, blank = no access. This is the default; hospitals can change it in custom roles.

| Role              | CORE | OPD | IPD | NUR | LAB   | RAD   | PHR | INV  | HRM | PAY | FIN |
|-------------------|------|-----|-----|-----|-------|-------|-----|------|-----|-----|-----|
| Hosp. Super Admin | CRUA | R   | R   | R   | R     | R     | R   | RA   | RA  | RA  | RA  |
| Hospital Admin    | CRU  | CRU | CRU | R   | CRU   | CRU   | CRU | CRU  | R   |     | R   |
| Consultant        | R    | CRU | CRU | R   | CR    | CR    | R   |      |     |     |     |
| Staff Nurse       | R    | R   | R   | CRU | R     | R     | CR  | C    |     |     |     |
| Front Office      | CRU  | CRU | CR  |     |       |       |     |      |     |     |     |
| Cashier           | R    | R   | R   |     | R     | R     | R   |      |     |     |     |
| Billing Manager   | RA   | R   | RUA |     | R     | R     | R   |      |     |     | R   |
| Lab Tech / Path.  | R    |     |     |     | CRU/A |       |     | C    |     |     |     |
| Radiologist       | R    |     |     |     |       | CRU/A |     |      |     |     |     |
| Pharmacist        | R    | R   | R   |     |       |       | CRU | C    |     |     |     |
| Store / Purchase  | R    |     |     |     |       |       | R   | CRUA |     |     | R   |

| Role            | CORE | OPD | IPD | NUR | LAB | RAD | PHR | INV | HRM  | PAY | FIN |
|-----------------|------|-----|-----|-----|-----|-----|-----|-----|------|-----|-----|
| HR Manager      | R    |     |     |     |     |     |     |     | CRUA | R   |     |
| Payroll Officer | R    |     |     |     |     |     |     |     | R    | CRU | R   |
| Accountant      | R    |     | R   |     |     |     | R   | R   |      | R   | CRU |
| Employee        | R    |     |     |     |     |     |     |     | own  | own |     |

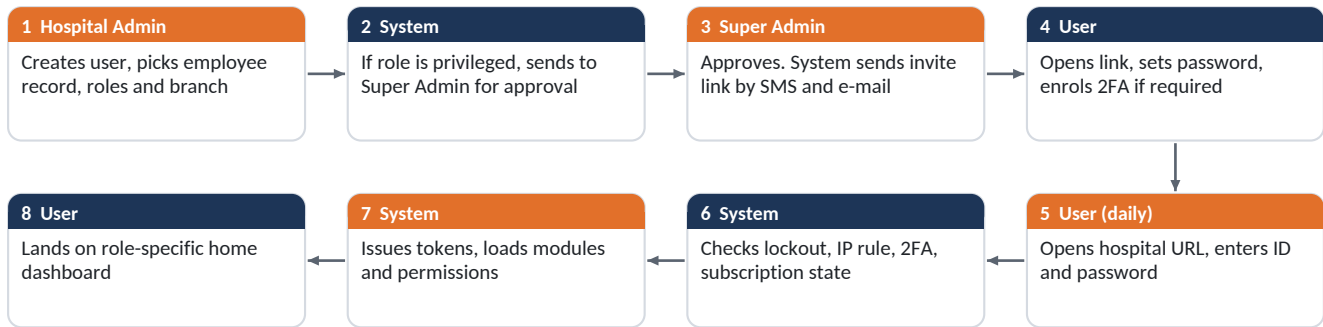
Support modules. Every staff login can also raise maintenance tickets (FAC) and report incidents (QLT).

| Role                                    | MRD         | DIET            | FAC | QLT  | CRM |
|-----------------------------------------|-------------|-----------------|-----|------|-----|
| Hospital Super Admin                    | RA          | R               | RA  | RA   | RA  |
| Hospital Admin                          | CRU         | CRU             | CRU | R    | CRU |
| Medical Superintendent                  | RA          | R               | R   | RA   | R   |
| Consultant / Resident                   | own records | C (diet orders) | C   | C    | R   |
| Staff Nurse                             | R           | CR              | C   | C    |     |
| MRD Officer / Coder                     | CRU         |                 | C   | C    |     |
| Dietitian / Kitchen Supervisor          |             | CRU             | C   | C    |     |
| Housekeeping / Maintenance / Biomedical |             |                 | CRU | C    |     |
| Quality Manager / Infection Control     | R           | R               | R   | CRUA | R   |
| Call Centre / CRM Executive             |             |                 | C   | C    | CRU |
| Front Office                            | R           |                 | C   | C    | CR  |
| Security Desk                           |             |                 | C   | C    |     |

## 4.4 Login and session security

| Control                | Rule                                                                                                                                           |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Login identifier       | Username or mobile number or e-mail, unique per tenant. Tenant is resolved from the sub-domain (e.g. citycare.example.com)                     |
| Password policy        | Minimum 10 characters, at least 3 of 4 character classes, not one of the last 5 passwords, stored with Argon2id                                |
| Two-factor             | OTP by SMS or authenticator app. Mandatory for Super Admin, Admin, Finance, HR and Payroll roles. Optional for others                          |
| Lockout                | 5 failed attempts locks the account for 15 minutes. Admin can unlock                                                                           |
| Session                | Access token 15 minutes, refresh token 12 hours for counters and 7 days for 'remember device', rotated on every use, stored as httpOnly cookie |
| Idle timeout           | 15 minutes for clinical and billing screens, configurable per hospital between 5 and 60 minutes                                                |
| Concurrent sessions    | Configurable per role. Default: 2 devices for doctors, 1 for cashiers                                                                          |
| IP / time restrictions | Optional. Restrict finance and payroll roles to hospital network IPs or working hours                                                          |
| Shared ward terminals  | Quick switch-user with 4-digit PIN after full login on the same shift, so nurses do not share accounts                                         |
| Password reset         | OTP to registered mobile or e-mail. Admin-initiated reset forces change at next login                                                          |

## User flow: first login and daily login

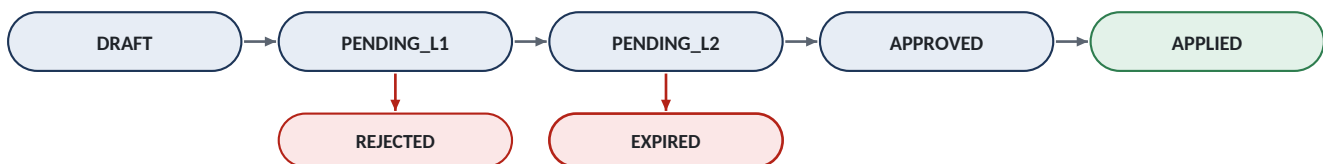


## 4.5 Maker-checker approvals

Maker-checker means one person proposes a change and a different person approves it before it takes effect. The proposal is stored as an **approval request**. The real record does not change until approval.

- The maker and checker must be different users. A user holding both roles still cannot approve their own request.
- Each rule defines the checker role, an optional second level, a threshold and an expiry. For example, discounts up to 10% need the Billing Manager, above 10% also need the Hospital Super Admin.
- Checkers see a single **Approvals inbox** across all modules, with a badge count, and get push notifications in the app plus optional SMS.
- The checker sees a side-by-side view of old and new values before deciding. A rejection needs a reason.
- Requests expire after a configured time (default 48 hours) and must be raised again.
- Every request, decision and comment is stored permanently in the audit log.

### Approval request lifecycle

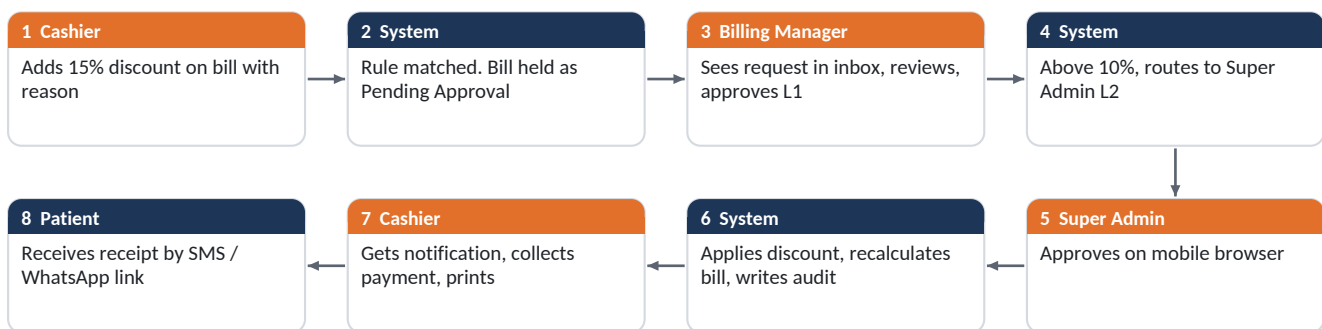


## Default maker-checker rules

| Action                       | Maker                     | Checker L1             | Checker L2 (if)                     |
|------------------------------|---------------------------|------------------------|-------------------------------------|
| Bill discount                | Cashier                   | Billing Manager        | Super Admin (> 10% or > INR 10,000) |
| Bill cancellation / refund   | Cashier                   | Billing Manager        | Finance Controller (> INR 25,000)   |
| Tariff / price change        | Hospital Admin            | Super Admin            | -                                   |
| Discharge with pending dues  | Billing                   | Medical Superintendent | -                                   |
| Lab result release           | Lab Technician            | Pathologist            | -                                   |
| Amend released lab report    | Pathologist               | Lab In-charge          | -                                   |
| Radiology report sign-off    | Resident / Tech           | Radiologist            | -                                   |
| Stock adjustment / write-off | Store Keeper / Pharmacist | Pharmacy In-charge     | Finance (> INR 5,000)               |
| Purchase order               | Purchase Manager          | HOD / Admin            | Super Admin (> INR 1,00,000)        |
| GRN with price variance      | Store Keeper              | Purchase Manager       | -                                   |
| Employee salary change       | HR Executive              | HR Manager             | Super Admin                         |

| Action                     | Maker                     | Checker L1             | Checker L2 (if)        |
|----------------------------|---------------------------|------------------------|------------------------|
| Payroll run release        | Payroll Officer           | HR Manager             | Finance Controller     |
| Manual journal entry       | Accountant                | Finance Controller     | -                      |
| Patient record merge       | Front Office              | Hospital Admin         | -                      |
| Department create / close  | Hospital Admin            | Super Admin            | -                      |
| User with privileged role  | Hospital Admin            | Super Admin            | -                      |
| Role permission change     | Hospital Admin            | Super Admin            | -                      |
| Medical record release     | MRD Officer               | Medical Superintendent | -                      |
| Linen / asset condemnation | Housekeeping / Biomedical | Hospital Admin         | -                      |
| Expense claim / petty cash | Employee / Custodian      | Manager                | Finance (> INR 10,000) |
| Incident closure           | Investigator              | Quality Manager        | -                      |

### User flow: discount approval



## 4.6 Super Admin approvals

The Hospital Super Admin is the owner-level account of a hospital. It is the final checker for anything that changes money rules, access rights or the structure of the hospital. A hospital can have up to three Super Admins so approvals do not stall when one is away.

- Creating, closing or merging departments and branches.
- Creating users with privileged roles, and any change to role permissions.
- Changes to tariffs, price lists, tax settings and discount policy.
- Reopening a closed financial period or a locked payroll month.
- High-value discounts, refunds, write-offs and purchase orders above thresholds.
- Approving data export requests and support-agent access to the tenant.
- Viewing subscription, usage and platform invoices. Requesting module changes.

**Safeguard.** Super Admin accounts must use two-factor authentication and cannot be deleted, only deactivated by another Super Admin. The last active Super Admin cannot be deactivated.

## 4.7 Audit trail

- Every create, update, delete, print, export, login and approval writes an audit event: who, when, from which IP and device, which record, and the before and after values of changed fields.
- Viewing a patient's clinical record is also logged (access log), so the hospital can answer 'who opened this record'.
- Audit events are append-only. No role, including Super Admin, can edit or delete them. They are kept for at least 7 years.
- Screens: audit search by user, patient, record or date; daily exception report of out-of-hours access, failed logins and bulk exports.

## SECTION 05

# Core Platform: Setup, Departments and Patients

## CORE Core Platform

|                      |                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Mandatory for every tenant                                                                                                                                                      |
| <b>Depends on</b>    | Nothing                                                                                                                                                                         |
| <b>Primary users</b> | Super Admin, Hospital Admin, Front Office, all staff                                                                                                                            |
| <b>Purpose</b>       | Holds everything every other module needs: hospital structure, departments, masters, patient identity, staff identity, billing engine, notifications, documents and dashboards. |

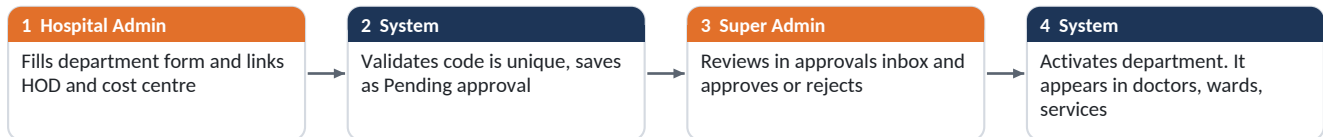
## 5.1 Hospital and branch setup

| Setting                 | Details                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Legal entity</b>     | Name, registration number, GSTIN, PAN, address, logo, letterhead, authorised signatory. One tenant can have several entities                                  |
| <b>Branches</b>         | Each branch has its own address, GSTIN, counters, stores, wards, numbering series and print headers                                                           |
| <b>Numbering series</b> | Prefix and format for UHID, visit, admission, bill, receipt, PO, GRN, employee ID. Reset yearly, monthly or never                                             |
| <b>Financial year</b>   | Start month, open and closed periods                                                                                                                          |
| <b>Working calendar</b> | Working days, OPD hours, public holidays per branch                                                                                                           |
| <b>Print templates</b>  | A4, A5 and 80 mm thermal layouts for bills, receipts, prescriptions, lab reports, discharge summary, payslips. Editable header, footer, signature and QR code |
| <b>Communication</b>    | SMS sender ID, e-mail domain, WhatsApp Business number, message templates in English and local languages                                                      |
| <b>Localisation</b>     | Currency, date format, time zone, languages (English, Hindi; others added as translation files)                                                               |

## 5.2 Department registration

Departments are the backbone of reporting, rostering and billing. Every doctor, nurse, service, ward, store and cost belongs to a department.

| Field                     | Description                                                                        |
|---------------------------|------------------------------------------------------------------------------------|
| <b>Code and name</b>      | Unique short code (e.g. CARD) and display name (Cardiology)                        |
| <b>Type</b>               | Clinical, Diagnostic, Support (e.g. CSSD, housekeeping) or Administrative          |
| <b>Parent</b>             | Optional parent for sub-departments (e.g. Paediatrics > NICU)                      |
| <b>Head of department</b> | Employee who approves rosters, leave and indents                                   |
| <b>Location</b>           | Branch, building, floor, room numbers                                              |
| <b>Services offered</b>   | OPD, IPD, procedures, diagnostics flags; drives which screens show this department |
| <b>Cost centre</b>        | Accounting cost centre for revenue and expense reports                             |
| <b>OPD timings</b>        | Days and hours when this department runs OPD                                       |
| <b>Status</b>             | Draft, Pending approval, Active, Inactive                                          |

**User flow: register a department**

- A department with active staff, beds or open bills cannot be deactivated. The system lists what must be moved first.
- Departments are never deleted, only deactivated, so history stays intact.

## 5.3 Master data

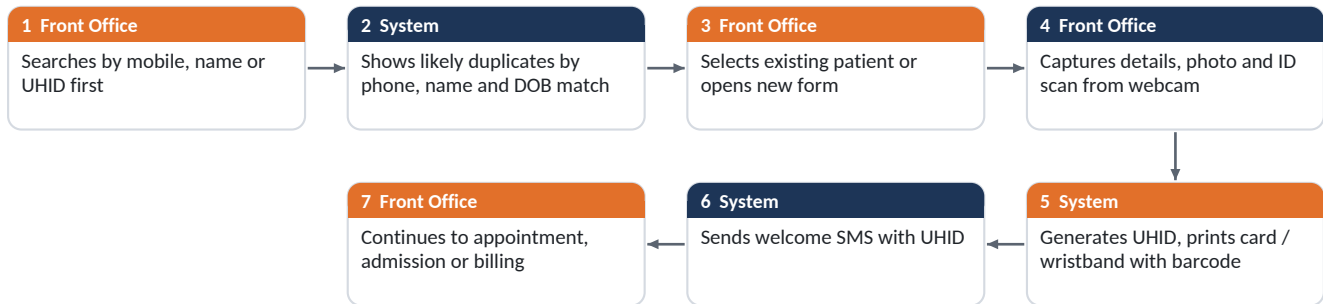
| Master                  | Used by                     | Notes                                                                                                                           |
|-------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Service / tariff master | Billing, OPD, IPD, LAB, RAD | Every chargeable item: consultation, procedure, bed day, test, package. Rate per price list, tax code, department, revenue head |
| Price lists             | Billing                     | General, staff, senior citizen, corporate. Insurance price lists added in Phase 2                                               |
| Packages                | Billing, IPD                | Bundled price for a set of services and days, with rules for exclusions and over-limit charges                                  |
| Tax codes               | Billing, PHR, INV           | GST rates, HSN / SAC codes, exempt healthcare services                                                                          |
| Payment modes           | Billing                     | Cash, card, UPI, cheque, bank transfer, wallet, advance adjustment                                                              |
| Diagnosis codes         | OPD, IPD                    | ICD-10 preloaded, searchable, with hospital favourites                                                                          |
| Referral sources        | Registration                | Doctors, camps, websites, walk-in, for referral reports                                                                         |
| Designations, grades    | HRM                         | Used in employee master and salary structure                                                                                    |
| Units of measure        | PHR, INV                    | Strip, tablet, bottle, box, with conversions                                                                                    |

**Go-live accelerator.** Bulk import from Excel is available for every master, with a validation preview before saving. This is how a hospital migrates from its old software in days, not weeks.

## 5.4 Patient registration

Each patient gets one permanent **UHID** (Unique Health ID) valid across all branches. Every visit, admission, bill, sample and prescription links to it.

| Group             | Fields                                                                                                   |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Identity          | Title, first, middle, last name, gender, date of birth or age, photo, blood group, marital status        |
| Contact           | Mobile (verified by OTP optional), alternate number, e-mail, address with PIN code lookup                |
| Documents         | ID type and number (Aadhaar, PAN, passport, voter ID), ABHA number optional, scanned copies stored on S3 |
| Emergency contact | Name, relation, phone                                                                                    |
| Clinical flags    | Allergies, chronic conditions, VIP flag, medico-legal case flag                                          |
| Commercial        | Patient category (general, staff, senior, corporate), default price list, corporate employer             |
| Source            | Referral source and referring doctor                                                                     |

**User flow: register a new patient****Business rules**

- Quick registration needs only name, gender, age and mobile, so a queue never stops. Missing fields are flagged until completed.
- Duplicate detection uses a score: same mobile and similar name, or same ID number. Score above threshold shows a warning, never a silent merge.
- Merging two UHIDs is a maker-checker action. All visits, bills and reports move to the surviving UHID and the merged one redirects to it.
- Age is stored as an estimated date of birth so it stays correct over time.
- Patient photos and documents are stored in the private S3 bucket and served only through short-lived signed links.

**5.5 Front office: enquiries and visitor passes**

| Feature                 | Details                                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Enquiry desk</b>     | Log phone and walk-in enquiries: doctor availability, tariffs, packages, bed availability. Convert an enquiry to an appointment in one click; enquiry-to-visit conversion report |
| <b>Attendant passes</b> | Issued at admission. Number allowed per bed category, photo and QR code, printed or sent by SMS. Revoked automatically at discharge                                              |
| <b>Visitor entry</b>    | Security desk scans the QR pass or records a visitor with ID, patient, ward and time in and out. Visiting-hours rules and ICU restrictions enforced                              |
| <b>Help desk</b>        | Patient information desk can look up a patient's ward and bed, but never clinical details; VIP and medico-legal patients hidden                                                  |
| <b>Counter tokens</b>   | One token system for registration, billing, pharmacy, sample collection and radiology counters, with TV display and voice call-out; average wait per counter                     |
| <b>Self check-in</b>    | Patients with a booking scan a QR code at a kiosk or on their phone to check in and get a token                                                                                  |

**5.6 ABDM integration (Ayushman Bharat Digital Mission)**

- Create or verify a patient's ABHA number with Aadhaar or mobile OTP at registration, and link it to the UHID.
- Scan and Share: patients scan the hospital's QR code with an ABHA app to share their profile, which creates a registration and a counter token instantly.
- Link prescriptions, lab reports and discharge summaries to the patient's ABHA so they appear in the patient's health locker, with consent.
- Records are mapped to FHIR resources as ABDM requires. The hospital registers in the Health Facility Registry and doctors in the Health Professional Registry.

## 5.7 Patient portal

A mobile-friendly web portal at {hospital}.example.com/my, built in React and hosted on S3 like the main app. Patients log in with an OTP on their registered mobile; one login can manage linked family members.

- Book, reschedule and cancel appointments; join tele-consultations.
- Download lab and radiology reports, prescriptions, discharge summaries and bills.
- Pay OPD fees, IPD deposits and outstanding dues online through the payment gateway.
- View upcoming follow-ups and medicine reminders; update address and e-mail.
- Give feedback and raise complaints, which go to the Quality module.
- Every portal download is written to the patient's access log.

## 5.8 Staff directory

Every person who logs in or appears on a document is an employee record. CORE keeps a light staff directory. The HRM module extends the same record with HR and payroll data.

- Search staff by name, department, designation, skill or phone. Shows photo, extension, current shift and on-duty status (when HRM is active).
- Doctor profile: qualifications, registration council and number, specialties, languages, signature image for reports, consultation fees per visit type.
- Visiting consultants and external doctors are supported with payout rules.
- Directory export and printable department-wise contact lists.

## 5.9 Notifications and documents

| Capability            | Description                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Channels</b>       | In-app bell, SMS, e-mail, WhatsApp Business API. Provider adapters so the hospital can change SMS vendor without code changes |
| <b>Templates</b>      | Per event and language, with variables such as {patientName}, {doctor}, {time}. DLT template IDs for India                    |
| <b>Patient events</b> | Registration, appointment booked, reminder, report ready with secure link, bill receipt, discharge, follow-up due             |
| <b>Staff events</b>   | Approval pending, roster published, leave approved, payslip ready, low stock, near expiry                                     |
| <b>Document store</b> | All generated PDFs and uploads go to S3 with tenant prefix. Patients get time-limited links, never public URLs                |
| <b>Delivery log</b>   | Status per message (queued, sent, delivered, failed) with retry                                                               |
| <b>Circulars</b>      | Hospital-wide or department circulars and notices with read acknowledgement tracking                                          |

## 5.10 Dashboards

| Role                       | Home dashboard widgets                                                                                |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Super Admin / Admin</b> | Today's OPD count, admissions, discharges, occupancy, collection, pending approvals, low stock alerts |
| <b>Doctor</b>              | My queue today, my in-patients, pending results, follow-ups due                                       |
| <b>Nurse</b>               | Ward census, meds due in next hour, vitals due, pending indents                                       |
| <b>Front Office</b>        | Appointments by doctor, waiting time, bed availability                                                |
| <b>Cashier</b>             | My collection by mode, pending bills, cash in hand                                                    |

| Role                    | Home dashboard widgets                                               |
|-------------------------|----------------------------------------------------------------------|
| Lab / Radiology         | Pending samples, in-process, awaiting validation, TAT breaches       |
| Pharmacy / Store        | Prescriptions waiting, near expiry, reorder, pending GRN             |
| HR / Payroll            | Headcount, absent today, leave pending, payroll status               |
| MRD                     | Records pending completion, deficiencies by doctor, release requests |
| Kitchen                 | Meal counts by diet for next meal, NBM patients, missed deliveries   |
| Housekeeping / Facility | Beds awaiting cleaning, open tickets, PM due, AMC renewals           |
| Quality                 | Open incidents and complaints, indicator trends, audits due          |
| CRM                     | Leads due for follow-up, campaign results, camp bookings, renewals   |

## SECTION 06

## Patient Operations: OPD, IPD and Beds

### OPD OPD and Appointments

|                      |                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                          |
| <b>Depends on</b>    | CORE                                                                                                                   |
| <b>Primary users</b> | Front Office, Doctors, Nurses (triage), Cashier, Patients (via SMS)                                                    |
| <b>Purpose</b>       | Runs the outpatient day: doctor schedules, booking, walk-ins, queue, vitals, consultation, prescription and follow-up. |

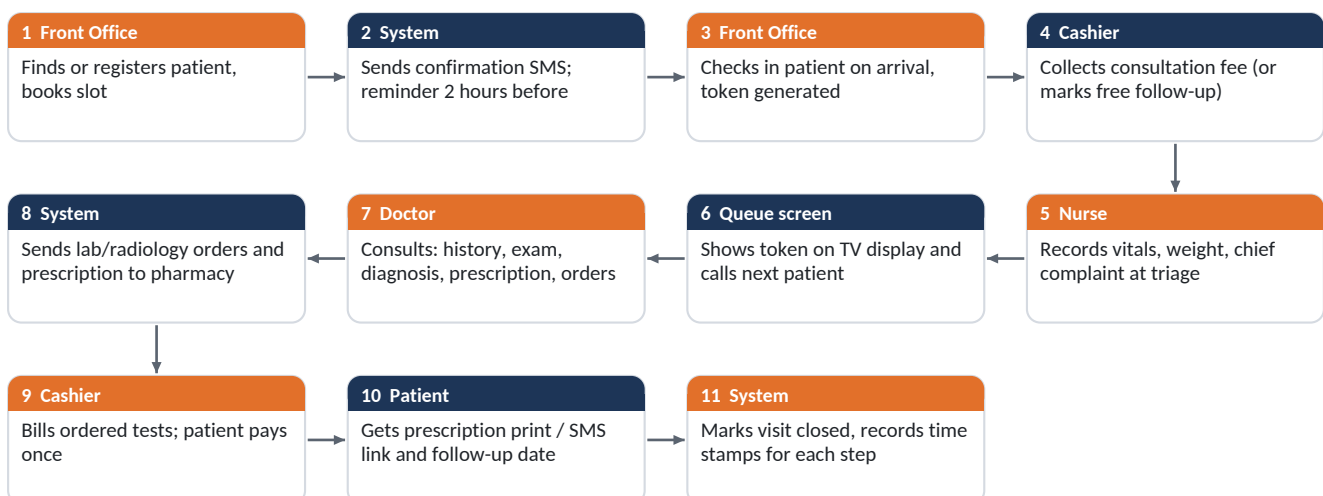
### 6.1 Doctor schedules and slots

- Schedule template per doctor: days, sessions (e.g. 10:00-13:00, 17:00-20:00), branch, room, slot length (e.g. 10 minutes), max patients per session.
- Visit types with own fee and slot length: new, follow-up (free within N days, configurable), review, tele-consultation link.
- Leave and blocks: doctor leave (from HRM if active) or ad-hoc blocks automatically remove slots and trigger reschedule messages to booked patients.
- Overbooking allowed up to a per-doctor limit. Emergency slots can be reserved.

### 6.2 Booking channels

| Channel                    | How it works                                                                                            |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Front desk</b>          | Pick department or doctor, see slot grid with colours, book in 3 clicks                                 |
| <b>Phone</b>               | Same screen as front desk, marked as phone booking, unpaid until arrival                                |
| <b>Walk-in</b>             | Token issued immediately in the doctor's queue                                                          |
| <b>Online booking page</b> | Public page per hospital (React, on S3) with OTP verification. Optional pre-payment via payment gateway |
| <b>Follow-up</b>           | Doctor sets follow-up date in prescription; system books or reminds                                     |

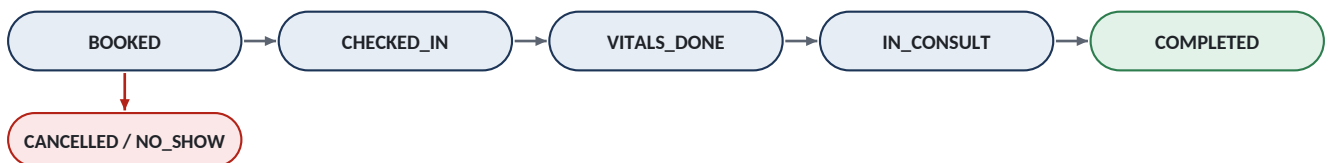
#### User flow: OPD visit end to end



## 6.3 Consultation screen (doctor)

- Left panel: patient banner with photo, age, allergies, VIP or medico-legal flags, past visits timeline, past reports and prescriptions.
- Centre: chief complaints, history, examination, vitals (from triage), provisional and final diagnosis with ICD-10 search, clinical notes.
- Orders: lab tests, radiology, procedures from service master, with favourites and order sets (e.g. 'Fever panel').
- E-prescription: drug search from pharmacy master with generic name, dose, frequency, duration, route, instructions in local language. Allergy and duplicate drug warnings. Templates per doctor.
- Advice, diet, follow-up date, referral to another doctor or admission request.
- Print or send prescription with the doctor's digital signature and QR code.

### Queue states



### Business rules

- Follow-up is free if within the configured window and same doctor; system applies it automatically.
- A doctor cannot close a visit without at least one diagnosis or a 'no diagnosis' reason, so morbidity reports stay accurate.
- No-show is set automatically at session end for un-arrived bookings.
- Time stamps (booked, arrived, vitals, consult start, consult end, billed) feed the Phase 2 'patient time spent in OPD' analytics.

## 6.4 Specialty templates and treatment plans

| Specialty                  | Template content                                                                                                                   |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Dental                     | Tooth chart (FDI numbering) with conditions per tooth, treatment plan per tooth, procedures done, lab work for crowns and dentures |
| Ophthalmology              | Visual acuity, refraction, IOP, slit lamp and fundus findings, spectacle prescription print                                        |
| Obstetrics and gynaecology | LMP and EDD calculator, antenatal card with visits, weight, BP, fundal height, scans; gravida and para                             |
| Paediatrics                | Growth charts (weight, height, head circumference against WHO curves), immunisation schedule with due dates and reminders          |
| Dermatology                | Body map to mark lesions, clinical photos with before and after comparison                                                         |
| Physiotherapy              | Assessment, goals, session-wise exercises and progress                                                                             |
| General                    | Hospitals can build their own templates with a form designer                                                                       |

- **Treatment plans.** Multi-session treatments such as dental work, physiotherapy, dressings and laser are planned with sessions, cost estimate and patient consent. Each session is tracked and billed per session or as a package.
- **Resource scheduling.** Procedure rooms, equipment (e.g. laser) and therapists can be booked like doctors, so a session needs the doctor, room and machine free.
- **Medical certificates.** Fitness, sick leave, medical and disability certificate templates filled from the consultation and signed digitally.

- **Drug interaction checks.** Optional licensed drug database adds drug-drug and drug-allergy interaction warnings and dose checks to e-prescriptions.

## 6.5 Health check-up packages

- Package master: tests, radiology, consultations and the order of stations, e.g. 'Executive Health Check' or 'Pre-employment'. Gender and age variants.
- Booking by date with daily capacity. Corporate bulk bookings by uploading an employee list, billed to the company.
- Check-in creates every order and a route sheet. A tracker shows which stations each person has finished, so the coordinator can move people along.
- When all results are in, a physician reviews and writes the summary. One consolidated report is released to the patient and, with consent, to the employer.

## 6.6 Tele-consultation

- Tele-consult is a visit type with its own fee and slots. Payment is taken online before the slot.
- A secure video link from an integrated video provider is sent by SMS and shown in the patient portal. The doctor uses the same consultation screen and e-prescription.
- Consent for tele-consultation is captured as required by the Telemedicine Practice Guidelines.

### Reports

- Daily OPD register
- Doctor-wise and department-wise footfall
- New vs follow-up
- Average waiting and consultation time
- No-show rate
- Diagnosis (morbidity) report
- Referral source report

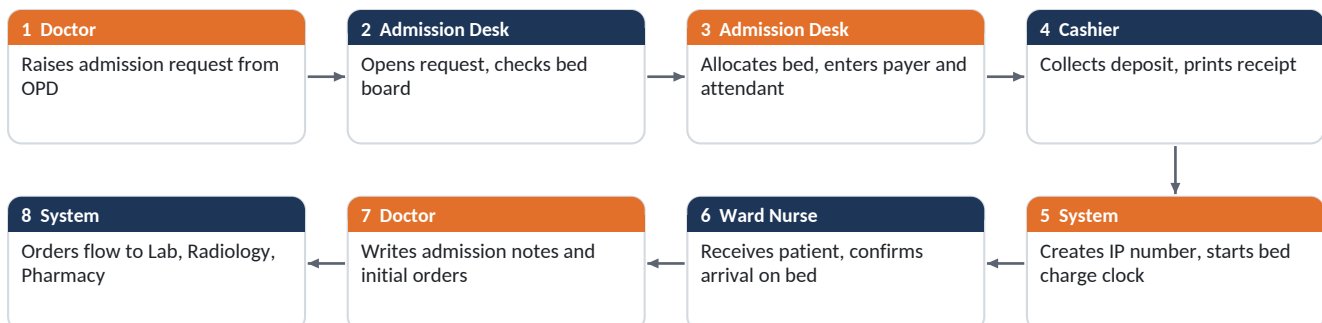
## IPD IPD: Admission, Transfer and Discharge

|                      |                                                                                                                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module (includes Bed Management)                                                                                                                                                               |
| <b>Depends on</b>    | CORE                                                                                                                                                                                                  |
| <b>Primary users</b> | Admission desk, Doctors, Ward In-charge, Billing, Medical Superintendent                                                                                                                              |
| <b>Purpose</b>       | Manages the in-patient stay from admission request to final bill and discharge, including bed allocation, a live running bill, cashless insurance and government schemes, day care and IPD analytics. |

## 6.7 Admission

| Step                     | Details                                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Admission request</b> | Raised by doctor from OPD consultation or directly at the admission desk. Captures reason, provisional diagnosis, expected stay, preferred bed category |
| <b>Bed allocation</b>    | Desk sees live bed board filtered by ward and category. Selected bed is locked for 15 minutes during admission                                          |
| <b>Admission details</b> | Admitting and attending consultant, department, admission type (planned, from OPD), payer (self, corporate), attendant details, medico-legal flag       |
| <b>Deposit</b>           | Suggested deposit based on bed category and package; receipt printed                                                                                    |
| <b>Consent</b>           | Consent forms printed for signature and scanned back, or captured on tablet with signature pad                                                          |
| <b>Wristband</b>         | Printed with name, UHID, IP number, barcode, allergy colour code                                                                                        |
| <b>Notifications</b>     | Ward In-charge and attending doctor notified; nursing station shows the new patient                                                                     |

### User flow: admission

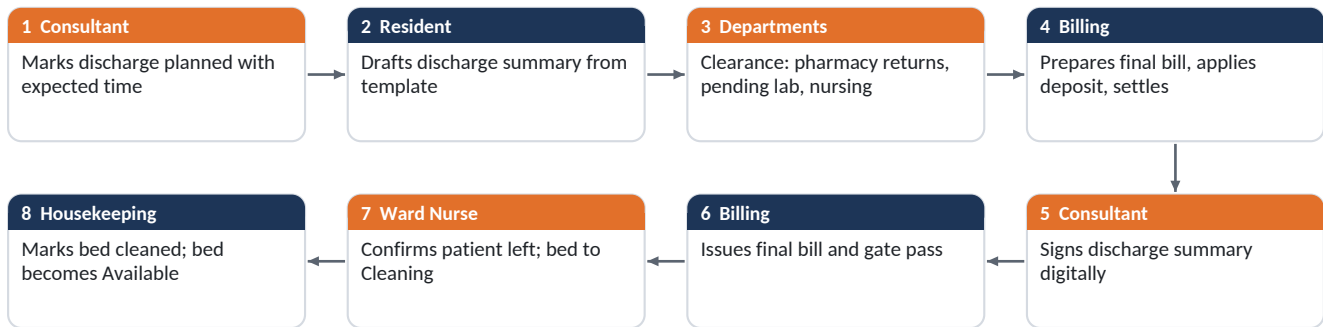


## 6.8 During the stay

- Daily doctor rounds with progress notes, orders and visit charges posted automatically to the running bill.
- Transfers: bed to bed, ward to ward or department to department. Each transfer records time so bed charges switch category at the right moment.
- Running bill visible to billing and, on request, to the patient's attendant. Alerts when bill crosses deposit by a set percentage.
- Additional deposits at any time; low-deposit SMS to attendant.
- Package admissions track inclusions and post only excluded services.
- Medication reconciliation at admission (home medicines recorded and continued, changed or stopped) and at discharge, as NABH requires.

## 6.9 Discharge

### User flow: discharge



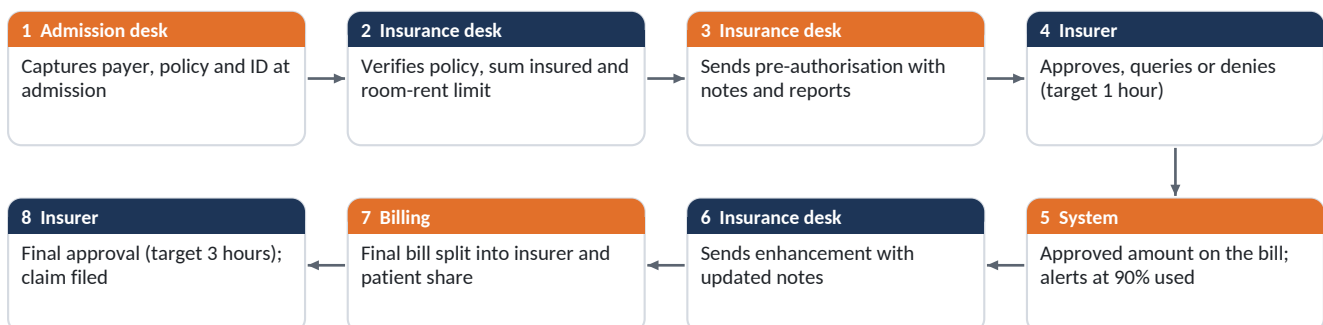
| Discharge type     | Rule                                                                                         |
|--------------------|----------------------------------------------------------------------------------------------|
| Normal             | All clearances and full settlement required                                                  |
| With dues (credit) | Maker-checker approval by Medical Superintendent; dues tracked in patient outstanding report |
| LAMA / DAMA        | Left against medical advice; signed form uploaded, mandatory reason                          |
| Transfer out       | Referral hospital captured, summary printed                                                  |
| Death              | Time and cause recorded, death summary, medico-legal steps if flagged                        |

**Discharge summary.** The discharge summary includes diagnosis, procedures, course in hospital, investigations, condition at discharge, medicines, advice and follow-up. Medicines are pulled from the active medication chart so nothing is retyped.

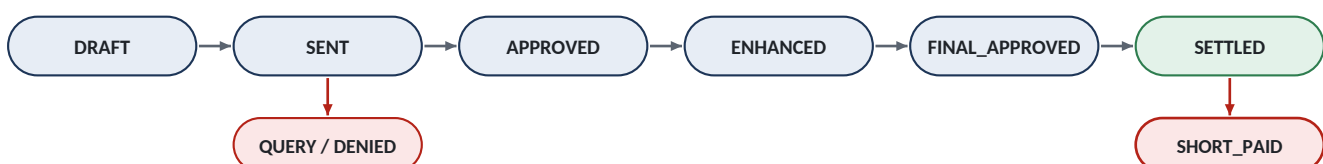
## 6.10 Insurance, TPA and government schemes

The insurance desk gets cashless approvals on time and turns them into paid claims. It works with insurers directly, through third-party administrators (TPAs), with corporates and with government schemes. Treatment never waits for an approval: emergencies are admitted first.

### User flow: cashless admission



### Pre-authorisation lifecycle



| Rule                      | Detail                                                                                                                                                                                     |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Payers</b>             | Cash, corporate, insurer (direct), TPA, PM-JAY, CGHS, ECHS. Each payer has a rate card and its own documents checklist.                                                                    |
| <b>Time limits</b>        | IRDAI Master Circular (May 2024): insurer decides a cashless request within 1 hour and final discharge approval within 3 hours. The desk sees both timers and escalates when they run out. |
| <b>Room-rent limit</b>    | A room above the policy limit shows the proportionate deduction warning and needs signed upgrade consent before transfer.                                                                  |
| <b>Enhancement</b>        | Suggested automatically at 90% of the approved amount.                                                                                                                                     |
| <b>Split bill</b>         | Every line is payable by insurer, patient or corporate; non-payable items and co-pay go to the patient share.                                                                              |
| <b>Government schemes</b> | Beneficiary check and package pre-authorisation are done on the scheme portal; the number and package are recorded here; scheme rates apply and the patient pays nothing for the package.  |
| <b>Claims</b>             | Filed with the documents checklist within the payer's window; payments are matched by UTR and short payments booked as deductions with reason.                                             |
| <b>Denial</b>             | Patient informed with reason; converted to cash or corporate; papers given for a reimbursement claim.                                                                                      |

## 6.11 Packages and day care

- Packages fix the price of a stay or procedure (for example normal delivery, knee arthroscopy, cataract). Items outside the package are billed separately and shown to the attendant every day; package variance is reported.
- Day care uses day-care beds for same-day admissions such as dialysis, chemotherapy and cataract: short admission, procedure, observation and same-day discharge on a day-care package, with the next session booked.

## 6.12 OPD and IPD analytics

- OPD: visits, booked versus walk-in, median wait per stage (arrival, check-in, triage, doctor, billing, pharmacy), no-show rate, doctor utilisation and late starts.
- IPD: occupancy by ward from the midnight census, average length of stay, bed turnover rate and interval, discharge and bed turnaround, revenue per occupied bed day, deaths, LAMA and readmissions within 30 days.
- Formulas, targets and the full module flows are in docs/modules/OPD.md and docs/modules/IPD.md.

## 6.13 Real-time bed management

The bed board is a live screen that every admission desk, ward and administrator sees at the same time. Changes appear within a second on every screen, using WebSockets.

| Bed status    | Colour    | Meaning                                             |
|---------------|-----------|-----------------------------------------------------|
| AVAILABLE     | Green     | Clean and free                                      |
| RESERVED      | Blue      | Held for a planned admission until a set time       |
| OCCUPIED      | Red       | Patient on bed; shows name, consultant, days stayed |
| DISCHARGE_DUE | Orange    | Discharge planned today; helps plan admissions      |
| CLEANING      | Yellow    | Patient left; housekeeping pending                  |
| MAINTENANCE   | Grey      | Out of service                                      |
| BLOCKED       | Dark grey | Blocked by admin (e.g. isolation)                   |

- Structure: Branch > Building > Floor > Ward > Room > Bed. Each bed has a category (General, Semi-private, Private, Deluxe, ICU, NICU, HDU) that drives the daily rate.
- Bed board views: floor map, ward grid, list; filters by category, gender, status.

- Occupancy summary by ward and category, with expected discharges today.
- Bed charges: charge per calendar day or 24-hour cycle (configurable), with half-day rule for short stays. ICU and nursing charges follow the bed category.
- Attendant bed and extra bed can be billed separately.
- Allocation uses an atomic database update, so two desks can never give the same bed to two patients.

### IPD reports

- Admission and discharge register
- Bed occupancy and average length of stay
- Census at midnight
- Deposit vs bill outstanding
- Doctor-wise admissions
- Discharge TAT (decision to exit)

## SECTION 07

# Clinical Workflows: Nursing, Doctors and Rosters

## NUR Nursing Station

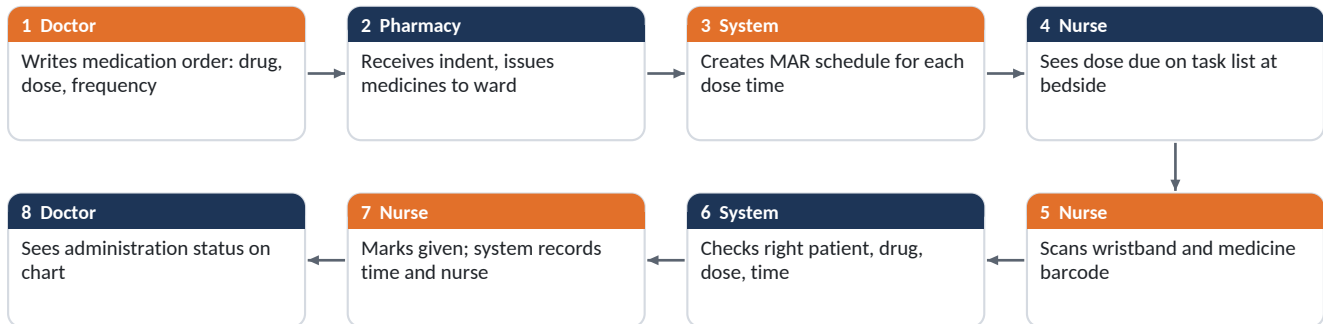
|                      |                                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                             |
| <b>Depends on</b>    | IPD                                                                                                                       |
| <b>Primary users</b> | Staff Nurses, Ward In-charge, Nursing Superintendent, Doctors (view)                                                      |
| <b>Purpose</b>       | Gives each ward a digital nursing station: census, tasks, vitals, medication administration, notes, handover and indents. |

## 7.1 Nursing station screen

- Ward census: every bed with patient name, age, consultant, diagnosis, days in, allergy and fall-risk icons, pending task count.
- Task list for the shift, generated from doctor orders: medicines due, vitals due, samples to collect, dressings, procedures. Overdue tasks turn red.
- Patient chart tabs: vitals graph, medication administration record (MAR), intake and output, nursing notes, assessments, orders, results.
- Works on tablets at the bedside, with barcode scan of the wristband to open the right patient.

## 7.2 Features

| Feature                                | Details                                                                                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vitals charting</b>                 | Temperature, pulse, BP, respiration, SpO2, pain score, GCS, blood sugar. Frequency set by order. Out-of-range values alert the duty doctor. Early warning score (NEWS2) calculated |
| <b>Medication administration (MAR)</b> | Each dose from the doctor's order shows as due. Nurse scans wristband and marks given, held or refused with reason. Late doses flagged                                             |
| <b>Nursing notes</b>                   | Free text plus templates (SOAP, shift note), time-stamped, signed. Cannot be edited after 24 hours, only addended                                                                  |
| <b>Assessments</b>                     | Admission assessment, fall risk (Morse), pressure ulcer risk (Braden), pain                                                                                                        |
| <b>Intake and output</b>               | Oral, IV, feeds vs urine, drains, vomit. Shift and 24-hour totals                                                                                                                  |
| <b>Care plans</b>                      | Nursing diagnosis, goals, interventions, evaluation                                                                                                                                |
| <b>ICU charting</b>                    | Hourly ICU flowsheet: vitals, ventilator settings, infusions and titrations, GCS, lines and drains with device days, fluid balance; severity scores (SOFA, APACHE II)              |
| <b>Handover</b>                        | Shift handover summary generated per patient (ISBAR format), acknowledged by incoming nurse                                                                                        |
| <b>Indents</b>                         | Pharmacy indent for patient medicines and ward stock indent to store, with status tracking                                                                                         |
| <b>Consumables charging</b>            | Consumables used on patient posted to running bill                                                                                                                                 |

**User flow: medication order to administration**

## 7.3 Doctor profiles and scheduling

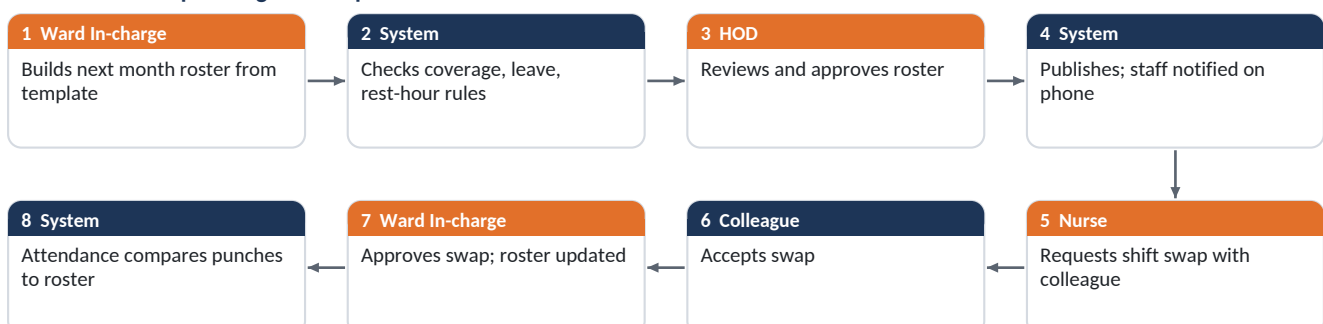
Doctor profiles are kept in CORE. Scheduling for OPD is covered in Section 6. This section covers the clinical side of a doctor's day.

- Profile: photo, qualifications, registration number and council, specialties, experience, languages, signature, consultation fees per visit type.
- Calendar view combining OPD sessions, IPD rounds, on-call duty and leave.
- In-patient list across wards with pending results and orders to review.
- Payout rules for visiting consultants: fixed per visit, percentage of collection or per procedure. Monthly doctor payout statement feeds Finance.
- Order sets and prescription templates owned by the doctor.

## 7.4 Shift rosters

Rosters are part of the HRM module but are used every day by wards, so they are described here. Without HRM, the hospital can still keep a simple duty list in CORE.

| Feature                | Details                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Shift master</b>    | Morning 07-14, Evening 14-21, Night 21-07, General 09-18, custom. Grace time and minimum hours per shift  |
| <b>Roster planner</b>  | Drag-and-drop weekly or monthly grid per department or ward. Copy last week, auto-fill rotating patterns  |
| <b>Coverage rules</b>  | Minimum staff per shift and skill (e.g. 1 ICU-trained nurse per 4 ICU beds). Planner warns on shortfall   |
| <b>Constraints</b>     | No more than N nights in a row, rest hours between shifts, weekly off. Approved leave blocks roster cells |
| <b>Publish</b>         | HOD or Ward In-charge publishes; staff get notification and see the roster in self-service                |
| <b>Swap requests</b>   | Staff request a swap with a colleague; colleague accepts; In-charge approves                              |
| <b>Attendance link</b> | Published roster is the expected attendance; punches are matched to it for late, early and overtime       |

**User flow: roster planning and swap**



## SECTION 08

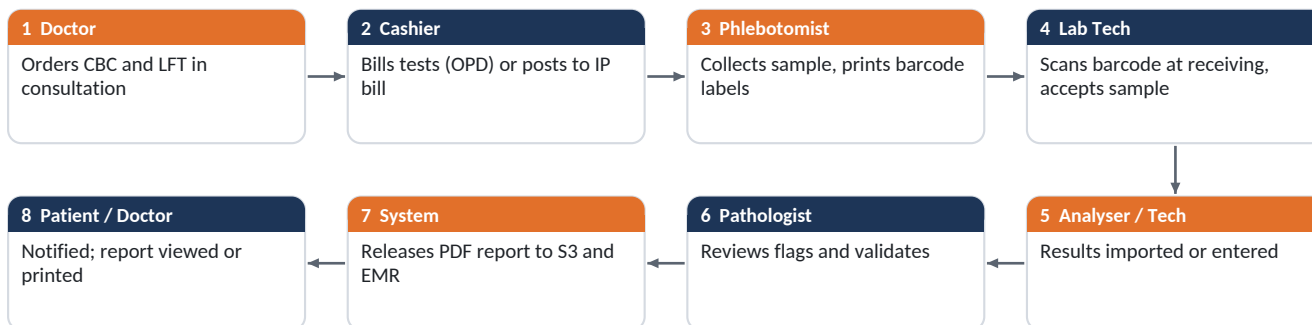
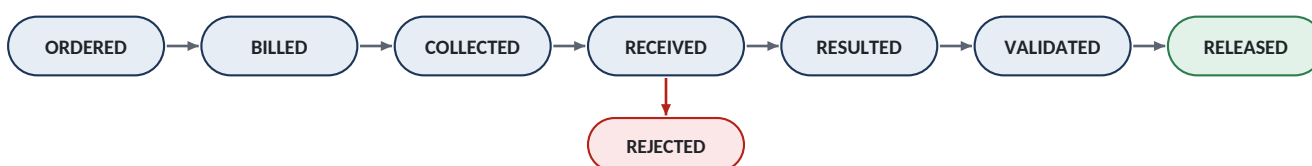
# Diagnostics and Pharmacy

## LAB Laboratory (LIS)

|                      |                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                          |
| <b>Depends on</b>    | CORE                                                                                                                   |
| <b>Primary users</b> | Phlebotomist, Lab Technician, Pathologist, Doctors, Front Office                                                       |
| <b>Purpose</b>       | Tracks every test from order to signed report, with barcodes, validation, turnaround time and patient report delivery. |

## 8.1 Laboratory features

| Feature                    | Details                                                                                                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test master</b>         | Test code, name, department (biochemistry, haematology, microbiology, pathology), sample type, container colour, volume, method, TAT target, price                                |
| <b>Parameters</b>          | Each test has parameters with units, reference ranges by age and gender, critical limits, formula fields (e.g. LDL calculated)                                                    |
| <b>Panels and profiles</b> | Groups of tests, e.g. Lipid profile, LFT, with one price                                                                                                                          |
| <b>Orders</b>              | From OPD, IPD, nursing or walk-in at lab counter. Billing rule: OPD pays before collection, IPD posts to running bill                                                             |
| <b>Sample collection</b>   | Worklist of pending collections; barcode label printed per container; collection time and collector recorded                                                                      |
| <b>Accession</b>           | Lab receives sample by scanning barcode; reject with reason (haemolysed, insufficient) triggers recollection                                                                      |
| <b>Result entry</b>        | Grid entry per worklist or per patient; machine results imported by file or connector; delta check against previous result                                                        |
| <b>Flags</b>               | High, low and critical flags; critical values notify the doctor immediately and must be acknowledged                                                                              |
| <b>Validation</b>          | Technician enters (maker), pathologist validates (checker). Only validated results are released                                                                                   |
| <b>Reports</b>             | Branded PDF with signature, QR verification code, interpretation notes; trend graph for repeat tests                                                                              |
| <b>Delivery</b>            | Doctor sees results instantly; patient gets SMS / WhatsApp link; printing at counter                                                                                              |
| <b>Outsourced tests</b>    | Send-out tests to partner labs tracked with expected date and uploaded report                                                                                                     |
| <b>Home collection</b>     | Book home sample collection online or by phone; assign phlebotomist by area and slot; mobile web app for collection, barcode printing and payment at home; live status to patient |
| <b>Collection centres</b>  | Partner clinics and collection centres with their own rate lists, credit billing, sample pick-up runs and monthly statements                                                      |
| <b>Quality control</b>     | QC lot entries and Levey-Jennings chart per analyser (basic in Phase 1)                                                                                                           |
| <b>Microbiology</b>        | Culture workflow with preliminary and final reports: organism, colony count, antibiotic sensitivity panel (S / I / R); antibiogram report for infection control                   |
| <b>Histopathology</b>      | Specimen accession, gross description, blocks and slides tracking, microscopic findings, diagnosis, synoptic templates; longer TAT targets                                        |

**User flow: lab order to report****Lab test status****Reports**

- Pending workload by department
- TAT by test and by stage
- Critical value log with acknowledgement
- Sample rejection report
- Test-wise volume and revenue
- Referral-doctor test report

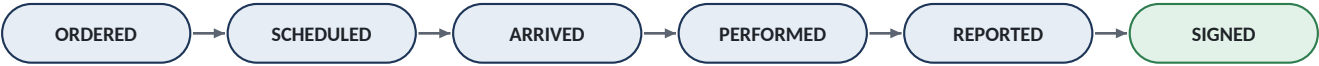
**8.2 Radiology workflows****RAD Radiology (RIS)**

|                      |                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                 |
| <b>Depends on</b>    | CORE                                                                                                          |
| <b>Primary users</b> | Radiology Front Desk, Technician, Radiologist, Doctors                                                        |
| <b>Purpose</b>       | Schedules and tracks imaging studies and produces signed radiology reports, with links to images in the PACS. |

| Feature                     | Details                                                                                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Modality master</b>      | X-Ray, USG, CT, MRI, Mammography, with rooms, machines and slot lengths                                                                                         |
| <b>Study master</b>         | Study name, modality, preparation instructions (e.g. fasting), contrast flag, price, report template                                                            |
| <b>Scheduling</b>           | Slot booking per modality; preparation SMS sent to patient                                                                                                      |
| <b>Study tracking</b>       | Arrived, in progress, completed, with technician and time                                                                                                       |
| <b>Contrast and consent</b> | Consent form and creatinine check prompt for contrast studies                                                                                                   |
| <b>Reporting</b>            | Template-based reporting with normal-report macros; draft by resident, sign-off by radiologist (maker-checker)                                                  |
| <b>Images</b>               | Accession number sent to PACS via DICOM worklist (optional); 'View images' button opens PACS web viewer. Small JPEG key images can be uploaded to S3 if no PACS |

| Feature           | Details                                                                                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Delivery          | Same as lab: EMR, SMS link, print, film issue register                                                                                                                                                                            |
| PCPNDT compliance | For ultrasound on pregnant women, Form F is mandatory under the PCPNDT Act: captured digitally with doctor declaration, patient signature, register kept for the statutory period and monthly report to the appropriate authority |

Radiology study status



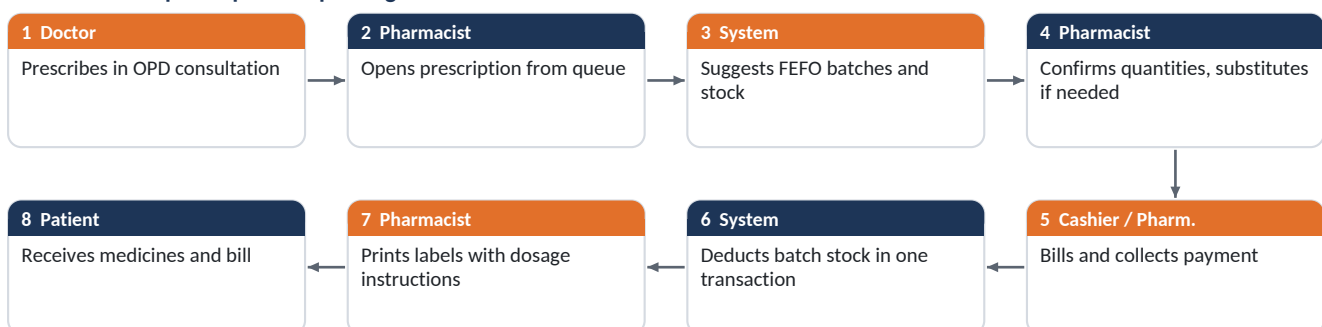
## PHR Pharmacy

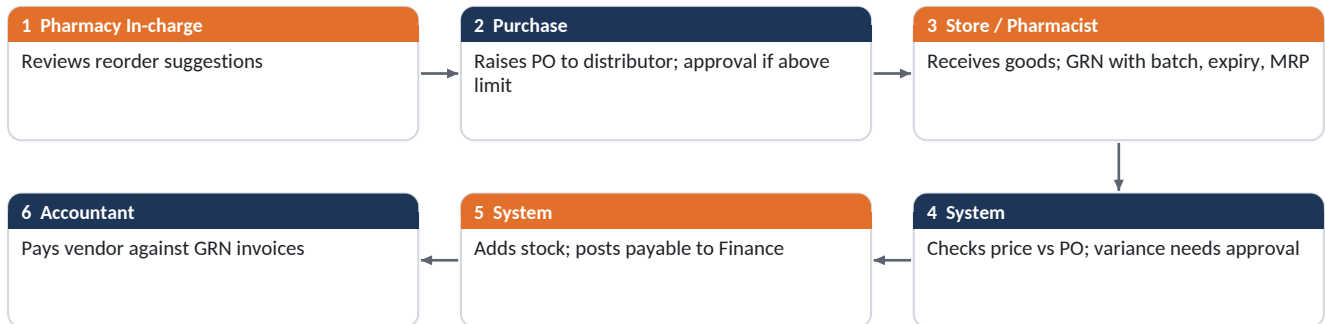
|                      |                                                                                                                                     |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                                       |
| <b>Depends on</b>    | CORE (IPD for ward issues)                                                                                                          |
| <b>Primary users</b> | Pharmacist, Pharmacy In-charge, Purchase, Nurses, Cashier                                                                           |
| <b>Purpose</b>       | Runs the hospital pharmacy back-end and counter: purchase, batch and expiry stock, dispensing, sales, returns and controlled drugs. |

## 8.3 Pharmacy features

| Feature                 | Details                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Drug master</b>      | Brand name, generic, strength, form, manufacturer, HSN, GST rate, schedule (H, H1, X, narcotic), pack size and units, rack location, reorder level, substitutes |
| <b>Multiple stores</b>  | Main pharmacy, OPD counter, IPD satellite, OT pharmacy; each with own stock and transfers between them                                                          |
| <b>Purchase</b>         | Purchase order to distributor, GRN against PO or direct, free quantity, scheme discounts, batch, expiry, MRP, purchase rate                                     |
| <b>Batch and expiry</b> | Stock is held per batch. Dispensing picks First-Expiry-First-Out automatically; pharmacist can override with reason                                             |
| <b>OPD dispensing</b>   | Prescription from consultation appears in pharmacy queue. Pharmacist checks, picks batches, bills, prints label and bill                                        |
| <b>Counter sales</b>    | Walk-in sale against external prescription (upload image)                                                                                                       |
| <b>IPD issues</b>       | Nurse indent or doctor order issued to patient; posted to running bill; returns credited back                                                                   |
| <b>Substitution</b>     | Generic substitution suggestions when out of stock, with doctor consent flag                                                                                    |
| <b>Controlled drugs</b> | Schedule H1 and narcotic register with prescriber, patient, quantity; mandatory prescription copy                                                               |
| <b>Returns</b>          | Patient returns (within policy), returns to vendor, expiry returns with credit notes                                                                            |
| <b>Alerts</b>           | Reorder level, near-expiry (90/60/30 days), slow-moving, dead stock                                                                                             |
| <b>Stock audit</b>      | Physical count by rack; variance goes to maker-checker adjustment                                                                                               |
| <b>Pricing</b>          | Sale at MRP or MRP minus discount by patient category; margin report                                                                                            |

### User flow: OPD prescription dispensing



**User flow: pharmacy purchase****Business rules**

- Expired batches are blocked from sale automatically at midnight of expiry date.
- Stock can never go negative. Sale and issue run in a database transaction that checks and deducts each batch together.
- Schedule H1 and narcotic drugs cannot be sold without prescriber details.
- Rates on GRN higher than PO rate by more than a set percentage need approval.

**Reports**

- Daily sales and collection
- Stock valuation (purchase and MRP)
- Expiry report
- Fast and slow moving
- Purchase register and GST input
- Sales GST register (HSN-wise)
- Schedule H1 / narcotic register
- Doctor-wise prescription value

## SECTION 09

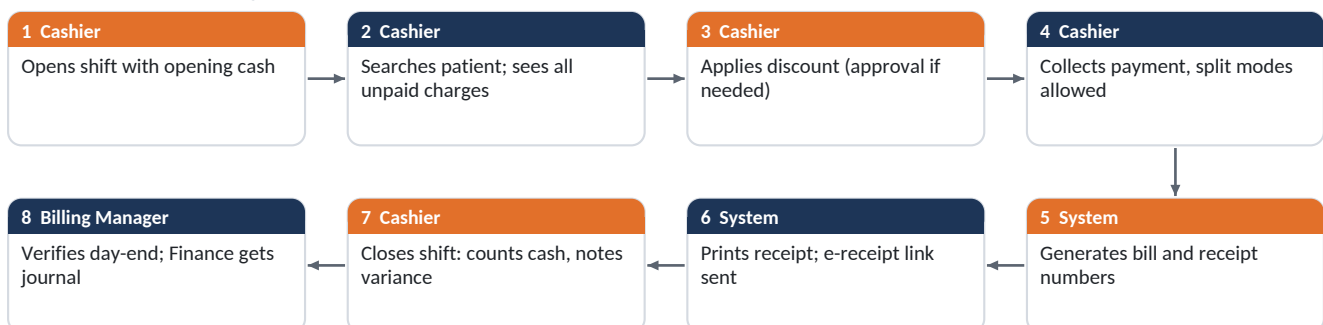
# Billing, Finance, Inventory, HR and Payroll

## 9.1 Cash, billing and printing (CORE)

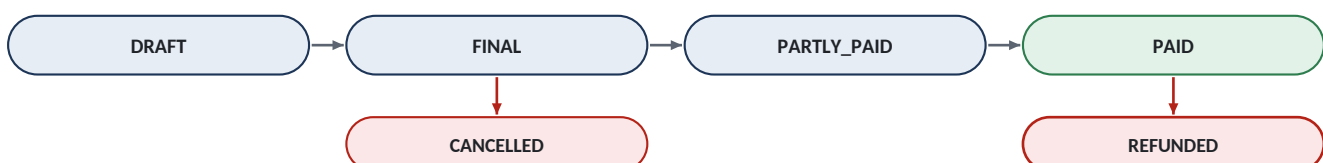
Billing is part of CORE because every module produces charges. One billing engine handles OPD, IPD, lab, radiology and pharmacy, so a patient can pay everything at one counter with one receipt.

| Feature                  | Details                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Bill types               | OPD bill, IPD interim and final bill, pharmacy bill, lab / radiology walk-in bill, package bill, miscellaneous bill                                 |
| Charge posting           | Services posted automatically from orders, bed days, visits, pharmacy issues and consumables; manual add with permission                            |
| Pricing                  | Rate from patient's price list; package inclusions; tax per item; rounding rules                                                                    |
| Discounts                | Item-level or bill-level, amount or percentage, with reason; maker-checker above thresholds                                                         |
| Deposits                 | Advance receipts for IPD, adjusted against final bill; refund of excess with approval                                                               |
| Payments                 | Cash, card, UPI (dynamic QR on screen), cheque, bank transfer; split payment across modes                                                           |
| Refunds and cancellation | Full or partial with reason and approval; credit note generated; original never deleted                                                             |
| Credit / corporate       | Bill to company with credit period; outstanding tracked in Finance receivables                                                                      |
| Printing                 | Thermal 80 mm for OPD, A4 for IPD with item detail or summary; duplicate print watermarked 'DUPLICATE'; reprint logged                              |
| E-receipts               | PDF on S3, link sent by SMS / WhatsApp / e-mail                                                                                                     |
| GST                      | Healthcare services exempt where applicable; pharmacy and some services taxable; GST breakup on bill; e-invoice for B2B (corporate) bills           |
| Cashier shift            | Opening cash, collections by mode, handover, closing count, variance with reason; cash deposited to bank recorded                                   |
| Payment devices          | Card machine and UPI QR integration so the amount is pushed to the device and the payment is matched automatically; daily settlement reconciliation |
| Wallet and memberships   | Prepaid patient or family wallet and membership discounts (from CRM) applied at any counter                                                         |

### User flow: counter billing and day-end



### Bill status



## 9.2 Financial reporting and accounts

### FIN Finance and Accounts

|                      |                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                       |
| <b>Depends on</b>    | CORE                                                                                                                |
| <b>Primary users</b> | Accountant, Finance Controller, Auditor, Super Admin                                                                |
| <b>Purpose</b>       | Replaces a separate accounting package. Every bill, receipt, GRN, payroll and payment posts journals automatically. |

| Feature                  | Details                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chart of accounts</b> | Pre-loaded hospital chart (revenue by department, receivables, payables, inventory, salaries, statutory). Editable groups and ledgers |
| <b>Auto-posting</b>      | Rules map each business event to debit and credit ledgers with cost centre: bill, receipt, refund, GRN, issue, payroll, payment       |
| <b>Vouchers</b>          | Payment, receipt, journal, contra; manual journal needs approval                                                                      |
| <b>Payables</b>          | Vendor bills from GRN, due dates, payment run, TDS deduction, vendor ledger                                                           |
| <b>Receivables</b>       | Corporate and patient outstanding, ageing, reminders                                                                                  |
| <b>Bank</b>              | Bank accounts, statement import (CSV), reconciliation                                                                                 |
| <b>Doctor payouts</b>    | Visiting consultant share computed from bills; payout voucher monthly                                                                 |
| <b>Period close</b>      | Month lock; reopening needs Super Admin approval                                                                                      |
| <b>Statements</b>        | Day book, ledger, trial balance, P&L, balance sheet, cash flow, cost-centre P&L by department                                         |
| <b>Tax reports</b>       | GSTR-1 and GSTR-3B data export, TDS summary                                                                                           |
| <b>Exports</b>           | Excel and PDF; Tally-compatible XML export as an option                                                                               |
| <b>Petty cash</b>        | Imprest per department or branch, petty cash vouchers with bills, replenishment request and approval                                  |
| <b>Expense claims</b>    | Employees claim travel, conference and other expenses with receipts; manager and finance approval; paid by bank or through payroll    |
| <b>Budgets</b>           | Annual budget per cost centre and ledger; actual vs budget report; warning when a purchase order exceeds the remaining budget         |

### Management financial reports (all hospitals, CORE)

- Daily collection summary by mode, counter and cashier
- Revenue by department, doctor, service and payer
- OPD vs IPD vs pharmacy vs diagnostics revenue mix
- Discount and refund register with approver
- Patient outstanding and deposit balance
- Month-on-month trend dashboard

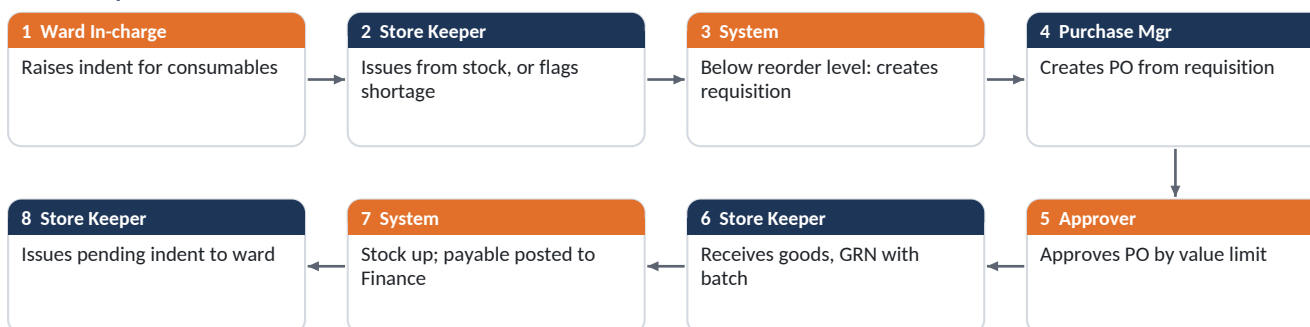
## INV Inventory and Purchase

|                      |                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                                         |
| <b>Depends on</b>    | CORE                                                                                                                                  |
| <b>Primary users</b> | Store Keeper, Purchase Manager, HODs, Ward In-charge, Finance                                                                         |
| <b>Purpose</b>       | Controls non-drug stock: consumables, surgical items, linen, stationery, housekeeping and equipment, from requisition to consumption. |

## 9.3 Inventory and stock tracking

| Feature                      | Details                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Item master</b>           | Code, name, category, UoM with conversions, HSN, GST, batch and expiry tracking flag, reorder level, preferred vendors |
| <b>Stores</b>                | Central store, sub-stores, ward stock points; each with own stock                                                      |
| <b>Vendors</b>               | Vendor master with GSTIN, bank details, rating, rate contracts                                                         |
| <b>Requisition to PO</b>     | Department requisition, quotation comparison (optional), PO with approval levels by value, PO e-mailed to vendor       |
| <b>GRN</b>                   | Against PO, partial receipt, rejection, batch and expiry capture, quality check flag                                   |
| <b>Indent and issue</b>      | Departments raise indents; store issues; partial issue and back-orders                                                 |
| <b>Transfers</b>             | Between stores with in-transit status and receive confirmation                                                         |
| <b>Consumption</b>           | Ward consumption entry; patient-chargeable items post to bill                                                          |
| <b>Valuation</b>             | Weighted average cost; stock value by store and category                                                               |
| <b>Physical verification</b> | Count sheets, variance, approved adjustment                                                                            |
| <b>Assets</b>                | Basic asset register: tag, location, warranty, AMC due date, depreciation (with FIN)                                   |

### User flow: procure-to-issue



## HRM HR, Rosters and Attendance

|                      |                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                     |
| <b>Depends on</b>    | CORE                                                                                                              |
| <b>Primary users</b> | HR Manager, HR Executive, HODs, all employees (self-service)                                                      |
| <b>Purpose</b>       | Manages the full employee lifecycle, attendance and leave, and feeds payroll. Rosters are described in Section 7. |

## 9.4 HR management

| Feature                     | Details                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Employee master</b>      | Personal details, contacts, family, education, experience, bank, PAN, Aadhaar, UAN, ESI number, photo                                                             |
| <b>Job details</b>          | Employee ID, department, designation, grade, employment type (permanent, contract, consultant, trainee), reporting manager, joining date, probation               |
| <b>Clinical credentials</b> | Medical / nursing registration number and expiry; alerts 60 days before expiry                                                                                    |
| <b>Documents</b>            | Offer letter, appointment letter, certificates, ID proofs stored on S3; letters generated from templates                                                          |
| <b>Onboarding</b>           | Checklist: documents, user account, uniform, ID card, training                                                                                                    |
| <b>Attendance</b>           | Biometric device import (CSV or connector), web punch with IP restriction, mobile punch with geo-fence. Matched to roster                                         |
| <b>Leave</b>                | Leave types (CL, SL, EL, maternity, comp-off), accrual rules, carry-forward and encashment, holiday calendar per branch                                           |
| <b>Self-service</b>         | Apply leave, regularise attendance, view roster, payslips, tax declarations, update profile (with HR approval)                                                    |
| <b>Movement</b>             | Transfers, promotions, increments with effective dates and history                                                                                                |
| <b>Recruitment</b>          | Manpower requisition with approval, job openings, candidate database, interview schedule and panel feedback, offer letter, convert selected candidate to employee |
| <b>Training</b>             | Training calendar, attendance, mandatory certifications (BLS, ACLS, fire safety, infection control) with expiry alerts                                            |
| <b>Credentialing</b>        | Doctors' clinical privileges granted by the credentialing committee, renewal dates, linked to what they may order or perform                                      |
| <b>Appraisal</b>            | Annual cycle with KRAs, self review, manager and HOD review, increment recommendation passed to Payroll                                                           |
| <b>Exit</b>                 | Resignation, notice period, clearance checklist across departments, full and final settlement via Payroll, experience letter                                      |

### User flow: leave application



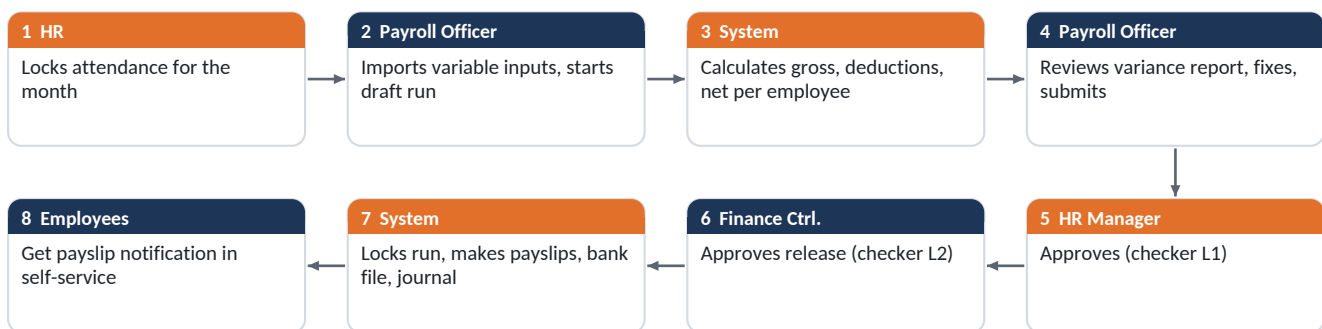
### PAY Payroll

|                      |                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                                   |
| <b>Depends on</b>    | HRM                                                                                                                             |
| <b>Primary users</b> | Payroll Officer, HR Manager, Finance Controller, employees                                                                      |
| <b>Purpose</b>       | Calculates monthly salaries from attendance, with statutory deductions, approvals, payslips, bank transfer file and accounting. |

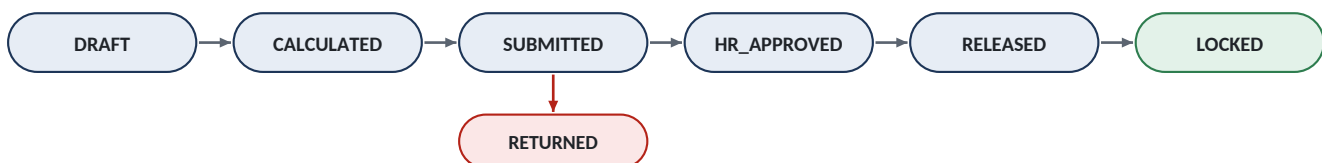
## 9.5 Payroll management

| Feature                   | Details                                                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Salary structure</b>   | Templates by grade: Basic, HRA, DA, conveyance, special, night shift allowance, on-call allowance. Formula-based components (e.g. HRA = 40% of Basic)                                                     |
| <b>Statutory (India)</b>  | PF (12% employee and employer, wage ceiling), ESI (0.75% / 3.25% under wage limit), Professional Tax by state slab, TDS under old or new regime from declarations, LWF. All rates in configuration tables |
| <b>Variable inputs</b>    | Overtime hours, night shifts, on-call, incentives, arrears, deductions (canteen, uniform), doctor variable pay                                                                                            |
| <b>Attendance link</b>    | Payable days = calendar days - LOP; LOP from absent days and unpaid leave in locked attendance                                                                                                            |
| <b>Loans and advances</b> | Salary advance and loans with EMI deduction schedule                                                                                                                                                      |
| <b>Payroll run</b>        | Draft run, review variances vs last month, approval, lock                                                                                                                                                 |
| <b>Outputs</b>            | Payslips (PDF on S3, e-mailed), bank transfer file (bank format), PF ECR file, ESI return, PT and TDS reports, Form 16 data                                                                               |
| <b>Accounting</b>         | Payroll journal by cost centre posted to FIN on lock                                                                                                                                                      |
| <b>Full and final</b>     | Notice recovery, leave encashment, gratuity, pending dues                                                                                                                                                 |

### User flow: monthly payroll



### Payroll run status



**Payroll lock.** A locked payroll month cannot be changed. Corrections are made as arrears in the next month. Reopening needs Super Admin approval and is audited.

## SECTION 10

## Support Services, Records, Quality and Patient Engagement

These modules cover the departments that keep a hospital running behind the wards and counters. Without them, hospitals fall back to paper registers or separate tools for records, kitchen, housekeeping, maintenance and quality.

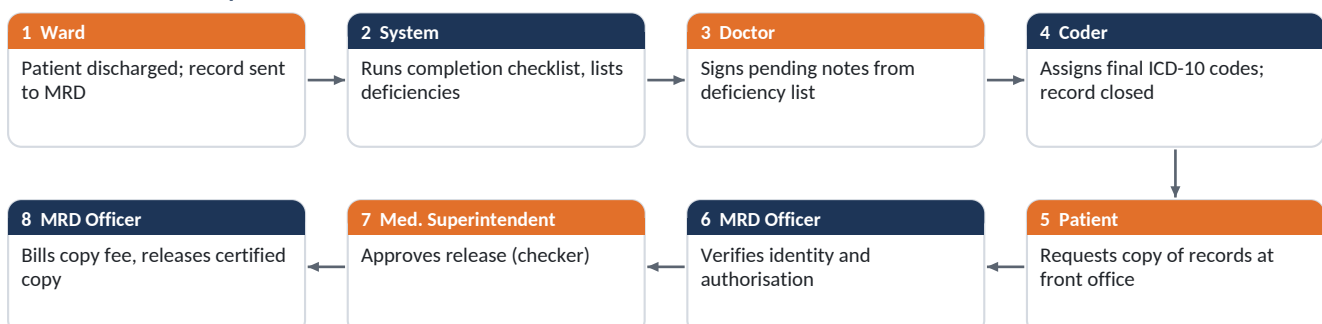
### MRD Medical Records and Certificates

|                      |                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                                                       |
| <b>Depends on</b>    | CORE (IPD recommended)                                                                                                                              |
| <b>Primary users</b> | MRD Officer, Medical Coder, Doctors, Front Office, Medical Superintendent                                                                           |
| <b>Purpose</b>       | Keeps the legal medical record complete, coded, tracked and released only with authorisation, and runs the birth, death and medico-legal registers. |

### 10.1 Medical records features

| Feature                       | Details                                                                                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Record completion</b>      | After discharge, a checklist of mandatory items per case (signed summary, consents, progress notes, MAR, investigations). Missing items create deficiencies assigned to the responsible doctor with reminders |
| <b>Clinical coding</b>        | Coder assigns final ICD-10 diagnoses and procedure codes; used in morbidity and mortality statistics                                                                                                          |
| <b>Physical file tracking</b> | For hospitals that still keep paper files: file number, rack location, issue to department or doctor, due date, return, overdue alerts, barcode on file jacket                                                |
| <b>Record release</b>         | Requests from patient, family, insurer, court or police. Identity and authorisation checked, maker-checker approval, copy fee billed, release logged, certified copies watermarked                            |
| <b>Birth register</b>         | Birth details (date, time, sex, weight, mother, father, place, type of delivery) captured from the ward. Generates the birth report for the Civil Registration System within the legal time limit             |
| <b>Death register</b>         | Date, time, cause; Medical Certificate of Cause of Death (MCCD) signed by the doctor; death report for the Civil Registration System; death summary                                                           |
| <b>Mortuary register</b>      | Body received, cold-storage slot, handover to relatives with ID proof and signature, or to police in medico-legal cases                                                                                       |
| <b>Medico-legal cases</b>     | MLC register with number, police station intimation, officer details, injury certificate, linked documents                                                                                                    |
| <b>Retention</b>              | Retention period per record type; archival and destruction lists with approval                                                                                                                                |
| <b>Statistics</b>             | Monthly hospital statistics: admissions, discharges, deaths, births, occupancy, average length of stay, gross and net death rate                                                                              |

#### User flow: record completion and release



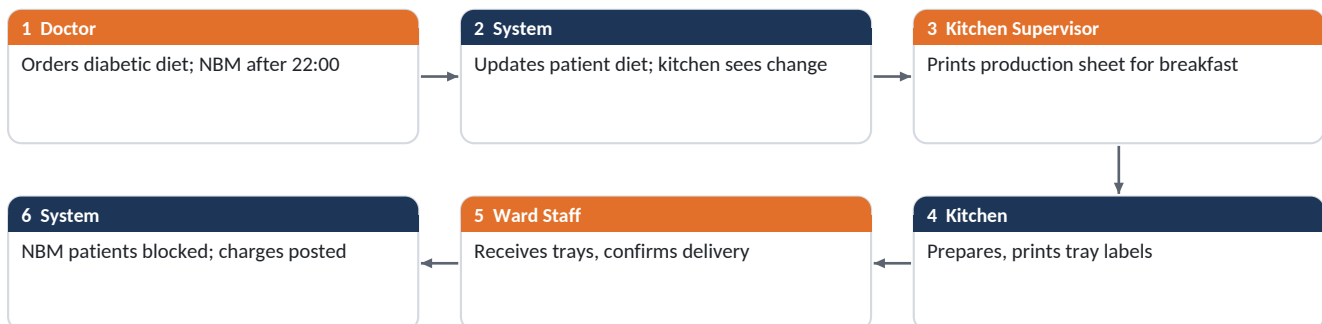
## DIET Diet and Kitchen

|                      |                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                                             |
| <b>Depends on</b>    | IPD, INV (kitchen store)                                                                                                                  |
| <b>Primary users</b> | Doctors, Dietitians, Nurses, Kitchen Supervisor, Billing                                                                                  |
| <b>Purpose</b>       | Turns diet orders for in-patients into a kitchen production plan, labelled trays and delivery tracking, and bills meals where chargeable. |

## 10.2 Diet and kitchen features

| Feature                 | Details                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diet master</b>      | Diet types (normal, soft, liquid, diabetic, renal, cardiac, low salt, high protein, tube feed), texture, meal schedule, menu cycle                                                     |
| <b>Diet orders</b>      | Ordered by doctor or dietitian per admission; default diet on admission; Nil By Mouth (NBM) with start and end time, e.g. before a procedure; allergies pulled from the patient record |
| <b>Dietitian</b>        | Nutrition assessment, counselling notes, diet plan at discharge; billable consultation                                                                                                 |
| <b>Production sheet</b> | Per meal: counts by diet type and ward, special requests, printed or shown on kitchen screen                                                                                           |
| <b>Tray labels</b>      | Patient name, bed, diet type, allergy warning, NBM block                                                                                                                               |
| <b>Delivery</b>         | Ward-wise delivery confirmation with time; missed meals reported                                                                                                                       |
| <b>Charges</b>          | Meals for private rooms and attendants posted to the running bill; staff canteen is out of scope                                                                                       |
| <b>Kitchen stock</b>    | Raw material indents and consumption through the Inventory kitchen store; cost per meal report                                                                                         |

### User flow: diet order to tray



## FAC Housekeeping, Linen and Facility

|                      |                                                                                                                      |
|----------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                        |
| <b>Depends on</b>    | CORE (IPD and INV recommended)                                                                                       |
| <b>Primary users</b> | Housekeeping Supervisor, Laundry, Maintenance Technicians, Biomedical Engineer, all staff (raise tickets)            |
| <b>Purpose</b>       | Runs housekeeping, linen and laundry, maintenance tickets, biomedical equipment upkeep and biomedical waste records. |

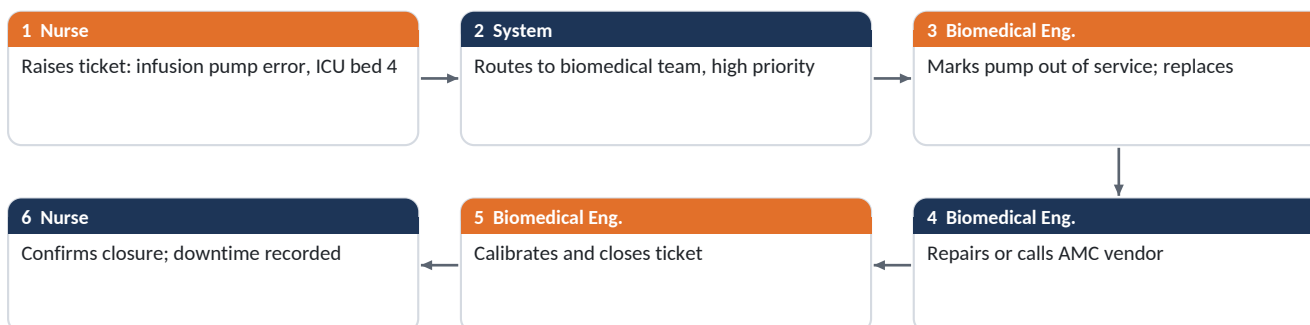
### 10.3 Housekeeping and linen

| Feature                   | Details                                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bed cleaning tasks</b> | Created automatically when a patient leaves a bed. Shown on the housekeeping board with a TAT target; supervisor verifies; bed returns to AVAILABLE on the live bed board |
| <b>Scheduled cleaning</b> | Area checklists (wards, toilets, OPD, ICU) by frequency, with staff assignment and supervisor sign-off                                                                    |
| <b>Linen par levels</b>   | Linen types and required stock per ward                                                                                                                                   |
| <b>Laundry cycle</b>      | Soiled linen collection count per ward, dispatch to in-house or outsourced laundry, receipt count, shortage tracking                                                      |
| <b>Condemnation</b>       | Damaged linen written off with approval; replacement indent                                                                                                               |
| <b>Patient transport</b>  | Porter requests for wheelchair or stretcher moves (ward to radiology, discharge to gate) with pickup and drop times                                                       |

### 10.4 Maintenance and biomedical equipment

| Feature                       | Details                                                                                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Maintenance tickets</b>    | Any user raises a ticket (electrical, plumbing, AC, civil, IT) with location and photo. Priority, assignment, TAT, closure with requester confirmation |
| <b>Equipment register</b>     | Medical equipment from the asset register with serial number, location, vendor, warranty, risk class                                                   |
| <b>Preventive maintenance</b> | PM and calibration schedules; due list; checklist completion; certificates uploaded to S3                                                              |
| <b>Breakdowns</b>             | Breakdown tickets with downtime tracked; equipment marked out of service so it is not scheduled (e.g. a radiology modality)                            |
| <b>AMC / CMC</b>              | Annual and comprehensive maintenance contracts, renewal alerts 60 days ahead, vendor visit log                                                         |

#### User flow: equipment breakdown



## 10.5 Biomedical waste

- Daily waste entry per ward by colour category (yellow, red, white, blue) with weight in kg, as required by the Bio-Medical Waste Management Rules, 2016.
- Handover to the authorised treatment facility with manifest number, vehicle and weights; mismatch alerts.
- Monthly and annual reports in the format the State Pollution Control Board asks for.
- Needle-stick and spill incidents link to the Quality module.

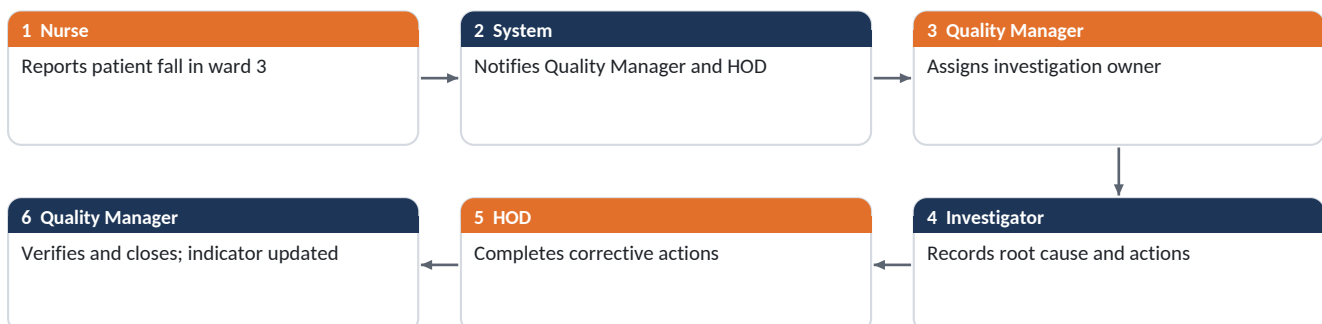
### QLT Quality, Feedback and Incidents

|                      |                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                                                                |
| <b>Depends on</b>    | CORE                                                                                                                                                         |
| <b>Primary users</b> | Quality Manager, Medical Superintendent, HODs, all staff, patients                                                                                           |
| <b>Purpose</b>       | Supports NABH accreditation: patient feedback, complaints, incident reporting with root cause analysis, quality indicators, audits and controlled documents. |

## 10.6 Quality features

| Feature                   | Details                                                                                                                                                                                                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Patient feedback</b>   | Short survey by SMS or WhatsApp link after OPD visit and at discharge, plus QR posters; ratings by department and Net Promoter Score                                                                                               |
| <b>Complaints</b>         | Complaint register from feedback, front office or portal; category, owner, TAT, escalation to Medical Superintendent, closure message to the patient                                                                               |
| <b>Incident reporting</b> | Any staff can report, anonymously if they choose: falls, medication errors, needle-stick injuries, equipment failure, near misses. Severity, investigation, root cause, corrective and preventive actions (CAPA), closure          |
| <b>Quality indicators</b> | Computed automatically from system data each month, e.g. OPD waiting time, discharge TAT, lab TAT, critical value reporting time, medication errors per 1,000 patient days, falls, bed occupancy, return to ICU within 48 hours    |
| <b>Audits</b>             | Checklist templates (hand hygiene, MAR compliance, consent, crash cart), scores, action items with owners                                                                                                                          |
| <b>Document control</b>   | Policies and SOPs with version, approval, review date and staff read-acknowledgement                                                                                                                                               |
| <b>Infection control</b>  | Surveillance of hospital-acquired infections (urinary catheter, central line, ventilator-associated, surgical site), device days from nursing records, infection rates, antibiogram from the lab, isolation flags on the bed board |

### User flow: incident to closure



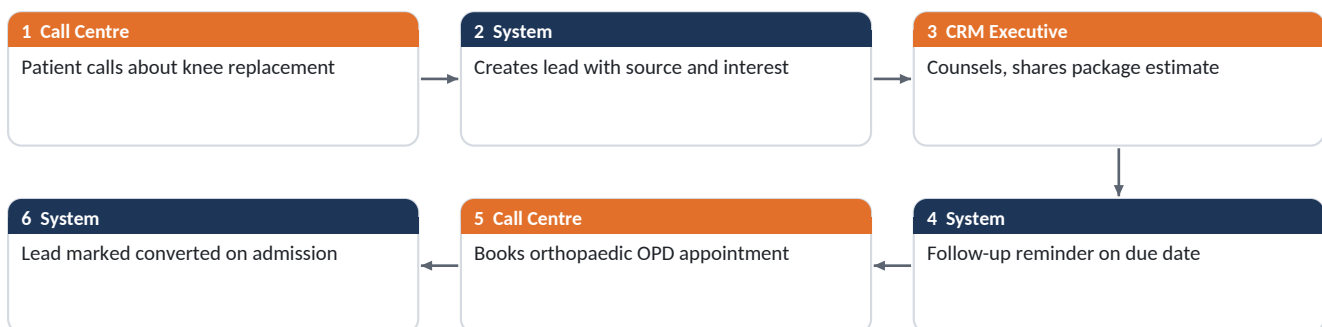
## CRM Patient CRM and Outreach

|                      |                                                                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Subscription</b>  | Add-on module                                                                                                                  |
| <b>Depends on</b>    | CORE                                                                                                                           |
| <b>Primary users</b> | Call Centre Executives, Marketing / CRM Executives, Front Office, Management                                                   |
| <b>Purpose</b>       | Brings patients in and keeps them coming back: call centre, leads, health camps, campaigns, memberships and referral partners. |

## 10.7 Patient CRM features

| Feature                  | Details                                                                                                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Call centre</b>       | Agent screen with caller lookup by phone number, appointment booking, enquiry logging and call notes. Optional cloud telephony integration (e.g. Exotel, Knowlarity) for click-to-call, call recording links and missed-call callbacks |
| <b>Leads</b>             | Enquiries for surgeries, packages and health check-ups tracked through stages (new, contacted, counselled, booked, lost) with follow-up dates and source                                                                               |
| <b>Health camps</b>      | Camp planning, on-site registration on a tablet, screening results, conversion of camp patients to visits, camp ROI report                                                                                                             |
| <b>Campaigns</b>         | SMS, WhatsApp and e-mail campaigns to filtered patient lists, e.g. diabetics due for HbA1c. Consent and opt-out respected; DLT templates                                                                                               |
| <b>Memberships</b>       | Family health cards and memberships with validity, member discounts, free visits and a prepaid wallet usable at all counters                                                                                                           |
| <b>Referral partners</b> | Referring doctors and clinics with visit history, revenue generated, agreed referral fee where permitted by policy, and monthly statements to Finance                                                                                  |
| <b>Reviews</b>           | After a good feedback score, the patient gets a link to post a public review (e.g. Google)                                                                                                                                             |
| <b>Reports</b>           | Lead conversion, campaign response, camp conversion, referral revenue, membership renewals                                                                                                                                             |

### User flow: lead to admission

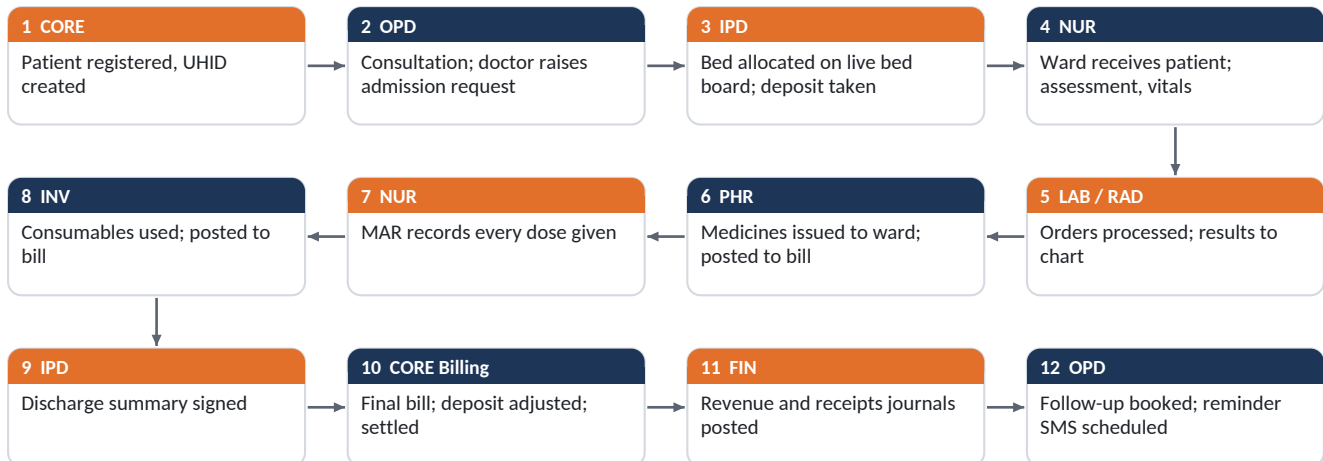


## SECTION 11

## End-to-End Journeys and Screen Map

Modules are useful only when they hand work to each other without re-entry. This section shows how data flows across modules in the four journeys a hospital runs every day.

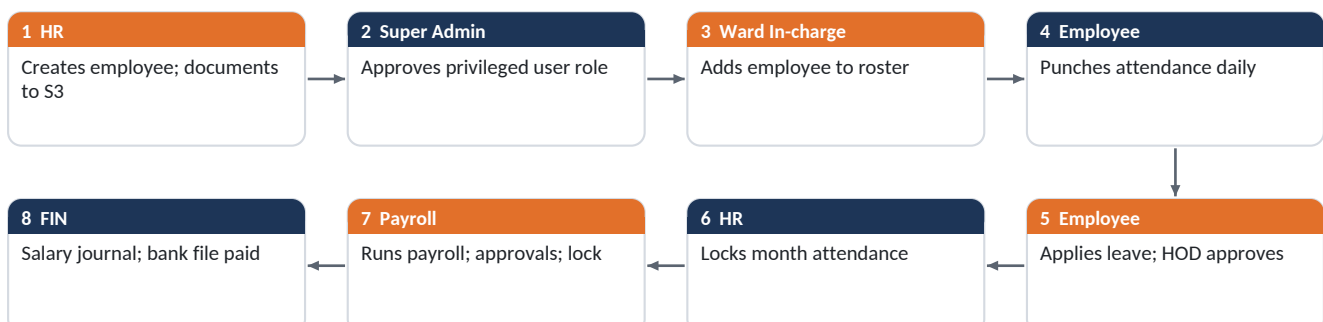
### 11.1 In-patient journey across modules



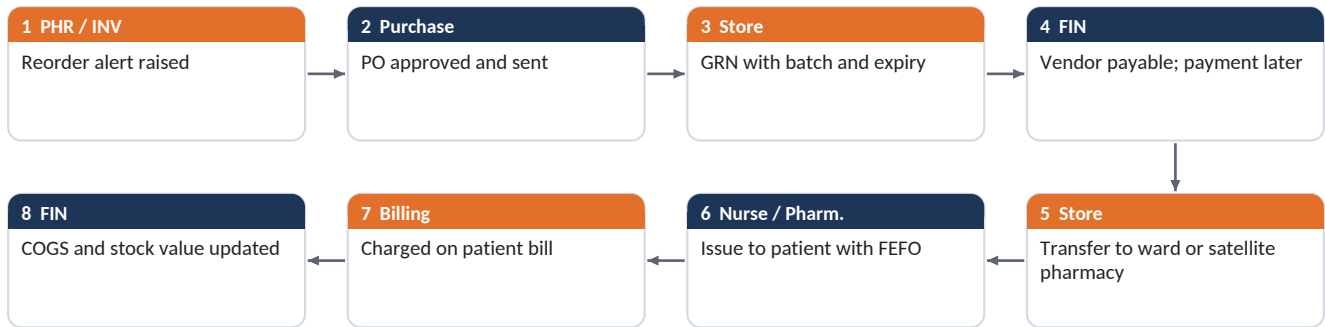
### 11.2 Money journey: charge to ledger

| Event            | Created by         | Bill effect          | Accounting entry (FIN)                                                      |
|------------------|--------------------|----------------------|-----------------------------------------------------------------------------|
| Consultation fee | OPD check-in       | OPD bill line        | Dr Cash / Cr OPD revenue                                                    |
| Bed day          | Nightly job        | IP running bill line | Dr Patient receivable / Cr Room revenue                                     |
| Lab test         | Order              | Bill line            | Dr Cash or receivable / Cr Lab revenue                                      |
| Pharmacy issue   | Pharmacy           | Bill line with GST   | Dr Receivable / Cr Pharmacy sales, Cr GST output; Dr COGS / Cr Inventory    |
| Deposit          | Cashier            | Advance receipt      | Dr Cash / Cr Patient advance                                                |
| Discount         | Cashier + approver | Negative line        | Dr Discount allowed / Cr Receivable                                         |
| GRN              | Store              | -                    | Dr Inventory, Dr GST input / Cr Vendor payable                              |
| Payroll lock     | Payroll            | -                    | Dr Salaries by cost centre / Cr Salary payable, Cr PF, ESI, PT, TDS payable |

### 11.3 People journey: hire to pay



## 11.4 Stock journey: purchase to patient



## 11.5 Screen map

The left menu is built from subscribed modules and the user's permissions. Each line below is a top-level menu with its main screens.

| Menu         | Screens                                                                                                     | Module     |
|--------------|-------------------------------------------------------------------------------------------------------------|------------|
| Home         | Role dashboard, approvals inbox, notifications, global search                                               | CORE       |
| Patients     | Register, search, patient profile (timeline, documents, bills), merge                                       | CORE       |
| Appointments | Slot calendar, book, check-in, queue display, doctor schedules                                              | OPD        |
| Consultation | My queue, consultation workspace, templates, order sets                                                     | OPD        |
| In-patients  | Admission desk, bed board, admitted list, transfers, discharge desk                                         | IPD        |
| Nursing      | Ward census, task list, patient chart, MAR, handover, indents                                               | NUR        |
| Laboratory   | Collection worklist, receiving, result entry, validation, reports, QC                                       | LAB        |
| Radiology    | Schedule, worklist, reporting, sign-off                                                                     | RAD        |
| Pharmacy     | Rx queue, sale, IPD issue, returns, purchase, GRN, stock, alerts                                            | PHR        |
| Billing      | Billing counter, IP billing, deposits, refunds, cashier shift                                               | CORE       |
| Inventory    | Indents, issues, requisitions, PO, GRN, transfers, stock, audit, assets                                     | INV        |
| HR           | Employees, onboarding, roster planner, attendance, leave, exit                                              | HRM        |
| Payroll      | Structures, inputs, payroll runs, payslips, statutory files                                                 | PAY        |
| Finance      | Vouchers, payables, receivables, bank, ledgers, statements, GST                                             | FIN        |
| Records      | Completion, deficiencies, coding, file tracking, release requests, birth, death, mortuary and MLC registers | MRD        |
| Diet         | Diet orders, NBM board, production sheet, tray labels, deliveries                                           | DIET       |
| Facility     | Housekeeping board, checklists, linen, tickets, equipment, PM, AMC, biomedical waste                        | FAC        |
| Quality      | Feedback, complaints, incidents, indicators, audits, documents                                              | QLT        |
| Front Office | Enquiries, counter tokens, attendant and visitor passes, help desk                                          | CORE       |
| CRM          | Call centre, leads, camps, campaigns, memberships, referral partners                                        | CRM        |
| Reports      | Every module's reports, export to Excel / PDF, scheduled e-mail                                             | All        |
| Settings     | Hospital, branches, departments, masters, users, roles, approval rules, templates, subscription             | CORE       |
| My Space     | My profile, roster, attendance, leave, payslips, password, 2FA                                              | CORE / HRM |

## SECTION 12

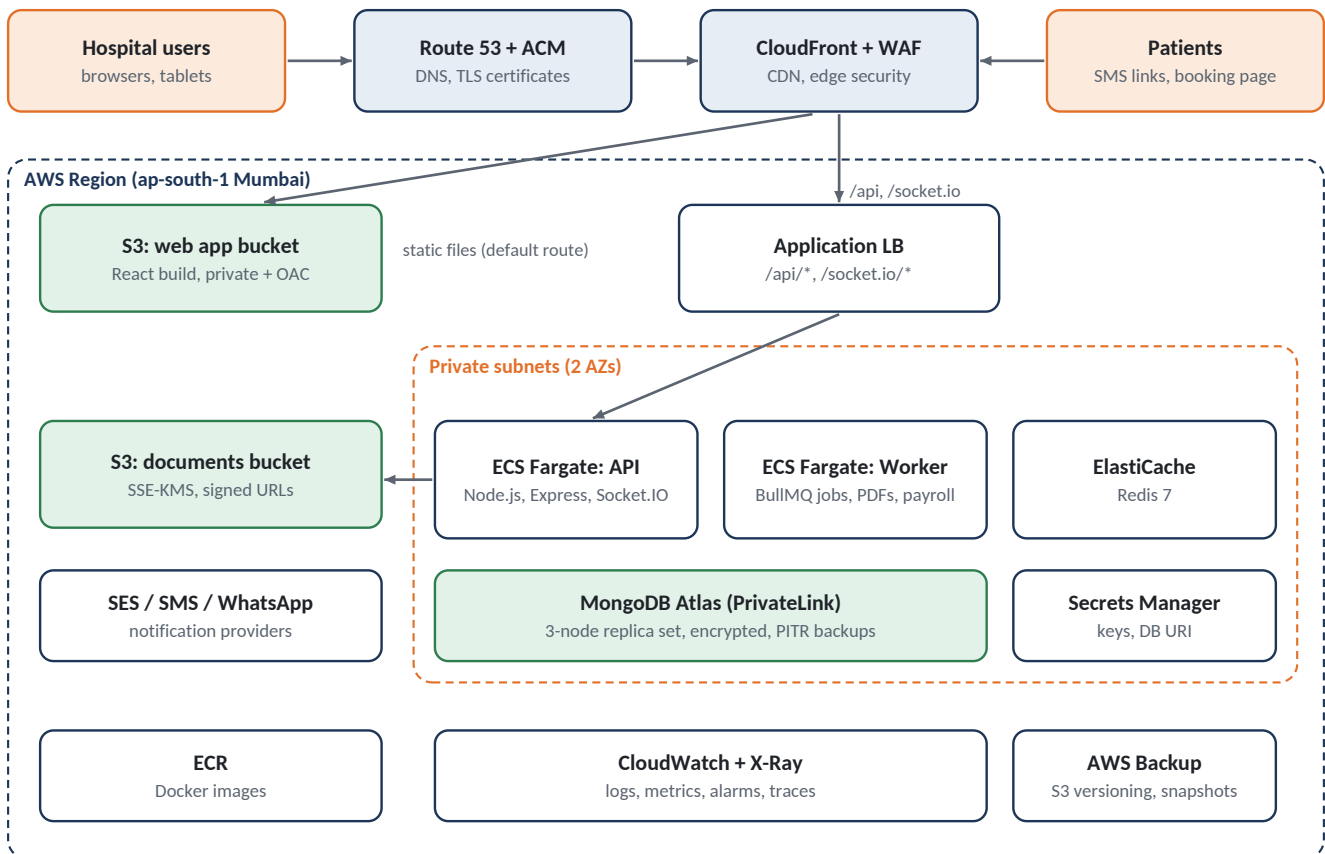
# Technical Architecture

## 12.1 Technology stack

| Layer            | Choice                                                 | Why                                                                                                                                       |
|------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Web app          | React 19, TypeScript, Vite                             | Fast builds, static output for S3, typed code shared with API                                                                             |
| UI kit           | Tailwind CSS v4 + Radix UI primitives + TanStack Table | Design tokens as CSS variables (light, dark, high contrast), accessible headless components, fast data tables for billing, stock, payroll |
| State and data   | Redux Toolkit + RTK Query                              | Caching, auto-refetch, typed API hooks                                                                                                    |
| Forms            | React Hook Form + Zod                                  | Same Zod schemas validate on client and server                                                                                            |
| Real-time        | Socket.IO client                                       | Bed board, queue screen, approvals, alerts                                                                                                |
| API              | Node.js 24 LTS, Express 5, TypeScript                  | Simple, widely known, large ecosystem                                                                                                     |
| Database         | MongoDB 8 (Atlas) + Mongoose 8                         | Flexible clinical documents, transactions on replica sets                                                                                 |
| Cache and queues | Redis 7 + BullMQ                                       | Sessions blacklist, rate limits, background jobs (PDFs, SMS, payroll, reports)                                                            |
| Documents        | Amazon S3 + pdfmake                                    | Generated PDFs and uploads in private bucket                                                                                              |
| Auth             | JWT (RS256) + refresh rotation, Argon2id, TOTP         | Stateless API, strong password hashing, 2FA                                                                                               |
| API docs         | OpenAPI 3.1 generated from Zod                         | Swagger UI at /api/docs, always in sync with code                                                                                         |
| Testing          | Vitest, Supertest, Playwright                          | Unit, API and end-to-end tests                                                                                                            |
| Hosting          | S3 + CloudFront (web), ECS Fargate (API, worker)       | Serverless containers, no servers to patch                                                                                                |
| CI/CD            | GitHub Actions                                         | Build, test, deploy on merge to main                                                                                                      |
| Monitoring       | CloudWatch, Sentry                                     | Logs, metrics, alarms, front-end errors                                                                                                   |

## 12.2 Deployment architecture on AWS

The React app is a set of static files in a private S3 bucket. CloudFront serves them worldwide over HTTPS and forwards /api and /socket.io traffic to the Node API behind an Application Load Balancer. Both web and API share one domain per tenant, so no CORS setup is needed.



| Component                  | Configuration (production)                                                                                                                                                               |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>S3 web bucket</b>       | Block all public access. CloudFront Origin Access Control. index.html cached 60 s, hashed assets cached 1 year                                                                           |
| <b>CloudFront</b>          | Wildcard certificate *.example.com in us-east-1. Behaviours: default to S3, /api/* and /socket.io/* to ALB with caching off. SPA fallback: 403/404 to /index.html. AWS WAF managed rules |
| <b>ECS API service</b>     | 2-6 Fargate tasks (1 vCPU, 2 GB), auto-scaling on CPU 60% and request count. Sticky sessions for Socket.IO plus Redis adapter                                                            |
| <b>ECS worker service</b>  | 1-4 tasks, scales on BullMQ queue depth                                                                                                                                                  |
| <b>MongoDB Atlas</b>       | M30 or higher, 3-node replica set across AZs in Mumbai, PrivateLink, encryption at rest, continuous backup with point-in-time restore                                                    |
| <b>Redis</b>               | ElastiCache, 2 nodes, Multi-AZ, encryption in transit                                                                                                                                    |
| <b>S3 documents bucket</b> | Versioning, SSE-KMS, lifecycle to Glacier after 1 year, access only via presigned URLs (5 minutes)                                                                                       |
| <b>Environments</b>        | dev, staging, production in separate AWS accounts                                                                                                                                        |

**Starter setup.** For a single small hospital or a pilot, run the API and worker with PM2 on one EC2 t3.medium behind Nginx, MongoDB Atlas M10 and the same S3 + CloudFront web hosting. The code is identical; only infrastructure changes. Expected cost is about USD 120-180 per month versus about USD 900-1,400 for the production layout above.

## 12.3 Multi-tenancy

- **Tenant resolution.** The sub-domain (e.g. citycare.example.com) or a custom domain maps to a tenant. The API reads the Host header, looks up the tenant in a cached registry, and puts it on the request context.
- **Data isolation.** Shared database by default. Every tenant-owned document has tenantId. A Mongoose plugin adds the filter to every query automatically and refuses queries that lack it.
- **Dedicated database.** Enterprise tenants can get their own database in the same or a separate cluster. The connection is chosen per request; code is unchanged.
- **Storage isolation.** S3 keys are prefixed with tenants/{tenantId}/. IAM policies on the API role restrict access to these buckets only.
- **Module gating.** Tenant document holds active modules and limits. Middleware rejects calls to unsubscribed modules with HTTP 402 and code MODULE\_NOT\_SUBSCRIBED.

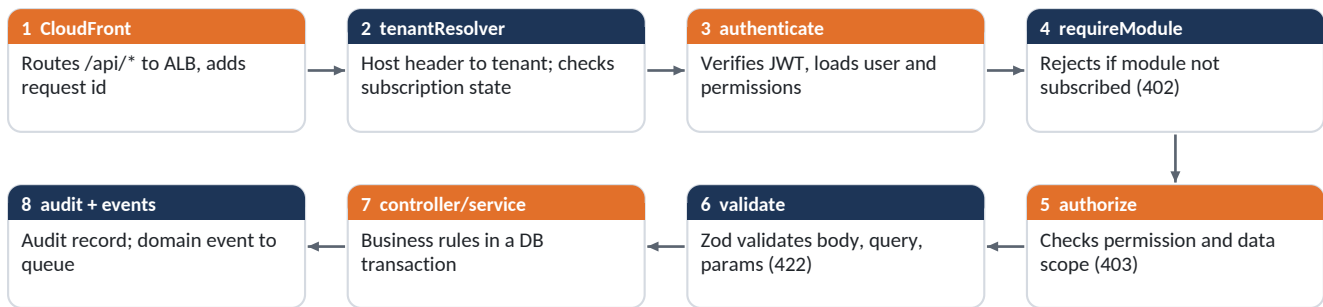
## 12.4 Backend structure: modular monolith

The API is one deployable application split into modules that mirror the subscription modules. Modules talk to each other only through service interfaces and domain events, never by reading each other's collections. This keeps Phase 1 simple to run and lets a module be split into its own service later if load demands it.

### Repository layout (pnpm monorepo)

```
apps/
  web/                React + Tailwind (hospital staff)    -> S3 + CloudFront
  portal/             Patient portal + public booking    -> S3 + CloudFront
  console/            React + Tailwind (platform owner)  -> S3 + CloudFront
  api/
    src/
      app.ts server.ts  Express app, HTTP + Socket.IO bootstrap
      worker.ts         BullMQ worker entry
      config/           env loading and validation (Zod)
      core/             shared kernel (never imports modules)
        tenancy/ auth/ rbac/ approvals/ audit/ sequences/ db/ events/
        files/ notify/ print/ realtime/ jobs/ security/ errors/ http/
      modules/
        platform/ setup/ users/ patients/ billing/ insurance/
        opd/ ipd/ nursing/ lab/ radiology/ pharmacy/ inventory/
        hr/ payroll/ finance/ mrd/ diet/ facility/ quality/ crm/
        portal/ abdm/ reports/
      <module>/
        index.ts        public surface: router, service API, events
        <module>.routes.ts module gate + permission + schema per route
        controllers/    HTTP in/out only
        services/       business rules, transactions
        models/         Mongoose models and indexes
        events.ts jobs.ts domain events, background jobs
        print/ reports/ seed/ PDF templates, reports, tenant defaults
        __tests__/      unit, API, isolation, concurrency
packages/
  shared/             Zod schemas, enums, permission keys (web + api)
  ui/                 Tailwind tokens + components (PatientBanner, DataTable)
  i18n/               English and Hindi translations
  config/             tsconfig, eslint, prettier, tailwind presets
  infra/              Terraform (S3, CloudFront, ECS, ALB, Redis, IAM), Docker
  e2e/ perf/          Playwright journeys, k6 load tests
  .github/workflows/  CI/CD pipelines
```

## 12.5 Request lifecycle



## 12.6 Domain events

Cross-module work happens through events on a BullMQ queue, published after the database transaction commits (outbox pattern). This keeps modules independent and makes retries safe.

| Event                                   | Published by | Consumed by                                |
|-----------------------------------------|--------------|--------------------------------------------|
| patient.registered                      | Patients     | Notify (welcome SMS)                       |
| opd.visit.checkedIn                     | OPD          | Billing (consult fee), Queue screen        |
| order.created                           | OPD / IPD    | LAB, RAD, PHR worklists; Billing charges   |
| lab.result.released                     | LAB          | EMR timeline, Notify, Doctor alerts        |
| ipd.admitted / transferred / discharged | IPD          | Beds, Nursing, Billing, Notify             |
| pharmacy.issued                         | PHR          | Billing, Finance (COGS)                    |
| billing.payment.received                | Billing      | Finance journal, Notify (e-receipt)        |
| inventory.grn.approved                  | INV / PHR    | Finance payable, stock alerts              |
| hr.leave.approved                       | HRM          | Roster, OPD slot blocking, Attendance      |
| payroll.run.locked                      | PAY          | Finance journal, Payslip PDFs, Notify      |
| approval.decided                        | Approvals    | Owning module applies or discards change   |
| ipd.bedVacated                          | IPD          | FAC (bed cleaning task), DIET (stop meals) |
| facility.bedCleaned                     | FAC          | IPD bed board (bed AVAILABLE)              |
| diet.orderChanged                       | DIET         | Kitchen production sheet, tray labels      |
| patient.died                            | IPD          | MRD death register, mortuary, Billing hold |
| quality.incidentReported                | QLT / any    | Quality Manager, HOD notifications         |

## SECTION 13

## Database Design (MongoDB)

### 13.1 Design rules

- Every tenant-owned collection has `tenantId` and `branchId` where relevant, and every index starts with `tenantId`.
- Common fields on every document: `createdAt`, `createdBy`, `updatedAt`, `updatedBy`, `isDeleted` (soft delete), `version` (optimistic locking).
- Money is stored as integers in paise (Decimal128 for rates with more precision), never as floating point.
- References use `ObjectId`; small, stable data needed for printing is copied (snapshot), e.g. patient name and UHID on a bill, so old bills print correctly after a name change.
- Writes that touch several documents (bill + payment + stock) use multi-document transactions, which need a replica set (Atlas provides this).
- Human-readable numbers (UHID, bill no.) come from an atomic counter collection per tenant, branch, series and year.
- Clinical records are never hard-deleted. Corrections create a new version with the previous one kept.

### 13.2 Collections by module

| Module   | Collection                                | Purpose and key fields                                                                                        |
|----------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Platform | tenants                                   | name, subdomain, status, plan, modules[{code, status, validTill}], limits{users, beds, storageGb}, dbStrategy |
| Platform | subscriptionInvoices                      | tenantId, period, lines, amount, status, gatewayRef                                                           |
| CORE     | branches, departments                     | code, name, type, parentId, hodId, costCentre, status                                                         |
| CORE     | users                                     | employeeId, username, mobile, passwordHash, roles[], branchIds[], mfa, lockedUntil, lastLoginAt               |
| CORE     | roles                                     | name, system, permissions[], dataScope                                                                        |
| CORE     | approvalRules, approvalRequests           | action, thresholds, levels; request: maker, payload, before, status, decisions[]                              |
| CORE     | auditLogs                                 | actor, action, entity, entityId, diff, ip, at (time-series, TTL none)                                         |
| CORE     | counters                                  | _id = tenant:branch:series:year, seq                                                                          |
| CORE     | patients                                  | uhid, name{}, gender, dob, mobile, ids[], allergies[], flags, category, mergedInto                            |
| CORE     | services, priceLists, packages, taxCodes  | Tariff and pricing masters                                                                                    |
| CORE     | bills, payments, deposits, refunds        | billNo, patient snapshot, visit / admission ref, lines[], totals, status; payment modes[]                     |
| CORE     | cashierShifts                             | user, counter, openedAt, opening, closing, variance                                                           |
| CORE     | employees                                 | empCode, name, department, designation, doctorProfile{}, contacts (HR fields added by HRM)                    |
| CORE     | files, notifications                      | S3 key, mime, size, owner entity; message channel, template, status                                           |
| OPD      | doctorSchedules, appointments, visits     | slots, visit status timeline, vitals, notes, diagnoses[], prescription{items[]}                               |
| OPD      | orders                                    | type (lab, rad, proc, drug), items[], source visit / admission, status                                        |
| IPD      | wards, rooms, beds                        | bed category, status, currentAdmissionId, statusChangedAt                                                     |
| IPD      | admissions, transfers, dischargeSummaries | ipNo, bed history[], consultants, payer, status                                                               |

| Module    | Collection                                                                       | Purpose and key fields                                                                  |
|-----------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| IPD       | preAuths, claims, settlements                                                    | admissionId, payerId, policyNo, requested, approved[], status, timers, documents[], utr |
| NUR       | vitals, marEntries, nursingNotes, ioCharts, handovers                            | admissionId, at, by, values                                                             |
| LAB       | labTests, labSamples, labResults                                                 | parameters, ranges; barcode, status history; values[], flags, validatedBy               |
| RAD       | radStudies, radReports                                                           | modality, schedule, accessionNo; template, signedBy                                     |
| PHR / INV | items, stores, stockBatches, stockLedger                                         | batchNo, expiry, mrp, cost, qty; every movement as ledger row                           |
| PHR / INV | vendors, purchaseOrders, grns, indents, transfers                                | procurement documents                                                                   |
| HRM       | shifts, rosters, attendance, leaveTypes, leaveRequests, leaveBalances            | roster cells per day; punches; accruals                                                 |
| PAY       | salaryStructures, employeeSalaries, payrollRuns, payslips, loans                 | components with formulas; run status; lines per employee                                |
| FIN       | ledgers, vouchers, journalEntries, bankStatements, periods                       | double-entry lines[], costCentre, source event ref                                      |
| FIN       | pettyCash, expenseClaims, budgets                                                | imprest, vouchers; claims with receipts; budget per cost centre                         |
| HRM       | requisitions, candidates, trainings, appraisals                                  | recruitment pipeline, certifications, reviews                                           |
| CORE      | enquiries, visitorPasses, circulars                                              | front office and notices                                                                |
| MRD       | recordFiles, deficiencies, releaseRequests, births, deaths, mlcCases, mortuary   | completion status, coding, legal registers                                              |
| DIET      | dietTypes, dietOrders, mealRuns, mealDeliveries                                  | per admission, NBM windows, counts                                                      |
| FAC       | hkTasks, linenTransactions, tickets, equipment, pmSchedules, bmwEntries          | TAT, downtime, waste weights                                                            |
| QLT       | feedback, complaints, incidents, indicators, audits, documents, haiEvents        | CAPA, monthly indicator values, infection events                                        |
| CRM       | leads, calls, camps, campaigns, memberships, wallets, referralPartners           | stages, consent, balances, partner statements                                           |
| OPD / LAB | specialtyForms, treatmentPlans, certificates, homeCollections, collectionCentres | templates, sessions, visits, rate lists                                                 |
| CORE      | tokens, abhaLinks                                                                | counter queues; ABHA to UHID links and consents                                         |

## 13.3 Critical indexes

### Index plan (all indexes lead with tenantId)

|                   |                                                                                     |                                                       |
|-------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------|
| patients:         | { tenantId: 1, uhid: 1 }                                                            | unique                                                |
|                   | { tenantId: 1, mobile: 1 }                                                          |                                                       |
|                   | { tenantId: 1, "name.searchKey": 1 }                                                | prefix search                                         |
|                   | { tenantId: 1, "ids.type": 1, "ids.number": 1 }                                     | duplicate check                                       |
| appointments:     | { tenantId: 1, doctorId: 1, slotStart: 1 }                                          | unique, partial(status!=CANCELLED)                    |
| visits:           | { tenantId: 1, branchId: 1, visitDate: -1, status: 1 }                              |                                                       |
| beds:             | { tenantId: 1, wardId: 1, status: 1 }                                               |                                                       |
| admissions:       | { tenantId: 1, ipNo: 1 }                                                            | unique ; { tenantId: 1, status: 1, wardId: 1 }        |
| bills:            | { tenantId: 1, billNo: 1 }                                                          | unique ; { tenantId: 1, patientId: 1, createdAt: -1 } |
| stockBatches:     | { tenantId: 1, storeId: 1, itemId: 1, expiry: 1 }                                   | FEFO lookup                                           |
| labSamples:       | { tenantId: 1, barcode: 1 }                                                         | unique ; { tenantId: 1, status: 1, deptId: 1 }        |
| attendance:       | { tenantId: 1, employeeId: 1, date: 1 }                                             | unique                                                |
| journalEntries:   | { tenantId: 1, entityId: 1, date: 1 } ; { tenantId: 1, "lines.ledgerId": 1 }        |                                                       |
| auditLogs:        | { tenantId: 1, entity: 1, entityId: 1, at: -1 } ; { tenantId: 1, actor: 1, at: -1 } |                                                       |
| approvalRequests: | { tenantId: 1, status: 1, "pendingWith.role": 1 }                                   |                                                       |

## 13.4 Sample documents

### tenants and bills

```
// tenants
{
  "_id": "6650f0...", "name": "CityCare Hospital", "subdomain": "citycare",
  "status": "ACTIVE", "plan": "HOSPITAL",
  "modules": [
    { "code": "CORE", "status": "ACTIVE", "validTill": "2027-03-31" },
    { "code": "OPD", "status": "ACTIVE", "validTill": "2027-03-31" },
    { "code": "PHR", "status": "ACTIVE", "validTill": "2027-03-31" },
    { "code": "PAY", "status": "DISABLED" }
  ],
  "limits": { "users": 100, "beds": 120, "branches": 3, "storageGb": 250 },
  "dbStrategy": "SHARED"
}

// bills (amounts in paise)
{
  "tenantId": "6650f0...", "branchId": "6650f1...", "billNo": "OP/26-27/000154",
  "type": "OPD", "status": "PAID",
  "patient": { "id": "6651aa...", "uhid": "CC000123", "name": "Ravi Kumar" },
  "visitId": "6652bb...",
  "lines": [
    { "serviceId": "...", "name": "Consultation - Cardiology", "qty": 1,
      "rate": 80000, "discount": 0, "taxRate": 0, "amount": 80000 },
    { "serviceId": "...", "name": "Lipid Profile", "qty": 1,
      "rate": 65000, "discount": 6500, "taxRate": 0, "amount": 58500 }
  ],
  "totals": { "gross": 145000, "discount": 6500, "tax": 0, "net": 138500, "paid": 138500 },
  "payments": [ { "mode": "UPI", "amount": 138500, "ref": "UPI3487..." } ],
  "createdBy": "6640cc...", "createdAt": "2026-10-09T05:42:11Z", "version": 3
}
```

## SECTION 14

# Reference Source Code

The code below is the reference implementation of the parts every module depends on: configuration, tenancy, authentication, subscription gating, permissions, maker-checker, numbering, and the transaction-heavy flows (bed allocation, pharmacy dispensing, billing, payroll). Developers build each module on these building blocks. All code is TypeScript.

## 14.1 Backend foundation

### apps/api/src/config/env.ts

```
import { z } from 'zod';

const Env = z.object({
  NODE_ENV: z.enum(['development', 'test', 'production']).default('development'),
  PORT: z.coerce.number().default(4000),
  MONGO_URI: z.string().url(),
  REDIS_URL: z.string().url(),
  JWT_PRIVATE_KEY: z.string().min(100), // RS256 PEM, from Secrets Manager
  JWT_PUBLIC_KEY: z.string().min(100),
  AWS_REGION: z.string().default('ap-south-1'),
  S3_DOCS_BUCKET: z.string(),
  ROOT_DOMAIN: z.string().default('example.com'),
  SMS_PROVIDER: z.enum(['msg91', 'twilio', 'console']).default('console'),
});

export const env = Env.parse(process.env); // crash on boot if config is wrong
```

### apps/api/src/core/tenancy/context.ts

```
import { AsyncLocalStorage } from 'node:async_hooks';
import type { ClientSession } from 'mongoose';

export interface RequestContext {
  tenantId: string;
  branchId?: string;
  userId?: string;
  permissions: Set<string>;
  modules: Set<string>;
  requestId: string;
  session?: ClientSession; // set inside withTransaction()
}

export const ctx = new AsyncLocalStorage<RequestContext>();

export function current(): RequestContext {
  const c = ctx.getStore();
  if (!c) throw new Error('No request context. Call inside a request or runAs().');
  return c;
}
```

### apps/api/src/core/tenancy/tenant.plugin.ts

```
import { Schema } from 'mongoose';
import { current } from './context';

/** Adds tenantId to every document and forces it into every query. */
export function tenantPlugin(schema: Schema) {
  schema.add({ tenantId: { type: Schema.Types.ObjectId, required: true, index: true } });

  const scoped = ['find', 'findOne', 'countDocuments', 'findOneAndUpdate',
    'updateOne', 'updateMany', 'deleteOne', 'deleteMany'] as const;

  for (const op of scoped) {
    schema.pre(op, function () {
      const { tenantId } = current();
      const filter = this.getFilter();
      if (filter.tenantId && String(filter.tenantId) !== tenantId) {
        throw new Error('Cross-tenant query blocked');
      }
      this.where({ tenantId });
    });
  }
}
```

```

    });
  }
  schema.pre('aggregate', function () {
    this.pipeline().unshift({ $match: { tenantId: toObjectId(current().tenantId) } });
  });
  schema.pre('validate', function () {
    if (!this.get('tenantId')) this.set('tenantId', current().tenantId);
  });
}

```

#### apps/api/src/core/tenancy/tenantResolver.ts

```

import type { RequestHandler } from 'express';
import { randomUUID } from 'node:crypto';
import { ctx } from '../context';
import { tenantRegistry } from '../tenant.registry'; // Redis-cached lookup, 60 s TTL
import { AppError } from '../errors';

const READ_ONLY_OK = new Set(['GET', 'HEAD', 'OPTIONS']);

export const tenantResolver: RequestHandler = async (req, _res, next) => {
  const host = req.hostname.toLowerCase();
  const tenant = await tenantRegistry.byHost(host);
  if (!tenant) return next(new AppError(404, 'TENANT_NOT_FOUND', 'Unknown hospital'));

  if (tenant.status === 'SUSPENDED' && !req.path.startsWith('/api/v1/subscription')) {
    return next(new AppError(402, 'TENANT_SUSPENDED', 'Subscription suspended'));
  }
  if (tenant.status === 'READ_ONLY' && !READ_ONLY_OK.has(req.method)) {
    return next(new AppError(402, 'TENANT_READ_ONLY', 'Subscription is read-only'));
  }

  const modules = new Set(tenant.modules
    .filter((m) => m.status === 'ACTIVE' && (!m.validTill || m.validTill > new Date()))
    .map((m) => m.code));

  ctx.run({ tenantId: String(tenant._id), modules, permissions: new Set(),
    requestId: String(req.headers['x-request-id'] ?? randomUUID()) }, () => next());
};

```

#### apps/api/src/core/auth/authenticate.ts

```

import type { RequestHandler } from 'express';
import jwt from 'jsonwebtoken';
import { env } from '../../config/env';
import { current } from '../tenancy/context';
import { AppError } from '../errors';
import { permissionCache } from '../rbac/permission.cache';
import { tokenBlacklist } from '../token.blacklist';

interface AccessClaims { sub: string; tid: string; bid?: string; jti: string; }

export const authenticate: RequestHandler = async (req, _res, next) => {
  const token = req.cookies?.access_token
    ?? req.headers.authorization?.replace(/^Bearer /, '');
  if (!token) return next(new AppError(401, 'UNAUTHENTICATED', 'Login required'));
  try {
    const claims = jwt.verify(token, env.JWT_PUBLIC_KEY,
      { algorithms: ['RS256'] }) as AccessClaims;
    const c = current();
    if (claims.tid !== c.tenantId) throw new Error('tenant mismatch');
    if (await tokenBlacklist.has(claims.jti)) throw new Error('revoked');
    c.userId = claims.sub;
    c.branchId = String(req.headers['x-branch-id'] ?? claims.bid ?? '');
    c.permissions = await permissionCache.forUser(c.tenantId, claims.sub);
    next();
  } catch {
    next(new AppError(401, 'TOKEN_INVALID', 'Session expired, please log in again'));
  }
};

```

#### apps/api/src/core/rbac/guards.ts

```

import type { RequestHandler } from 'express';
import { current } from '../tenancy/context';
import { AppError } from '../errors';

export type ModuleCode = 'CORE' | 'OPD' | 'IPD' | 'NUR' | 'LAB' | 'RAD' | 'PHR'

```

```

    | 'INV' | 'HRM' | 'PAY' | 'FIN' | 'MRD' | 'DIET' | 'FAC' | 'QLT' | 'CRM';

/** 402 if the hospital has not subscribed to the module. */
export const requireModule = (code: ModuleCode): RequestHandler => (_req, _res, next) =>
  current().modules.has(code)
    ? next()
    : next(new AppError(402, 'MODULE_NOT_SUBSCRIBED',
      `The ${code} module is not part of your subscription`));

/** 403 unless the user has the permission, e.g. 'billing:invoice:create'. */
export const authorize = (permission: string): RequestHandler => (_req, _res, next) => {
  const { permissions } = current();
  const [mod, res] = permission.split(':');
  const ok = permissions.has(permission)
    || permissions.has(`${mod}:${res}:*`) || permissions.has(`${mod}:*`);
  return ok ? next() : next(new AppError(403, 'FORBIDDEN', `Missing ${permission}`));
};

```

#### apps/api/src/app.ts

```

import express from 'express';
import helmet from 'helmet';
import cookieParser from 'cookie-parser';
import { pinoHttp } from 'pino-http';
import { rateLimiter } from './core/security/rateLimiter';
import { tenantResolver } from './core/tenancy/tenantResolver';
import { authenticate } from './core/auth/authenticate';
import { errorHandler } from './core/errors';
import { authRoutes } from './core/auth/auth.routes';
import { mountModules } from './modules';
import { openApiRouter } from './core/openapi';

export function createApp() {
  const app = express();
  app.set('trust proxy', 2); // CloudFront + ALB
  app.use(helmet());
  app.use(express.json({ limit: '1mb' }));
  app.use(cookieParser());
  app.use(pinoHttp({ redact: ['req.headers.authorization', 'req.headers.cookie'] }));
  app.get('/api/health', (_req, res) => res.json({ ok: true }));

  app.use('/api', rateLimiter, tenantResolver);
  app.use('/api/v1/auth', authRoutes); // login, refresh, logout, 2FA
  app.use('/api/docs', openApiRouter);
  app.use('/api/v1', authenticate);
  mountModules(app); // each module mounts its own router
  app.use(errorHandler); // { error: { code, message, details } }
  return app;
}

```

#### apps/api/src/modules/billing/billing.routes.ts

```

import { Router } from 'express';
import { requireModule, authorize } from '../../core/rbac/guards';
import { validate } from '../../core/http/validate';
import * as c from './billing.controller';
import * as s from './billing.schemas';

export const billingRoutes = Router();
billingRoutes.use(requireModule('CORE'));

billingRoutes.get('/bills', authorize('billing:bill:read'),
  validate({ query: s.ListBillsQuery }), c.listBills);
billingRoutes.post('/bills', authorize('billing:bill:create'),
  validate({ body: s.CreateBillBody }), c.createBill);
billingRoutes.post('/bills/:id/payments', authorize('billing:payment:create'),
  validate({ params: s.IdParam, body: s.PaymentBody }), c.addPayment);
billingRoutes.post('/bills/:id/discount', authorize('billing:discount:request'),
  validate({ params: s.IdParam, body: s.DiscountBody }), c.requestDiscount);
billingRoutes.post('/bills/:id/cancel', authorize('billing:bill:cancel'),
  validate({ params: s.IdParam, body: s.ReasonBody }), c.requestCancel);
billingRoutes.get('/bills/:id/pdf', authorize('billing:bill:print'), c.billPdf);

```

## 14.2 Numbering, transactions and approvals

### apps/api/src/core/sequences/sequence.service.ts

```
import mongoose from 'mongoose';
import { current } from '../tenancy/context';

const Counter = mongoose.model('Counter',
  new mongoose.Schema({ _id: String, seq: Number }), 'counters');

/** Atomic, gap-free per series. e.g. next('BILL_OP', 'OP/{FY}/{SEQ:6}') -> OP/26-27/000154 */
export async function nextNumber(series: string, pattern: string, at = new Date()) {
  const { tenantId, branchId, session } = current();
  const fy = financialYear(at); // '26-27'
  const key = `${tenantId}:${branchId}:${series}:${fy}`;
  const doc = await Counter.findOneAndUpdate(
    { _id: key }, { $inc: { seq: 1 } }, { upsert: true, new: true, session },
  );
  return pattern
    .replace('{FY}', fy)
    .replace(/\{SEQ:(\d+)\}/, (_, w) => String(doc!.seq).padStart(Number(w), '0'));
}

function financialYear(d: Date) {
  const y = d.getMonth() >= 3 ? d.getFullYear() : d.getFullYear() - 1; // April start
  return `${String(y).slice(2)}-${String(y + 1).slice(2)}`;
}
```

### apps/api/src/core/db/withTransaction.ts

```
import mongoose from 'mongoose';
import { ctx, current } from '../tenancy/context';
import { outbox } from '../events/outbox';

/** Runs fn in a MongoDB transaction; events are published only after commit. */
export async function withTransaction<T>(fn: () => Promise<T>): Promise<T> {
  const session = await mongoose.startSession();
  try {
    let result!: T;
    await session.withTransaction(async () => {
      result = await ctx.run({ ...current(), session }, fn);
    }, { readConcern: { level: 'snapshot' }, writeConcern: { w: 'majority' } });
    await outbox.flush(session.id); // publish queued domain events
    return result;
  } finally {
    await session.endSession();
  }
}
```

### apps/api/src/core/approvals/approval.service.ts

```
import { ApprovalRule, ApprovalRequest } from './approval.models';
import { current } from '../tenancy/context';
import { AppError } from '../errors';
import { events } from '../events/bus';

/** Returns null if no approval needed, otherwise creates a pending request. */
export async function requireApproval(input: {
  action: string; // e.g. 'billing.discount'
  entity: string; entityId: string;
  amount?: number; percent?: number;
  before: unknown; after: unknown; reason: string;
}) {
  const rule = await ApprovalRule.findOne({ action: input.action, active: true });
  if (!rule) return null;
  const levels = rule.levels.filter((l) =>
    (l.minAmount == null || (input.amount ?? 0) > l.minAmount) &&
    (l.minPercent == null || (input.percent ?? 0) > l.minPercent));
  if (levels.length === 0) return null;

  const req = await ApprovalRequest.create({
    ...input, maker: current().userId, status: 'PENDING', levelIndex: 0,
    levels: levels.map((l) => ({ role: l.role })),
    expiresAt: new Date(Date.now() + rule.expiryHours * 3600_000),
  });
  await events.publish('approval.requested', { id: req.id, role: levels[0].role });
  return req;
}
```

```

export async function decide(id: string, decision: 'APPROVE' | 'REJECT', comment?: string) {
  const { userId, permissions } = current();
  const req = await ApprovalRequest.findById(id);
  if (!req || req.status !== 'PENDING') throw new AppError(409, 'NOT_PENDING', 'Not pending');
  if (req.expiresAt < new Date()) throw new AppError(409, 'EXPIRED', 'Request expired');
  if (String(req.maker) === userId) {
    throw new AppError(403, 'MAKER_CANNOT_CHECK', 'You cannot approve your own request');
  }
  const level = req.levels[req.levelIndex];
  if (!permissions.has(`approvals:${level.role}:decide`)) {
    throw new AppError(403, 'FORBIDDEN', 'Not an approver for this level');
  }
  if (decision === 'REJECT' && !comment) {
    throw new AppError(422, 'REASON_REQUIRED', 'Reason required');
  }

  level.decidedBy = userId; level.decision = decision; level.at = new Date();
  level.comment = comment;
  if (decision === 'REJECT') req.status = 'REJECTED';
  else if (req.levelIndex + 1 < req.levels.length) req.levelIndex += 1;
  else req.status = 'APPROVED';
  await req.save();
  // The owning module listens and applies the change (or notifies the maker).
  await events.publish('approval.decided', { id: req.id, action: req.action,
    status: req.status, entityId: req.entityId });
  return req;
}

```

## 14.3 Patient, bed and pharmacy services

### apps/api/src/modules/patients/models/patient.model.ts

```

import { Schema, model } from 'mongoose';
import { tenantPlugin } from '../../../../core/tenancy/tenant.plugin';
import { auditPlugin } from '../../../../core/audit/audit.plugin';

const PatientSchema = new Schema({
  uhid: { type: String, required: true },
  name: {
    title: String, first: { type: String, required: true }, middle: String, last: String,
    searchKey: { type: String, index: true }, // lower-case 'first last' for prefix search
  },
  gender: { type: String, enum: ['M', 'F', 'O'], required: true },
  dob: { type: Date, required: true },
  dobEstimated: { type: Boolean, default: false },
  mobile: { type: String, required: true },
  email: String,
  address: { line1: String, city: String, state: String, pin: String },
  ids: [{ type: { type: String }, number: String, fileId: Schema.Types.ObjectId }],
  bloodGroup: String,
  allergies: [{ substance: String, reaction: String, severity: String }],
  flags: { vip: Boolean, mlc: Boolean },
  category: { type: String, default: 'GENERAL' },
  pricelistId: Schema.Types.ObjectId,
  emergencyContact: { name: String, relation: String, mobile: String },
  photoFileId: Schema.Types.ObjectId,
  mergedInto: Schema.Types.ObjectId,
}, { timestamps: true, optimisticConcurrency: true });

PatientSchema.plugin(tenantPlugin);
PatientSchema.plugin(auditPlugin, { entity: 'patient' });
PatientSchema.index({ tenantId: 1, uhid: 1 }, { unique: true });
PatientSchema.index({ tenantId: 1, mobile: 1 });
PatientSchema.pre('save', function () {
  this.name.searchKey = `${this.name.first} ${this.name.last ?? ''}`.trim().toLowerCase();
});

export const Patient = model('Patient', PatientSchema);

```

### apps/api/src/modules/patients/patient.service.ts (excerpt)

```

export async function registerPatient(input: RegisterPatientInput) {
  const dupes = await findLikelyDuplicates(input); // mobile + name, or ID number
  if (dupes.length && !input.confirmNotDuplicate) {
    throw new AppError(409, 'POSSIBLE_DUPLICATE', 'Similar patients found', { dupes });
  }
}

```

```

return withTransaction(async () => {
  const uhid = await nextNumber('UHID', 'CC{SEQ:7}');
  const [patient] = await Patient.create([ { ...input, uhid }, { session: current().session } ]);
  outbox.add('patient.registered', { patientId: patient.id, uhid });
  return patient;
});
}

```

#### apps/api/src/modules/ipd/bed.service.ts (excerpt)

```

/** Allocates a bed atomically: only succeeds if the bed is still AVAILABLE or RESERVED for us. */
export async function allocateBed(bedId: string, admissionId: string, reservationId?: string) {
  const bed = await Bed.findOneAndUpdate(
    {
      _id: bedId,
      $or: [{ status: 'AVAILABLE' },
        { status: 'RESERVED', reservationId: reservationId ?? '__none__' }],
    },
    { $set: { status: 'OCCUPIED', currentAdmissionId: admissionId,
      statusChangedAt: new Date(), reservationId: null } },
    { new: true, session: current().session },
  );
  if (!bed) throw new AppError(409, 'BED_NOT_AVAILABLE', 'Bed was just taken, pick another');
  outbox.add('bed.statusChanged', { bedId, wardId: bed.wardId, status: bed.status });
  return bed;
}

// Worker: pushes bed changes to every open bed board of the tenant in < 1 s
events.on('bed.statusChanged', async (e, meta) => {
  io.to(`tenant:${meta.tenantId}:beds`).emit('bed:update', e);
});

```

#### apps/api/src/modules/pharmacy/dispense.service.ts

```

/** Dispenses items using First-Expiry-First-Out across batches, in one transaction. */
export async function dispense(storeId: string, lines: { itemId: string; qty: number }[],
  ref: { type: 'OPD_SALE' | 'IPD_ISSUE'; id: string }) {
  return withTransaction(async () => {
    const { session } = current();
    const picked: PickedBatch[] = [];
    for (const line of lines) {
      let remaining = line.qty;
      const batches = await StockBatch.find(
        { storeId, itemId: line.itemId, qty: { $gt: 0 }, expiry: { $gt: new Date() },
      }).sort({ expiry: 1 }).session(session!);

      for (const b of batches) {
        if (remaining === 0) break;
        const take = Math.min(b.qty, remaining);
        // conditional decrement protects against a concurrent sale of the same batch
        const res = await StockBatch.updateOne(
          { _id: b._id, qty: { $gte: take } }, { $inc: { qty: -take } }, { session });
        if (res.modifiedCount !== 1) throw new AppError(409, 'STOCK_CHANGED', 'Retry');
        await StockLedger.create([ { batchId: b._id, itemId: line.itemId, storeId,
          qty: -take, refType: ref.type, refId: ref.id }, { session } ]);
        picked.push({ itemId: line.itemId, batchId: b.id, batchNo: b.batchNo,
          expiry: b.expiry, qty: take, mrp: b.mrp, cost: b.cost });
        remaining -= take;
      }
      if (remaining > 0) {
        throw new AppError(422, 'INSUFFICIENT_STOCK', 'Not enough stock',
          { itemId: line.itemId, short: remaining });
      }
    }
    outbox.add('pharmacy.issued', { ref, picked });
    return picked;
  });
}

```

## 14.4 Billing and payroll services

### apps/api/src/modules/billing/payment.service.ts

```
export async function addPayment(billId: string, input: PaymentInput) {
  return withTransaction(async () => {
    const { session, userId } = current();
    const shift = await CashierShift.findOne({ userId, status: 'OPEN' }).session(session!);
    if (!shift) throw new AppError(409, 'NO_OPEN_SHIFT', 'Open your cashier shift first');

    const bill = await Bill.findById(billId).session(session!);
    if (!bill || ![ 'FINAL', 'PARTLY_PAID' ].includes(bill.status)) {
      throw new AppError(409, 'BILL_NOT_PAYABLE', 'Bill is not open for payment');
    }
    const due = bill.totals.net - bill.totals.paid;
    const amount = input.modes.reduce((sum, m) => sum + m.amount, 0); // paise
    if (amount <= 0 || amount > due) throw new AppError(422, 'BAD_AMOUNT', `Due is ${due}`);

    const receiptNo = await nextNumber('RECEIPT', 'RC/{FY}/{SEQ:6}');
    const [payment] = await Payment.create([
      { billId, receiptNo, modes: input.modes, amount, shiftId: shift.id },
      { session }
    ]);

    bill.totals.paid += amount;
    bill.status = bill.totals.paid === bill.totals.net ? 'PAID' : 'PARTLY_PAID';
    await bill.save({ session });
    await CashierShift.updateOne(
      { _id: shift.id },
      { $inc: Object.fromEntries(input.modes.map((m) => [`collected.${m.mode}`, m.amount])) },
      { session }
    );

    outbox.add('billing.payment.received', { billId, paymentId: payment.id, amount });
    return { bill, payment };
  });
}
```

### apps/api/src/modules/payroll/compute.ts

```
/** Pure function: one employee's salary for one month. Rates come from config tables. */
export function computePayslip(emp: EmployeeSalary, att: MonthAttendance,
  inputs: VariableInputs, rules: StatutoryRules): PayslipLines {
  const ratio = att.payableDays / att.calendarDays;
  const earnings: Line[] = emp.components
    .filter((c) => c.kind === 'EARNING')
    .map((c) => ({ code: c.code, amount: round(evalFormula(c, emp) * (c.prorate ? ratio : 1)) }));

  earnings.push({ code: 'OT', amount: round(inputs.otHours * emp.otRatePerHour) });
  earnings.push({ code: 'NIGHT', amount: inputs.nightShifts * emp.nightAllowance });
  const gross = sum(earnings);

  const basicDa = amountOf(earnings, 'BASIC') + amountOf(earnings, 'DA');
  const pfWage = Math.min(basicDa, rules.pf.wageCeiling);
  const deductions: Line[] = [
    { code: 'PF', amount: emp.pfApplicable ? round(pfWage * rules.pf.employeeRate) : 0 },
    { code: 'ESI', amount: gross <= rules.esi.wageLimit
      ? Math.ceil(gross * rules.esi.employeeRate) : 0 },
    { code: 'PT', amount: professionalTax(gross, emp.state, att.month, rules.pt) },
    { code: 'TDS', amount: monthlyTds(emp, gross, rules.tds) },
    { code: 'LOAN', amount: inputs.loanEmi },
    ...inputs.otherDeductions,
  ];
  const employer = {
    pf: emp.pfApplicable ? round(pfWage * rules.pf.employerRate) : 0,
    esi: gross <= rules.esi.wageLimit ? Math.ceil(gross * rules.esi.employerRate) : 0,
  };
  return { earnings, deductions, gross, net: gross - sum(deductions), employer };
}
```

### apps/api/src/core/files/s3.service.ts

```
import { S3Client, PutObjectCommand, GetObjectCommand } from '@aws-sdk/client-s3';
import { getSignedUrl } from '@aws-sdk/s3-request-presigner';

const s3 = new S3Client({ region: env.AWS_REGION }); // IAM task role, no keys in code

export async function uploadUrl(kind: FileKind, mime: string, size: number) {
  if (!ALLOWED_MIME[kind].includes(mime) || size > MAX_BYTES[kind]) {
    throw new AppError(422, 'FILE_REJECTED', 'File type or size not allowed');
  }
  const { tenantId } = current();
```

```

const file = await FileRecord.create({ kind, mime, size, status: 'PENDING' });
const key = `tenants/${tenantId}/${kind}/${file.id}`;
file.key = key; await file.save();
const url = await getSignedUrl(s3, new PutObjectCommand({
  Bucket: env.S3_DOCS_BUCKET, Key: key, ContentType: mime,
  ServerSideEncryption: 'aws:kms',
}), { expiresIn: 300 });
return { fileId: file.id, url }; // browser PUTs the file directly
}

export async function downloadUrl(fileId: string) {
  const file = await FileRecord.findById(fileId); // tenant plugin scopes this
  if (!file) throw new AppError(404, 'NOT_FOUND', 'File not found');
  await audit('file.download', file.id);
  return getSignedUrl(s3, new GetObjectCommand({ Bucket: env.S3_DOCS_BUCKET, Key: file.key }),
    { expiresIn: 300 });
}

```

## 14.5 Frontend foundation

### apps/web/src/app/api.ts

```

import { createApi, fetchBaseQuery, type BaseQueryFn } from '@reduxjs/toolkit/query/react';
import { Mutex } from 'async-mutex';

const mutex = new Mutex();
const raw = fetchBaseQuery({
  baseUrl: '/api/v1', // same origin through CloudFront
  credentials: 'include', // httpOnly cookies
  prepareHeaders: (h, { getState }) => {
    const branch = (getState() as RootState).session.branchId;
    if (branch) h.set('x-branch-id', branch);
    return h;
  },
});

/** Refreshes the access token once on 401 and retries; logs out if refresh fails. */
const baseQuery: BaseQueryFn = async (args, api, extra) => {
  await mutex.waitForUnlock();
  let res = await raw(args, api, extra);
  if (res.error?.status === 401) {
    if (!mutex.isLocked()) {
      const release = await mutex.acquire();
      try {
        const r = await raw({ url: '/auth/refresh', method: 'POST' }, api, extra);
        if (r.error) api.dispatch(sessionExpired());
      } finally { release(); }
    } else {
      await mutex.waitForUnlock();
    }
    res = await raw(args, api, extra);
  }
  if (res.error?.status === 402) api.dispatch(showUpgradeDialog(res.error.data));
  return res;
};

export const api = createApi({
  reducerPath: 'api',
  baseQuery,
  tagTypes: ['Patient', 'Appointment', 'Bed', 'Bill', 'Approval', 'Stock', 'Payroll'],
  endpoints: () => ({}), // each module injects endpoints
});

```

### apps/web/src/modules/patients/patients.api.ts

```

export const patientsApi = api.injectEndpoints({
  endpoints: (b) => ({
    searchPatients: b.query<Page<PatientSummary>, { q: string; page?: number }>({
      query: (params) => ({ url: '/patients', params }),
      providesTags: ['Patient'],
    }),
    registerPatient: b.mutation<Patient, RegisterPatientInput>({
      query: (body) => ({ url: '/patients', method: 'POST', body }),
      invalidatesTags: ['Patient'],
    }),
  }),
});

```

```
});
export const { useSearchPatientsQuery, useRegisterPatientMutation } = patientsApi;
```

#### apps/web/src/app/access.tsx

```
import { useSession } from '@app/session';

/** Renders children only if the module is subscribed and the user holds the permission. */
export function Can({ module, perm, children, fallback = null }: CanProps) {
  const { modules, permissions } = useSession();
  if (module && !modules.includes(module)) return fallback;
  if (perm && !hasPermission(permissions, perm)) return fallback;
  return <>{children}</>;
}

// Menu is data; items disappear when the module or permission is missing.
export const MENU: MenuItem[] = [
  { label: 'Patients', icon: PeopleIcon, path: '/patients', module: 'CORE',
    perm: 'patients:patient:read' },
  { label: 'Appointments', icon: EventIcon, path: '/opd/appointments', module: 'OPD',
    perm: 'opd:appointment:read' },
  { label: 'Bed Board', icon: BedIcon, path: '/ipd/beds', module: 'IPD', perm: 'ipd:bed:read' },
  { label: 'Pharmacy', icon: PharmacyIcon, path: '/pharmacy', module: 'PHR',
    perm: 'pharmacy:sale:read' },
  { label: 'Payroll', icon: PaymentsIcon, path: '/payroll', module: 'PAY',
    perm: 'payroll:run:read' },
  // ...
];

// Route-level guard (React Router 6 data routes)
{ path: '/pharmacy/*', element: <ModuleRoute module="PHR"><PharmacyRoutes /></ModuleRoute> }
```

#### apps/web/src/modules/ipd/useLiveBeds.ts

```
import { useEffect } from 'react';
import { io } from 'socket.io-client';
import { api } from '@app/api';

/** Keeps the RTK Query bed cache in sync with server pushes. */
export function useLiveBeds(wardId?: string) {
  const dispatch = useAppDispatch();
  const query = useGetBedsQuery({ wardId });
  useEffect(() => {
    const socket = io({ path: '/socket.io', withCredentials: true });
    socket.emit('subscribe', { channel: 'beds' });
    socket.on('bed:update', (e: BedUpdate) => {
      dispatch(bedsApi.util.updateQueryData('getBeds', { wardId }, (draft) => {
        const bed = draft.items.find((b) => b.id === e.bedId);
        if (bed) Object.assign(bed, e);
      }));
    });
    socket.on('connect', () => query.refetch()); // resync after reconnect
    return () => { socket.disconnect(); };
  }, [wardId]);
  return query;
}
```

## SECTION 15

# REST API Documentation

## 15.1 Conventions

|                           |                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Base URL</b>           | https://{hospital}.example.com/api/v1 (same origin as the web app)                                                           |
| <b>Format</b>             | JSON, UTF-8. Dates in ISO 8601 UTC. Money in paise as integers                                                               |
| <b>Authentication</b>     | httpOnly cookie access_token (web) or Authorization: Bearer <token> (integrations). Refresh via POST /auth/refresh           |
| <b>Branch</b>             | Header x-branch-id selects the working branch for multi-branch users                                                         |
| <b>Idempotency</b>        | POST endpoints that create money or stock records accept Idempotency-Key header; repeats within 24 h return the first result |
| <b>Pagination</b>         | ?page=1&limit=25 (max 100). Response: { items, page, limit, total }                                                          |
| <b>Filtering, sorting</b> | ?status=PAID&from=2026-10-01&sort=-createdAt                                                                                 |
| <b>Concurrency</b>        | Updates send version; mismatch returns 409 VERSION_CONFLICT                                                                  |
| <b>Rate limits</b>        | Login: 10/min per IP. API: 600/min per user. Headers RateLimit-Remaining, RateLimit-Reset                                    |
| <b>Versioning</b>         | /api/v1. Breaking changes go to /api/v2 with 6 months overlap                                                                |
| <b>Live docs</b>          | OpenAPI 3.1 at /api/docs (Swagger UI) and /api/docs/openapi.json                                                             |

### Error format and status codes

HTTP/1.1 422 Unprocessable Entity

```
{
  "error": {
    "code": "VALIDATION_FAILED",
    "message": "Some fields are invalid",
    "details": [ { "path": "mobile", "message": "Must be 10 digits" } ],
    "requestId": "c1b7e0a2-4f0e-4c3e-9a51-0d8f3f8e2a11"
  }
}
```

| Status | Code examples                                           | Meaning                                      |
|--------|---------------------------------------------------------|----------------------------------------------|
| 400    | BAD_REQUEST                                             | Malformed request                            |
| 401    | UNAUTHENTICATED, TOKEN_INVALID                          | Not logged in or session expired             |
| 402    | MODULE_NOT_SUBSCRIBED, TENANT_READ_ONLY, LIMIT_REACHED  | Subscription does not allow this             |
| 403    | FORBIDDEN, MAKER_CANNOT_CHECK                           | Logged in but not permitted                  |
| 404    | NOT_FOUND, TENANT_NOT_FOUND                             | Record or hospital not found                 |
| 409    | VERSION_CONFLICT, BED_NOT_AVAILABLE, POSSIBLE_DUPLICATE | State conflict; reload and retry             |
| 422    | VALIDATION_FAILED, INSUFFICIENT_STOCK                   | Business or field validation                 |
| 202    | APPROVAL_PENDING                                        | Accepted, waiting for maker-checker approval |
| 429    | RATE_LIMITED                                            | Too many requests                            |
| 500    | INTERNAL                                                | Unexpected error; requestId helps support    |

## 15.2 Authentication

| Method | Path                  | Permission          | Description                                 |
|--------|-----------------------|---------------------|---------------------------------------------|
| POST   | /auth/login           | public              | Username or mobile + password               |
| POST   | /auth/2fa/verify      | public (temp token) | Verify OTP or TOTP                          |
| POST   | /auth/refresh         | refresh cookie      | Rotate tokens                               |
| POST   | /auth/logout          | any                 | Revoke current session                      |
| POST   | /auth/password/forgot | public              | Send reset OTP                              |
| POST   | /auth/password/reset  | public (OTP)        | Set new password                            |
| GET    | /auth/me              | any                 | User, roles, permissions, modules, branches |
| POST   | /auth/switch-user     | any (ward PIN)      | Quick switch on shared terminal             |

### Login with two-factor authentication

```

POST /api/v1/auth/login
{ "username": "anita.nurse", "password": "*****" }

200 OK (Set-Cookie: access_token=...; HttpOnly; Secure; SameSite=Strict)
{ "mfaRequired": true, "tempToken": "eyJhbGciOi...", "channels": ["SMS", "TOTP"] }

POST /api/v1/auth/2fa/verify
{ "tempToken": "eyJhbGciOi...", "code": "482913" }

200 OK
{
  "user": { "id": "6640cc...", "name": "Anita Sharma", "employeeCode": "EMP0042" },
  "roles": ["STAFF_NURSE"], "branches": [{ "id": "6650f1...", "name": "Main" }],
  "modules": ["CORE", "OPD", "IPD", "NUR", "LAB", "PHR"],
  "permissions": ["nursing:vitals:create", "nursing:mar:create", "ipd:bed:read", "..."]
}

```

## 15.3 Platform, settings and approvals

| Method | Path                          | Permission                 | Description                           |
|--------|-------------------------------|----------------------------|---------------------------------------|
| GET    | /subscription                 | settings:subscription:read | Plan, modules, limits, usage          |
| POST   | /subscription/module-requests | settings:subscription:edit | Request module add or removal         |
| CRUD   | /branches                     | settings:branch:*          | Branches                              |
| CRUD   | /departments                  | settings:department:*      | Create goes to approval               |
| CRUD   | /users                        | settings:user:*            | Users; privileged roles need approval |
| CRUD   | /roles                        | settings:role:*            | Custom roles and permissions          |
| CRUD   | /masters/{type}               | settings:master:*          | Services, price lists, tax codes...   |
| POST   | /masters/{type}/import        | settings:master:import     | Excel import with preview             |
| GET    | /approvals?status=PENDING     | approvals:inbox:read       | My approvals inbox                    |
| POST   | /approvals/{id}/decision      | approvals:{role}:decide    | Approve or reject                     |
| GET    | /audit                        | audit:log:read             | Search audit log                      |

| Method | Path                     | Permission        | Description               |
|--------|--------------------------|-------------------|---------------------------|
| POST   | /files/upload-url        | files:file:create | Presigned S3 upload URL   |
| GET    | /files/{id}/download-url | files:file:read   | Presigned S3 download URL |

#### Approve a request (moves to level 2)

```
POST /api/v1/approvals/66alf0.../decision
{ "decision": "APPROVE", "comment": "Senior citizen, approved as per policy" }
```

```
200 OK
{ "id": "66alf0...", "action": "billing.discount", "status": "PENDING",
  "levelIndex": 1, "levels": [
    { "role": "BILLING_MANAGER", "decision": "APPROVE", "decidedBy": "Rahul M",
      "at": "2026-10-09T06:10:00Z" },
    { "role": "SUPER_ADMIN" } ] }
```

## 15.4 Patients

| Method | Path                    | Permission              | Description                        |
|--------|-------------------------|-------------------------|------------------------------------|
| GET    | /patients?q=            | patients:patient:read   | Search by UHID, mobile, name       |
| POST   | /patients               | patients:patient:create | Register (duplicate check)         |
| GET    | /patients/{id}          | patients:patient:read   | Profile (access logged)            |
| PATCH  | /patients/{id}          | patients:patient:update | Update with version                |
| GET    | /patients/{id}/timeline | patients:patient:read   | Visits, admissions, reports, bills |
| POST   | /patients/merge         | patients:patient:merge  | Merge (approval)                   |

#### Register a patient

```
POST /api/v1/patients
{
  "name": { "title": "Mr", "first": "Ravi", "last": "Kumar" },
  "gender": "M", "dob": "1979-05-14", "mobile": "9876543210",
  "address": { "line1": "12 MG Road", "city": "Pune", "state": "MH", "pin": "411001" },
  "ids": [ { "type": "AADHAAR", "number": "XXXX-XXXX-4321" } ],
  "allergies": [ { "substance": "Penicillin", "reaction": "Rash", "severity": "MODERATE" } ],
  "category": "GENERAL"
}
```

```
201 Created
{ "id": "6651aa...", "uhid": "CC0000123", "name": { "first": "Ravi", "last": "Kumar" },
  "age": "47Y", "createdAt": "2026-10-09T05:30:00Z", "version": 0 }
```

```
409 Conflict (likely duplicate; resend with "confirmNotDuplicate": true)
{ "error": { "code": "POSSIBLE_DUPLICATE", "message": "Similar patients found",
  "details": { "dups": [ { "id": "65ff...", "uhid": "CC0000077", "score": 0.91 } ] } } }
```

## 15.5 OPD and appointments

| Method | Path                            | Permission             | Description               |
|--------|---------------------------------|------------------------|---------------------------|
| CRUD   | /opd/schedules                  | opd:schedule:*         | Doctor schedule templates |
| GET    | /opd/slots?doctorId=&date=      | opd:appointment:read   | Free slots                |
| POST   | /opd/appointments               | opd:appointment:create | Book                      |
| PATCH  | /opd/appointments/{id}          | opd:appointment:update | Reschedule / cancel       |
| POST   | /opd/appointments/{id}/check-in | opd:visit:create       | Check in, token           |

| Method | Path                              | Permission             | Description                  |
|--------|-----------------------------------|------------------------|------------------------------|
| GET    | /opd/queue?doctorId=              | opd:visit:read         | Live queue                   |
| POST   | /opd/visits/{id}/vitals           | opd:vitals:create      | Triage vitals                |
| PUT    | /opd/visits/{id}/consultation     | opd:consultation:write | Notes, diagnoses, Rx, orders |
| POST   | /opd/visits/{id}/complete         | opd:consultation:write | Close visit                  |
| GET    | /opd/visits/{id}/prescription.pdf | opd:visit:read         | Prescription PDF             |

### Book an appointment

```
POST /api/v1/opd/appointments
{ "patientId": "6651aa...", "doctorId": "6640dd...", "slotStart": "2026-10-10T04:30:00Z",
  "visitType": "NEW", "channel": "FRONT_DESK" }
```

```
201 Created
{ "id": "6653cc...", "status": "BOOKED", "slotStart": "2026-10-10T04:30:00Z",
  "slotEnd": "2026-10-10T04:40:00Z", "doctor": { "name": "Dr. Meera Iyer" },
  "fee": 80000, "notifications": ["SMS_CONFIRMATION_QUEUED"] }
```

### Save consultation

```
PUT /api/v1/opd/visits/6654dd.../consultation
{
  "complaints": [{ "text": "Chest pain on exertion", "duration": "2 weeks" }],
  "diagnoses": [{ "icd10": "I20.9", "text": "Angina pectoris, unspecified", "type": "PROVISIONAL" }],
  "prescription": { "items": [
    { "itemId": "66a0...", "drug": "Aspirin 75 mg tab", "dose": "1", "frequency": "OD",
      "durationDays": 30, "route": "ORAL", "instructions": "After lunch" } ] },
  "orders": [ { "type": "LAB", "serviceId": "65e1...", "name": "Lipid Profile" },
    { "type": "RAD", "serviceId": "65e2...", "name": "ECG" } ],
  "followUp": { "date": "2026-10-24" }, "version": 2
}
```

## 15.6 IPD and beds

| Method | Path                                    | Permission             | Description                  |
|--------|-----------------------------------------|------------------------|------------------------------|
| GET    | /ipd/beds?wardId=&status=               | ipd:bed:read           | Bed board (plus socket)      |
| POST   | /ipd/beds/{id}/reserve                  | ipd:bed:reserve        | Hold bed until a time        |
| PATCH  | /ipd/beds/{id}/status                   | ipd:bed:update         | Cleaning, maintenance, block |
| POST   | /ipd/admission-requests                 | ipd:admission:request  | From OPD                     |
| POST   | /ipd/admissions                         | ipd:admission:create   | Admit and allocate bed       |
| POST   | /ipd/admissions/{id}/transfer           | ipd:admission:transfer | Bed/ward transfer            |
| GET    | /ipd/admissions/{id}/running-bill       | billing:bill:read      | Live bill                    |
| POST   | /ipd/admissions/{id}/discharge/initiate | ipd:discharge:initiate | Start clearances             |
| PUT    | /ipd/admissions/{id}/discharge-summary  | ipd:summary:write      | Draft or sign                |
| POST   | /ipd/admissions/{id}/discharge/complete | ipd:discharge:complete | Exit; bed to CLEANING        |
| POST   | /ipd/admissions/{id}/preauths           | ipd:preauth:create     | Send pre-authorisation       |
| POST   | /ipd/preauths/{id}/replies              | ipd:preauth:update     | Approval, query or denial    |
| POST   | /ipd/preauths/{id}/enhancements         | ipd:preauth:create     | Request more                 |
| POST   | /ipd/preauths/{id}/final                | ipd:preauth:create     | Final bill for approval      |
| POST   | /ipd/claims                             | ipd:claim:create       | File claim with documents    |

| Method | Path                         | Permission       | Description               |
|--------|------------------------------|------------------|---------------------------|
| POST   | /ipd/claims/{id}/settlements | ipd:claim:settle | Match payment, deductions |
| GET    | /reports/ipd-indicators      | reports:ipd:read | Occupancy, stay, turnover |

### Admit a patient

POST /api/v1/ipd/admissions

Idempotency-Key: 5d9alc2e-...

```
{ "patientId": "6651aa...", "admissionRequestId": "66b1...", "bedId": "6620be...",
  "attendingDoctorId": "6640dd...", "departmentId": "6610ca...", "payer": { "type": "SELF" },
  "attendant": { "name": "Sunita Kumar", "relation": "Wife", "mobile": "9876500000" },
  "deposit": { "amount": 2000000, "modes": [{ "mode": "CARD", "amount": 2000000 } ] }
```

201 Created

```
{ "id": "66b2...", "ipNo": "IP/26-27/000871", "status": "ADMITTED",
  "bed": { "id": "6620be...", "label": "W2-204-B", "category": "SEMI_PRIVATE" },
  "depositReceiptNo": "RC/26-27/004512", "wristbandPdf": "/api/v1/files/66b3.../download-url" }
```

409 Conflict

```
{ "error": { "code": "BED_NOT_AVAILABLE", "message": "Bed was just taken, pick another" } }
```

## 15.7 Nursing

| Method | Path                             | Permission              | Description              |
|--------|----------------------------------|-------------------------|--------------------------|
| GET    | /nursing/wards/{id}/census       | nursing:ward:read       | Ward patients and tasks  |
| POST   | /nursing/admissions/{id}/vitals  | nursing:vitals:create   | Record vitals            |
| GET    | /nursing/admissions/{id}/mar     | nursing:mar:read        | Medication chart         |
| POST   | /nursing/mar/{doseId}/administer | nursing:mar:create      | Given / held / refused   |
| POST   | /nursing/admissions/{id}/notes   | nursing:note:create     | Nursing note             |
| POST   | /nursing/admissions/{id}/io      | nursing:io:create       | Intake / output          |
| POST   | /nursing/handovers               | nursing:handover:create | Shift handover           |
| POST   | /nursing/indents                 | nursing:indent:create   | Pharmacy or store indent |

## 15.8 Laboratory and radiology

| Method | Path                            | Permission          | Description                |
|--------|---------------------------------|---------------------|----------------------------|
| GET    | /lab/worklists?stage=           | lab:sample:read     | Collect / receive / result |
| POST   | /lab/samples/collect            | lab:sample:collect  | Collect, print barcodes    |
| POST   | /lab/samples/{barcode}/receive  | lab:sample:receive  | Accept or reject           |
| PUT    | /lab/tests/{id}/results         | lab:result:enter    | Enter results (maker)      |
| POST   | /lab/results/import             | lab:result:import   | Analyser file import       |
| POST   | /lab/tests/{id}/validate        | lab:result:validate | Validate (checker)         |
| GET    | /lab/reports/{orderId}.pdf      | lab:report:read     | Signed report              |
| GET    | /rad/schedule?modalityId=&date= | rad:study:read      | Modality schedule          |
| PATCH  | /rad/studies/{id}/status        | rad:study:update    | Arrived / performed        |
| PUT    | /rad/studies/{id}/report        | rad:report:write    | Draft report               |
| POST   | /rad/studies/{id}/sign          | rad:report:sign     | Radiologist sign-off       |

### Enter lab results with automatic flags

```

PUT /api/v1/lab/tests/66c4.../results
{ "values": [
  { "parameterId": "hb", "value": 9.1 },
  { "parameterId": "wbc", "value": 14200 },
  { "parameterId": "plt", "value": 38000 } ],
  "remarks": "Platelets verified on smear", "version": 1 }

200 OK
{ "status": "RESULTED", "flags": [
  { "parameterId": "hb", "flag": "LOW", "range": "13.0-17.0 g/dL" },
  { "parameterId": "wbc", "flag": "HIGH", "range": "4000-11000 /uL" },
  { "parameterId": "plt", "flag": "CRITICAL_LOW", "range": "150000-410000 /uL" } ],
  "criticalAlertSentTo": ["Dr. Meera Iyer"], "awaiting": "PATHOLOGIST_VALIDATION" }

```

## 15.9 Pharmacy and inventory

| Method | Path                              | Permission                | Description               |
|--------|-----------------------------------|---------------------------|---------------------------|
| CRUD   | /inventory/items                  | inventory:item:*          | Drug and item master      |
| GET    | /pharmacy/rx-queue                | pharmacy:sale:read        | Prescriptions to dispense |
| POST   | /pharmacy/sales                   | pharmacy:sale:create      | OPD / counter sale (FEFO) |
| POST   | /pharmacy/ipd-issues              | pharmacy:issue:create     | Issue to in-patient       |
| POST   | /pharmacy/returns                 | pharmacy:return:create    | Patient / vendor returns  |
| GET    | /inventory/stock?storeId=&itemId= | inventory:stock:read      | Stock by batch            |
| POST   | /inventory/purchase-orders        | inventory:po:create       | PO (approval by value)    |
| POST   | /inventory/grns                   | inventory:grn:create      | Goods receipt             |
| POST   | /inventory/indents                | inventory:indent:create   | Department indent         |
| POST   | /inventory/issues                 | inventory:issue:create    | Issue against indent      |
| POST   | /inventory/transfers              | inventory:transfer:create | Store to store            |
| POST   | /inventory/adjustments            | inventory:stock:adjust    | Adjustment (approval)     |
| GET    | /inventory/alerts                 | inventory:stock:read      | Reorder and expiry alerts |

### Pharmacy sale with FEFO batch selection

```

POST /api/v1/pharmacy/sales
Idempotency-Key: 0b7e...
{ "storeId": "66d0...", "patientId": "6651aa...", "prescriptionId": "6654dd...",
  "lines": [ { "itemId": "66a0...", "qty": 30 } ],
  "payment": { "modes": [{ "mode": "UPI", "amount": 4350 } ] } }

201 Created
{ "billNo": "PH/26-27/010233", "lines": [
  { "itemId": "66a0...", "name": "Aspirin 75 mg tab",
    "batches": [ { "batchNo": "AS2401", "expiry": "2027-01-31", "qty": 20 },
                  { "batchNo": "AS2407", "expiry": "2027-07-31", "qty": 10 } ],
    "mrp": 145, "amount": 4350, "gstRate": 12 } ],
  "totals": { "net": 4350, "paid": 4350 }, "receiptPdf": "/api/v1/files/66d1.../download-url" }

```

## 15.10 Billing

| Method | Path                            | Permission          | Description              |
|--------|---------------------------------|---------------------|--------------------------|
| POST   | /billing/shifts/open            | billing:shift:open  | Open cashier shift       |
| POST   | /billing/shifts/close           | billing:shift:close | Close with cash count    |
| GET    | /billing/patients/{id}/unbilled | billing:bill:read   | Pending charges          |
| POST   | /billing/bills                  | billing:bill:create | Create bill from charges |

| Method | Path                                      | Permission               | Description       |
|--------|-------------------------------------------|--------------------------|-------------------|
| POST   | /billing/bills/{id}/discount              | billing:discount:request | 202 if approval   |
| POST   | /billing/bills/{id}/payments              | billing:payment:create   | Collect payment   |
| POST   | /billing/deposits                         | billing:deposit:create   | IPD advance       |
| POST   | /billing/bills/{id}/cancel                | billing:bill:cancel      | Approval required |
| POST   | /billing/refunds                          | billing:refund:request   | Approval required |
| GET    | /billing/bills/{id}/pdf?format=A4 THERMAL | billing:bill:print       | Print / reprint   |

#### Discount that triggers maker-checker

```
POST /api/v1/billing/bills/66e1.../discount
{ "type": "PERCENT", "value": 15, "reason": "Senior citizen, financial hardship" }

202 Accepted
{ "code": "APPROVAL_PENDING", "approvalId": "66a1f0...",
  "pendingWith": "BILLING_MANAGER", "levels": 2, "expiresAt": "2026-10-11T06:00:00Z" }
```

## 15.11 HR, attendance and payroll

| Method | Path                             | Permission           | Description               |
|--------|----------------------------------|----------------------|---------------------------|
| CRUD   | /hr/employees                    | hr:employee:*        | Employee master           |
| CRUD   | /hr/shifts                       | hr:shift:*           | Shift master              |
| GET    | /hr/rosters?deptId=&month=       | hr:roster:read       | Roster grid               |
| PUT    | /hr/rosters                      | hr:roster:write      | Save draft roster         |
| POST   | /hr/rosters/{id}/publish         | hr:roster:publish    | Publish                   |
| POST   | /hr/roster-swaps                 | self                 | Request swap              |
| POST   | /hr/attendance/punches/import    | hr:attendance:import | Biometric file            |
| POST   | /hr/attendance/punch             | self                 | Web / mobile punch (geo)  |
| POST   | /hr/attendance/lock              | hr:attendance:lock   | Lock month                |
| POST   | /hr/leave-requests               | self                 | Apply leave               |
| POST   | /hr/leave-requests/{id}/decision | hr:leave:approve     | Approve / reject          |
| CRUD   | /payroll/structures              | payroll:structure:*  | Salary templates          |
| POST   | /payroll/runs                    | payroll:run:create   | Start draft run for month |
| GET    | /payroll/runs/{id}/variance      | payroll:run:read     | Compare with last month   |
| POST   | /payroll/runs/{id}/submit        | payroll:run:create   | Submit for approval       |
| GET    | /payroll/runs/{id}/bank-file     | payroll:run:export   | Bank transfer file        |
| GET    | /payroll/payslips/me             | self                 | My payslips               |

#### Payroll run

```
POST /api/v1/payroll/runs
{ "month": "2026-09", "branchIds": ["6650f1..."] }

202 Accepted (calculation runs in the background worker)
{ "id": "66f0...", "status": "DRAFT", "jobId": "payroll:66f0...", "employees": 412 }

GET /api/v1/payroll/runs/66f0...
{ "id": "66f0...", "month": "2026-09", "status": "CALCULATED",
  "summary": { "employees": 412, "gross": 1842345000, "deductions": 211230000,
               "net": 1631115000, "employerPf": 98450000, "employerEsi": 12345000 },
  "warnings": [ { "employeeCode": "EMP0187", "message": "Net pay changed by +38%" } ] }
```

## 15.12 Finance and reports

| Method | Path                               | Permission              | Description               |
|--------|------------------------------------|-------------------------|---------------------------|
| CRUD   | /finance/ledgers                   | finance:ledger:*        | Chart of accounts         |
| POST   | /finance/vouchers                  | finance:voucher:create  | Manual voucher (approval) |
| GET    | /finance/ledgers/{id}/statement    | finance:ledger:read     | Ledger statement          |
| GET    | /finance/trial-balance?as0n=       | finance:report:read     | Trial balance             |
| GET    | /finance/pnl?from=&to=&costCentre= | finance:report:read     | P&L                       |
| GET    | /finance/balance-sheet?as0n=       | finance:report:read     | Balance sheet             |
| GET    | /finance/gst/gstr1?month=          | finance:gst:read        | GSTR-1 data               |
| POST   | /finance/periods/{id}/close        | finance:period:close    | Close month               |
| GET    | /reports/{code}?params             | reports:{module}:read   | Any module report         |
| POST   | /reports/{code}/export             | reports:{module}:export | Async Excel / PDF to S3   |

## 15.13 Medical records, diet, facility and quality

| Method | Path                              | Permission              | Description                |
|--------|-----------------------------------|-------------------------|----------------------------|
| GET    | /mrd/deficiencies?doctorId=       | mrd:record:read         | Pending record items       |
| PUT    | /mrd/records/{admissionId}/coding | mrd:coding:write        | Final ICD-10 codes         |
| POST   | /mrd/files/{id}/issue             | mrd:file:issue          | Issue physical file        |
| POST   | /mrd/release-requests             | mrd:release:request     | Record copy (approval)     |
| POST   | /mrd/births                       | mrd:birth:create        | Birth register entry       |
| POST   | /mrd/deaths                       | mrd:death:create        | Death register entry       |
| GET    | /mrd/deaths/{id}/mccd.pdf         | mrd:death:read          | Cause of death certificate |
| CRUD   | /mrd/mlc                          | mrd:mlc:*               | Medico-legal register      |
| PUT    | /diet/orders/{admissionId}        | diet:order:write        | Diet and NBM               |
| GET    | /diet/production?date=&meal=      | diet:kitchen:read       | Production sheet           |
| POST   | /diet/deliveries                  | diet:kitchen:write      | Confirm ward delivery      |
| GET    | /facility/hk-tasks?status=        | facility:hk:read        | Housekeeping board         |
| POST   | /facility/hk-tasks/{id}/done      | facility:hk:write       | Bed or area cleaned        |
| POST   | /facility/linen/transactions      | facility:linen:write    | Issue / laundry            |
| POST   | /facility/tickets                 | any staff               | Raise maintenance ticket   |
| PATCH  | /facility/tickets/{id}            | facility:ticket:update  | Assign, resolve            |
| CRUD   | /facility/equipment               | facility:equipment:*    | Equipment, PM, AMC         |
| POST   | /facility/bmw/entries             | facility:bmw:write      | Biomedical waste weights   |
| POST   | /quality/incidents                | any staff               | Report incident            |
| PATCH  | /quality/incidents/{id}           | quality:incident:update | RCA, CAPA, close           |
| CRUD   | /quality/complaints               | quality:complaint:*     | Complaint register         |
| GET    | /quality/indicators?month=        | quality:indicator:read  | Indicator values           |
| POST   | /public/feedback/{token}          | patient link            | Submit feedback            |

## 15.14 CRM, ABDM and clinic features

| Method | Path                               | Permission               | Description                   |
|--------|------------------------------------|--------------------------|-------------------------------|
| GET    | /crm/callers/{phone}               | crm:call:read            | Caller lookup                 |
| CRUD   | /crm/leads                         | crm:lead:*               | Leads and follow-ups          |
| CRUD   | /crm/camps                         | crm:camp:*               | Health camps                  |
| POST   | /crm/campaigns                     | crm:campaign:create      | Send campaign (consent-aware) |
| CRUD   | /crm/memberships                   | crm:membership:*         | Memberships and wallet        |
| CRUD   | /crm/referral-partners             | crm:partner:*            | Referral partners             |
| POST   | /abdm/abha/verify                  | patients:abha:link       | Create or verify ABHA         |
| POST   | /abdm/scan-share/callback          | ABDM gateway (signed)    | Scan and Share                |
| POST   | /abdm/care-contexts/link           | patients:abha:link       | Link records to ABHA          |
| POST   | /front-office/tokens               | frontoffice:token:create | Issue counter token           |
| CRUD   | /opd/specialty-forms               | opd:template:*           | Specialty templates           |
| CRUD   | /opd/treatment-plans               | opd:plan:*               | Multi-session plans           |
| POST   | /opd/visits/{id}/certificates      | opd:certificate:create   | Medical certificate           |
| CRUD   | /lab/home-collections              | lab:home:*               | Home sample visits            |
| CRUD   | /lab/collection-centres            | lab:centre:*             | Partner centres               |
| POST   | /nursing/admissions/{id}/icu-chart | nursing:icu:write        | ICU flowsheet                 |
| POST   | /facility/transport-requests       | any staff                | Porter request                |

## 15.15 Front office, portal and other additions

| Method | Path                                  | Permission                       | Description               |
|--------|---------------------------------------|----------------------------------|---------------------------|
| CRUD   | /front-office/enquiries               | frontoffice:enquiry:*            | Enquiry log               |
| POST   | /front-office/passes                  | frontoffice:pass:create          | Attendant / visitor pass  |
| POST   | /front-office/passes/{code}/scan      | frontoffice:pass:scan            | Gate in / out             |
| POST   | /portal/auth/otp                      | patient (public)                 | Portal login OTP          |
| GET    | /portal/me/documents                  | patient                          | Reports, bills, summaries |
| POST   | /portal/me/payments                   | patient                          | Pay dues online           |
| CRUD   | /opd/health-check-packages            | opd:package:*                    | Check-up packages         |
| POST   | /opd/health-checks                    | opd:healthcheck:create           | Book and route sheet      |
| POST   | /rad/studies/{id}/form-f              | rad:pcpndt:write                 | PCPNDT Form F             |
| CRUD   | /hr/requisitions, /hr/candidates      | hr:recruitment:*                 | Recruitment               |
| CRUD   | /hr/trainings, /hr/appraisals         | hr:training:*,<br>hr:appraisal:* | Training, appraisal       |
| POST   | /finance/expense-claims               | self                             | Claim with receipts       |
| CRUD   | /finance/petty-cash, /finance/budgets | finance:pettycash:*              | Petty cash, budgets       |

## 15.16 WebSocket events

| Channel / event          | Payload                                | Who receives                              |
|--------------------------|----------------------------------------|-------------------------------------------|
| beds / bed:update        | bedId, wardId, status, patient summary | Bed board viewers                         |
| queue / queue:update     | doctorId, token, status                | Doctor screen, TV display                 |
| approvals / approval:new | approvalId, action, summary            | Users with checker role                   |
| alerts / lab:critical    | patient, test, value                   | Ordering doctor, ward                     |
| nursing / task:due       | admissionId, task, dueAt               | Assigned ward nurses                      |
| jobs / job:done          | jobId, fileId                          | User who started an export or payroll run |

## SECTION 16

## Deployment on AWS (S3, CloudFront, ECS)

### 16.1 Environments

| Environment | URL pattern                    | Purpose                   | Deploys                         |
|-------------|--------------------------------|---------------------------|---------------------------------|
| dev         | *.dev.example.com              | Developer integration     | Every merge to develop          |
| staging     | *.staging.example.com          | QA, UAT with client, demo | Release branch                  |
| production  | *.example.com + custom domains | Live hospitals            | Tagged release, manual approval |

### 16.2 One-time AWS setup

1. Create three AWS accounts (dev, staging, prod) under AWS Organizations; enable CloudTrail, GuardDuty and AWS Config in all.
2. Register domain in Route 53. Request a wildcard ACM certificate for \*.example.com in **us-east-1** (required by CloudFront) and one in ap-south-1 for the ALB.
3. Apply Terraform in infra/ to create: VPC (2 public, 2 private subnets), S3 web buckets, S3 documents bucket with KMS key, CloudFront distributions with OAC, ALB, ECS cluster and services, ECR repositories, ElastiCache Redis, Secrets Manager entries, IAM roles, CloudWatch alarms, WAF.
4. Create the MongoDB Atlas project in Mumbai, an M30 cluster, PrivateLink endpoint into the VPC, database user with least privilege, continuous backup on.
5. Store secrets (Mongo URI, JWT keys, SMS and payment keys) in Secrets Manager; ECS injects them as environment variables at task start.
6. Create a GitHub OIDC role per account so GitHub Actions deploys without long-lived AWS keys.

### 16.3 React app on S3 and CloudFront

#### Manual deployment commands

```
# Build
pnpm --filter web build # outputs apps/web/dist (index.html + hashed assets)

# Upload hashed assets with long cache, then index.html with no cache
aws s3 sync apps/web/dist s3://hms-web-prod \
  --delete --exclude index.html \
  --cache-control "public,max-age=31536000,immutable"

aws s3 cp apps/web/dist/index.html s3://hms-web-prod/index.html \
  --cache-control "no-cache,no-store,must-revalidate" --content-type text/html

# Only index.html needs invalidation because assets have content hashes
aws cloudfront create-invalidation --distribution-id E1ABCDEF2GHIJK --paths "/index.html"
```

#### S3 bucket policy: only CloudFront can read (bucket stays private)

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Sid": "AllowCloudFrontOAC",
    "Effect": "Allow",
    "Principal": { "Service": "cloudfront.amazonaws.com" },
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::hms-web-prod/*",
    "Condition": { "StringEquals": {
      "AWS:SourceArn": "arn:aws:cloudfront::123456789012:distribution/E1ABCDEF2GHIJK" } }
  }]
}
```

| CloudFront setting | Value                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| Origins            | S3 web bucket (OAC); ALB (HTTPS only, custom header secret so ALB rejects direct traffic)                            |
| Default behaviour  | S3, CachingOptimized, redirect HTTP to HTTPS, compress                                                               |
| /api/*             | ALB, CachingDisabled, origin request policy AllViewer (forwards Host header for tenant resolution), all HTTP methods |
| /socket.io/*       | ALB, CachingDisabled, WebSocket allowed                                                                              |
| Custom errors      | 403 and 404 from S3 return /index.html with 200 so React Router deep links work                                      |
| Security headers   | Response headers policy: HSTS, CSP, X-Frame-Options DENY, Referrer-Policy                                            |
| Alternate names    | *.example.com plus each hospital's custom domain                                                                     |
| WAF                | AWS managed core rule set, SQLi, known bad inputs, IP rate limiting                                                  |

## 16.4 API and worker containers

### apps/api/Dockerfile

```
FROM node:24-alpine AS build
WORKDIR /app
RUN corepack enable
COPY pnpm-lock.yaml pnpm-workspace.yaml package.json ./
COPY apps/api/package.json apps/api/
COPY packages/shared/package.json packages/shared/
RUN pnpm install --frozen-lockfile
COPY . .
RUN pnpm --filter api build && pnpm --filter api deploy --prod /out

FROM node:24-alpine
ENV NODE_ENV=production
WORKDIR /app
RUN addgroup -S app && adduser -S app -G app
COPY --from=build /out .
USER app
EXPOSE 4000
HEALTHCHECK CMD wget -qO- http://localhost:4000/api/health || exit 1
CMD ["node", "dist/server.js"] # worker task overrides with dist/worker.js
```

## 16.5 CI/CD with GitHub Actions

### .github/workflows/deploy-web.yml

```
name: deploy-web
on:
  push:
    branches: [main]
    paths: ['apps/web/**', 'packages/**']
permissions: { id-token: write, contents: read }
jobs:
  deploy:
    runs-on: ubuntu-latest
    environment: production # requires manual approval in GitHub
    steps:
      - uses: actions/checkout@v4
      - uses: pnpm/action-setup@v4
      - uses: actions/setup-node@v4
        with: { node-version: 24, cache: pnpm }
      - run: pnpm install --frozen-lockfile
      - run: pnpm --filter web lint && pnpm --filter web test -- --run
      - run: pnpm --filter web build
      - uses: aws-actions/configure-aws-credentials@v4
        with:
          role-to-assume: arn:aws:iam::123456789012:role/github-deploy-web
          aws-region: ap-south-1
      - run: |
        aws s3 sync apps/web/dist s3://${{ vars.WEB_BUCKET }} --delete \
          --exclude index.html --cache-control "public,max-age=31536000,immutable"
        aws s3 cp apps/web/dist/index.html s3://${{ vars.WEB_BUCKET }}/index.html \
          --cache-control "no-cache,no-store,must-revalidate"
```

```
aws cloudfront create-invalidation \
  --distribution-id ${vars.CF_DISTRIBUTION_ID} --paths "/index.html"
```

### .github/workflows/deploy-api.yml

```
name: deploy-api
on:
  push:
    branches: [main]
    paths: ['apps/api/**', 'packages/**']
permissions: { id-token: write, contents: read }
jobs:
  test:
    runs-on: ubuntu-latest
    services:
      mongo: { image: 'mongo:8', ports: ['27017:27017'], options: '--replSet rs0' }
      redis: { image: 'redis:7', ports: ['6379:6379'] }
    steps:
      - uses: actions/checkout@v4
      - uses: pnpm/action-setup@v4
      - uses: actions/setup-node@v4
        with: { node-version: 24, cache: pnpm }
      - run: pnpm install --frozen-lockfile
      - run: docker exec $(docker ps -qf ancestor=mongo:8) mongosh --eval 'rs.initiate()'
      - run: pnpm --filter api lint && pnpm --filter api typecheck
      - run: pnpm --filter api test -- --run
  deploy:
    needs: test
    runs-on: ubuntu-latest
    environment: production
    steps:
      - uses: actions/checkout@v4
      - uses: aws-actions/configure-aws-credentials@v4
        with:
          role-to-assume: arn:aws:iam::123456789012:role/github-deploy-api
          aws-region: ap-south-1
      - id: ecr
        uses: aws-actions/amazon-ecr-login@v2
      - run: |
          IMAGE=${{ steps.ecr.outputs.registry }}/hms-api:${{ github.sha }}
          docker build -f apps/api/Dockerfile -t $IMAGE .
          docker push $IMAGE
          echo "IMAGE=$IMAGE" >> $GITHUB_ENV
      - run: pnpm --filter api migrate:up # idempotent index + data migrations
      - run: |
          for svc in api worker; do
            ./infra/scripts/ecs-deploy.sh hms-prod hms-$svc "$IMAGE"
          done
          # rolling update, circuit breaker rollback
```

## 16.6 Environment variables

| Variable                          | Example / source        | Notes                        |
|-----------------------------------|-------------------------|------------------------------|
| NODE_ENV                          | production              |                              |
| MONGO_URI                         | Secrets Manager         | Atlas PrivateLink SRV string |
| REDIS_URL                         | rediss://...:6379       | TLS                          |
| JWT_PRIVATE_KEY / JWT_PUBLIC_KEY  | Secrets Manager         | RS256, rotated yearly        |
| S3_DOCS_BUCKET                    | hms-docs-prod           | Access via ECS task role     |
| ROOT_DOMAIN                       | example.com             | Tenant resolution            |
| SMS_PROVIDER, SMS_API_KEY         | msg91 / Secrets Manager | DLT templates in DB          |
| SES_FROM                          | no-reply@example.com    | Verified domain              |
| RAZORPAY_KEY_ID / SECRET          | Secrets Manager         | Patient and SaaS payments    |
| SENTRY_DSN                        | Secrets Manager         | Error tracking               |
| VITE_SENTRY_DSN, VITE_APP_VERSION | Build-time (web)        | No secrets in web build      |

## 16.7 Release and rollback

- Database changes are backwards compatible (expand, then contract), so the old API version can run against the new schema during rolling deploys.
- ECS deployment circuit breaker rolls back automatically if new tasks fail health checks.
- Web rollback: re-run the deploy job for the previous Git tag; S3 versioning also allows restoring the previous index.html in seconds.
- Feature flags per tenant allow releasing to one pilot hospital first.
- Production deploys happen in a low-traffic window (02:00-05:00 IST) with a status page notice for planned downtime, which should be zero for normal releases.

## SECTION 17

# Security, Compliance and Operations

## 17.1 Security controls

| Area           | Control                                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transport      | TLS 1.2+ everywhere, HSTS, internal traffic in private subnets                                                                                        |
| Data at rest   | Atlas encryption, S3 SSE-KMS, encrypted EBS / Redis; field-level encryption for Aadhaar, PAN and bank account numbers                                 |
| Application    | OWASP ASVS L2 checklist; Helmet headers; Zod validation on every input; Mongo operator injection blocked (sanitize \$ keys); output encoding in React |
| Authentication | Argon2id, 2FA, lockout, refresh token rotation with reuse detection, session revocation on password change                                            |
| Authorization  | Server-side RBAC on every route; tenant plugin on every query; automated tests that try cross-tenant access                                           |
| Files          | Type and size allow-list, antivirus scan (ClamAV Lambda) before file is marked available, presigned URLs only                                         |
| Secrets        | Secrets Manager; no secrets in code, images or web build                                                                                              |
| Dependencies   | Dependabot, npm audit in CI, container image scanning in ECR                                                                                          |
| Testing        | Annual third-party penetration test; OWASP ZAP baseline in CI                                                                                         |
| People         | Least-privilege IAM, SSO with MFA for engineers, production access via break-glass role with alerts                                                   |

## 17.2 Compliance

- **DPDP Act 2023 (India).** Consent capture at registration, purpose limitation, data principal rights (access, correction, erasure where law permits), breach notification process, data processing agreement with each hospital.
- **Clinical record retention.** Medical records retained per state rules and hospital policy (commonly 3 years for OPD, longer for IPD and medico-legal cases). Retention is configurable per record type.
- **ABDM readiness.** ABHA number captured; Phase 1 data model maps to FHIR resources so ABDM health-record sharing can be added.
- **NABH support.** Audit trails, consent, medication safety checks, critical value alerts and indicator reports support NABH documentation.
- **HIPAA-style safeguards.** Access logs, minimum necessary access and encryption make the product suitable for international clients later.
- **Data residency.** All Indian tenant data stays in the Mumbai region.

## 17.3 Backup and disaster recovery

| Item           | Policy                                                                                                                                  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| MongoDB        | Continuous backup with point-in-time restore (last 7 days), daily snapshots kept 35 days, monthly kept 1 year; copy to Hyderabad region |
| S3 documents   | Versioning, cross-region replication to ap-south-2, object lock for audit exports                                                       |
| Targets        | RPO 15 minutes, RTO 4 hours for full region loss; RTO 30 minutes for a single-AZ failure (automatic)                                    |
| Restore drills | Quarterly restore test into staging, documented                                                                                         |
| Tenant export  | Self-service full export for the hospital, any time                                                                                     |

## 17.4 Monitoring and support

- Structured JSON logs (Pino) with requestId and tenantId to CloudWatch Logs; PHI is never written to logs.
- Alarms: API 5xx above 1%, p95 latency above 800 ms, queue depth, failed jobs, Mongo CPU and connections, low disk, certificate expiry.
- Sentry for front-end and back-end errors with release tracking.
- Uptime checks on each region endpoint; public status page.
- Support tiers: L1 hospital super user, L2 product support, L3 engineering. Severity 1 (hospital cannot bill or admit) response in 30 minutes, 24x7.

## SECTION 18

# Non-Functional Requirements and Testing

## 18.1 Non-functional requirements

| Category      | Requirement                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Availability  | 99.9% monthly for production (about 43 minutes downtime a month)                                                                                                       |
| Performance   | p95 API under 300 ms for reads and 600 ms for writes at 200 concurrent users per tenant; patient search under 500 ms on 1 million patients; bed board update under 1 s |
| Scalability   | 500 tenants and 20,000 concurrent users on the shared cluster by scaling ECS tasks and Atlas tier; no code change                                                      |
| Front-end     | First load under 3 s on 4G; route chunks lazy-loaded; works on Chrome, Edge, Firefox, Safari (last 2 versions); tablet 768 px+                                         |
| Usability     | Common counter tasks in 3 clicks or fewer; full keyboard use for billing and pharmacy; English and Hindi                                                               |
| Accessibility | WCAG 2.1 AA for colour contrast and keyboard navigation                                                                                                                |
| Printing      | Bills and receipts print in under 2 s; thermal and A4 supported                                                                                                        |
| Offline       | Short network drops tolerated: forms keep unsaved data locally and retry; full offline mode is not in Phase 1                                                          |
| Auditability  | 100% of create, update, approve, print and export actions logged                                                                                                       |

## 18.2 Testing strategy

| Level            | Tool                                    | Scope and target                                                                         |
|------------------|-----------------------------------------|------------------------------------------------------------------------------------------|
| Unit             | Vitest                                  | Services and pure functions (payroll, pricing, FEFO, tax): 80% line coverage on services |
| API integration  | Supertest + Mongo replica set in Docker | Every route: auth, permission, module gate, validation, happy path                       |
| Tenant isolation | Custom suite                            | Every endpoint called with another tenant's IDs must return 404                          |
| Concurrency      | Custom suite                            | Parallel bed allocation, stock sale and counter increment never double-allocate          |
| End-to-end       | Playwright                              | The journeys in Section 11 run on every release candidate                                |
| Performance      | k6                                      | Load profile of a 300-bed hospital at 2x peak                                            |
| Security         | OWASP ZAP, npm audit, pen test          | No high findings open at release                                                         |
| UAT              | Pilot hospital                          | Signed test scripts per module with client users                                         |

## SECTION 19

# Phase 1 Delivery Plan, Commercials and Acceptance

## 19.1 Delivery summary

|               |                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------|
| Scope         | Everything in Sections 2 to 19 of this document: all Phase 1 modules, the SaaS platform and console, API, AWS deployment |
| Delivery time | 1 month (4 weeks) from kickoff to production go-live                                                                     |
| Price         | INR 1,00,000 one-time for Phase 1 delivery, plus GST                                                                     |
| Go-live       | Production on AWS with the first hospital live at the end of week 4                                                      |
| Support       | 30 days of post go-live support (hyper-care) included                                                                    |

## 19.2 Delivery squads

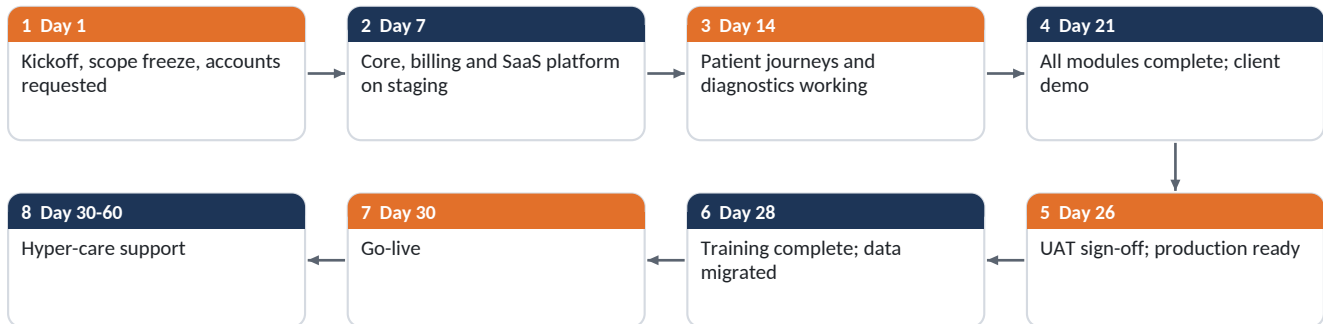
Work runs in parallel squads that build on the shared Core from week 1. Each squad owns its modules end to end: API, screens, tests and reports.

| Squad                 | Owns                                                                                                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| Platform              | SaaS platform and console, tenancy, auth, roles, approvals, audit, subscription billing, AWS, CI/CD             |
| Core and Front Office | Hospital setup, departments, masters, registration, billing engine, printing, front office, patient portal, CRM |
| Clinical              | OPD, IPD and beds, nursing, doctor scheduling, rosters, diet, medical records                                   |
| Diagnostics and Stock | Laboratory, radiology, pharmacy, inventory and purchase                                                         |
| People and Money      | HR, attendance, leave, payroll, finance and accounts                                                            |
| Support and Quality   | Housekeeping and facility, quality and incidents                                                                |
| QA and Release        | Test automation from day 1, performance, security, UAT, deployment                                              |

## 19.3 Four-week plan

| Week | Days  | Deliverables                                                                                                                                                                                                                                                    | Exit check                                         |
|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 1    | 1-7   | Kickoff and scope freeze; AWS, domain, CI/CD, staging live; design system; SaaS tenancy, signup, auth, roles, maker-checker, audit; hospital setup, departments, masters with Excel import; patient registration; billing engine and printing                   | Register a patient and print a bill on staging     |
| 2    | 8-14  | OPD scheduling, queue, consultation, e-prescription, specialty templates; IPD admission, transfer, discharge, live bed board, insurance and TPA desk, packages and day care; nursing station; laboratory; radiology; pharmacy with FEFO; inventory and purchase | Full OPD and IPD patient journey on staging        |
| 3    | 15-21 | HR, rosters, attendance, leave; payroll with statutory outputs; finance auto-posting and statements; medical records; diet; housekeeping and facility; quality; CRM; patient portal; subscription billing; all module reports; OPD and IPD analytics            | Feature complete; demo to hospital; masters loaded |
| 4    | 22-30 | UAT with hospital super users; bug fixing; performance and security tests; production setup and restore drill; training; data migration; go-live and hyper-care start                                                                                           | Acceptance criteria met; hospital live             |

## Milestones



## 19.4 Delivery assumptions

The one-month timeline holds when the following are in place. Delays on these items move the go-live date by the same number of days.

1. Scope is frozen at kickoff as described in this document. New requests are logged and delivered after go-live as change requests.
2. The hospital fills the Excel import templates (tariffs, drugs and items with opening stock, employees and salaries, chart of accounts, opening balances) by day 10.
3. One decision-maker on the hospital side answers questions and approves designs within 24 hours.
4. Accounts are opened in week 1: AWS, domain, MongoDB Atlas, SMS provider with DLT template registration, WhatsApp Business, payment gateway, and video provider if tele-consultation is used. DLT and WhatsApp approvals can take several days, so they start on day 1.
5. Hospital super users are available for UAT and training in week 4.
6. Hardware (printers, barcode scanners, biometric devices, queue TVs) is installed on site by day 21.

## 19.5 Commercial

| Item                                  | Terms                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 1 price</b>                  | INR 1,00,000 one-time, plus GST at 18%                                                                                                            |
| <b>Included</b>                       | Design, development, testing, AWS setup, deployment, data import of masters, training of super users, documentation, 30 days hyper-care           |
| <b>Not included (paid at actuals)</b> | AWS hosting, MongoDB Atlas, domain, SMS and WhatsApp messages, payment gateway fees, video provider, licensed drug-interaction database, hardware |
| <b>Payment milestones</b>             | 40% at kickoff, 30% at week 3 demo, 30% at go-live                                                                                                |
| <b>Change requests</b>                | Estimated and quoted separately; scheduled after go-live                                                                                          |
| <b>After hyper-care</b>               | Annual maintenance and support on a separate agreement, or covered by the SaaS subscription for hosted hospitals                                  |

## 19.6 Go-live checklist for each hospital

1. Tenant created with plan, modules, limits and Super Admins.
2. Hospital profile, branches, departments, wards and beds configured.
3. Masters imported: services and tariffs, drugs and items with opening stock, employees, salary structures, chart of accounts with opening balances.
4. Users created, roles assigned, 2FA enrolled for privileged users.
5. Print templates and SMS templates approved by the hospital.
6. Printers, barcode scanners, biometric devices and TV queue screens tested.

7. Role-wise training completed, with sign-off; super users identified per department.
8. Parallel run of 2-3 days for billing and pharmacy if migrating from old software.
9. Go-live day support on site; hyper-care for 30 days.

## 19.7 Phase 1 acceptance criteria

- Every user flow in Sections 3 to 11 runs end to end on staging with the pilot hospital's own masters.
- A module that is not subscribed is invisible in the menu and its API returns 402.
- No maker can approve their own request; all rules in Section 4 are enforced.
- Bills, receipts, lab reports, discharge summaries and payslips print correctly on A4 and, where specified, thermal printers.
- Payroll for one month matches the hospital's manual calculation for a sample of 50 employees, including PF, ESI, PT and TDS.
- Trial balance balances after a full test month of transactions.
- Performance targets in Section 18 met; no open high-severity security findings.
- Production on AWS with web on S3 + CloudFront, automated deploys, backups and alarms verified by a restore drill.

## 19.8 Glossary

| Term                | Meaning                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------|
| UHID                | Unique Health ID, the hospital's permanent patient number                                        |
| OPD / IPD           | Out-patient department / in-patient department                                                   |
| ADT                 | Admission, discharge and transfer                                                                |
| MAR                 | Medication administration record                                                                 |
| LIS / RIS / PACS    | Laboratory, radiology information systems; picture archiving and communication system for images |
| FEFO                | First-expiry-first-out stock picking                                                             |
| GRN                 | Goods receipt note                                                                               |
| LOP                 | Loss of pay (unpaid days)                                                                        |
| PF / ESI / PT / TDS | Provident fund, employee state insurance, professional tax, tax deducted at source               |
| Maker-checker       | Two-person rule: one proposes, another approves                                                  |
| Tenant              | One hospital or hospital group using the SaaS                                                    |
| OAC                 | CloudFront Origin Access Control, lets CloudFront read a private S3 bucket                       |
| RPO / RTO           | Maximum data loss / maximum downtime after a disaster                                            |